Applied Statistics

LaTeX by Min Soo Jeon
mjeon@uc.cl
Professor Piercesare Secchi
Politecnico di Milano

Semester 2025-1

Disclaimer:

Version 1.0.0

Introduction

These notes are based on the lectures delivered by Professor Piercesare Secchi during the Applied Statistics course at Politecnico di Milano during Semester 2025-1. Official course information can be found on the official website (*Accessed: 2025-02-15*, 2024). Some sections, particularly the practical components, were taught by Teaching Assistant Guillaume Koechlin. Further details on the practical part can be found in the provided resources. Section 15 was taught by Professor Francesca Leva. Section 17 was taught by Professor Alessandra Menafoglio. Section 18 was taught by Professor Laura Sangalli. This material was prepared independently by the author(s) and has not been reviewed or approved by the professors named herein.
This material may be outdated, the latest version is available in the repository. To report an error, please contact mjeon@uc.cl.

Contents

1 Introduction

1.1 Definitions

Every data analysis starts with a problem rather than data. Data is the tool used to solve the questions we formulate based on the problem. Data is able to solve a limited set of questions, so we should find out what are the answers we can extract from data.

Definition 1.1: Dataframe

A set of n data units with p different dimensions, each notated as $X_1, X_2, \dots, X_p$. The data of the unit i in the dimension j is notated as: x_{ij}

Each value in x_{ij} could be categorical, ordinal, and numerical

Notate it as:

$$\mathbb{X} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ & \dots & \dots & \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

In order to analyze a dataset, it is necessary to preprocess it to prepare it for the statistical analysis. Anyway, this is not a linear process. While we do our analysis, we gain knowledge of the data, and we may preprocess it again based on new knowledge, being a non-linear process.

Anyway, our start point analysis will be from the 'clean' dataset.

It is said row perspective when we manage and think of data as rows,
i.e. $\underline{x}_i \in \mathbb{R}^p, \quad i = 1, \dots, n$. It provides us with a global perspective of the data and how it is inter-related, allowing us to find patterns in the data.

It is said column perspective when we manage and think of data as columns,
i.e. $\underline{c}_j \in \mathbb{R}^n, \quad j = 1, \dots, p$.

We are particularly interested in the vector $\underline{1} = (1, \dots, 1)^T \in \mathbb{R}^n$, whose span represents the situation where the data has no variability (and no informative statistic), i.e. if we know the data of 1 unit, we know the data of every unit in the population.

The best approximation from a vector c_j to $\mathcal{L}_1 = span(\underline{1})$, is the projection $\Pi_{\underline{1}}(c_j)$. The deviation of this approximation is $\underline{d}_1 = \underline{c}_1 - \Pi_{\underline{1}}(c_j)$. Note that $\dim(\Pi_{\underline{1}}(c_j)) = \dim(\mathcal{L}_\infty)$, and $\dim(\underline{d}_1) = \dim(\mathcal{L}_\infty^\perp)$.

1.2 Hilbert Space

When we work with data, we work on Hilbert spaces ($\mathbb{R}^n$), so remember:

- $\underline{n}, \underline{v} \in \mathbb{R}^n$
- $\|\underline{u}\| = \sqrt{\underline{u}^T \underline{u}} = \sum_{i=1}^n u_i^2 = \sqrt{\langle \underline{u}, \underline{u} \rangle}$

- $\|u\|^2 = \langle u, u\rangle = u^T u$
- The angle θ between two vectors is:

$$\cos\theta = \frac{\langle u, v\rangle}{\|u\|\|v\|} = \frac{\sum_{i=1}^n u_i v_i}{\sqrt{\sum_{i=1}^n u_i^2 \sum_{i=1}^n v_i^2}}$$

The projection of a vector is calculated by:

$$\begin{aligned}
\Pi_{\underline{v}}(\underline{u}) &= \|\underline{u}\| \cos\theta \frac{\underline{v}}{\|\underline{v}\|} \\
&= \|\underline{u}\| \frac{\langle u, v\rangle}{\|u\|\|v\|} \frac{\underline{v}}{\|\underline{v}\|} = \frac{\langle u, v\rangle}{\|v\|^2}\underline{v} \\
&= \frac{\underline{v}^T \underline{u}}{\underline{v}^T \underline{v}} \cdot \underline{v} = \frac{\underline{v} \cdot \underline{v}^T}{\underline{v}^T \underline{v}} \cdot \underline{u}
\end{aligned}$$

Note that, by property of Hilbert Spaces:

$$\|u\|^2 = \|\Pi_{\underline{v}}(\underline{u})\|^2 + \|u - \Pi_{\underline{v}}(\underline{u})\|^2$$

1.3 Derivation of mean and variance

Applying the projection formula:

$$\Pi_{\underline{1}}(\underline{c}_1) = \frac{\underline{1} \cdot \underline{1}^T}{\underline{1}^T \underline{1}} \cdot \underline{c}_1 = \frac{\sum_{i=1}^n x_{i1}}{n} \cdot \underline{1} = \bar{x}_1 \cdot \underline{1}$$

Using this for the deviation of the approximation:

$$\underline{d}_1 = \underline{c}_1 - \bar{x}_1 \cdot \underline{1} = \begin{pmatrix} x_{1i} - \bar{x}_1 \\ x_{2i} - \bar{x}_1 \\ \dots \\ x_{pi} - \bar{x}_1 \end{pmatrix}$$

Then:

$$\begin{aligned}
\|d_1\|^2 = \underline{d}_1^T \underline{d}_1 &= \sum_{i=1}^n (x_{i1} - \bar{x}_1)^2 \\
&= (n-1) \underbrace{\frac{1}{n-1} \sum_{i=1}^n (x_{i1} - \bar{x})^2}_{\text{sample variance}} = (n-1) S_{11}^2
\end{aligned}$$

Note that we define the standard deviation as:

$$\text{std. dev.} := \sqrt{S_{11}}$$

This is an important value, since when we analyze a data point, a statistical description is how many standard deviations it is from the mean. The variability is the key to describing the data properly.

1.4 Chebychev Inequality

Let be X a random variable with mean and standard deviation μ, σ, then:

$$P\left[\mu - k\sigma < x < \mu + k\sigma\right] > 1 - \frac{1}{k^2}$$

1.5 Correlation derivation

Now take two different columns:

$$\underline{c}_1 = \Pi_{\underline{c}_2}(\underline{c}_1) + \underline{d}_{12}$$

$$\underline{d}_{12} = \underline{c}_1 - \Pi_{\underline{c}_2}(\underline{c}_1)$$

The angle between $\underline{c}_1$ and $\underline{c}_2$ is very relevant.

On the other hand, we can compare $\underline{d}_1$ and $\underline{d}_2$

$$\underline{d}_1 = \Pi_{\underline{d}_2}(\underline{d}_1) + (\underline{d}_1 - \Pi_{\underline{d}_2}(\underline{d}_1))$$

if $\theta = 0 \qquad \Longrightarrow \qquad \underline{d}_1 = \beta \underline{d}_2$
:

$$\begin{aligned}
&\Longrightarrow \underline{c}_1 - \bar{x}_1 \cdot \underline{1} = \beta(\underline{c}_2 - \bar{x}_2 \cdot \underline{1}) \\
&\Longrightarrow \underline{c}_1 = (\bar{x}_1 - \beta \bar{x}_2) \cdot \underline{1} + \beta \underline{c}_2
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\cos\theta_{12} &= \frac{\langle \underline{d}_1, \underline{d}_2 \rangle}{\|\underline{d}_1\| \|\underline{d}_2\|} = \frac{\underline{d}_1^T \underline{d}_2}{\sqrt{\underline{d}_1^T \underline{d}_1 \cdot \underline{d}_2^T \underline{d}_2}} \\
&= \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2))}{\sqrt{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2 \sum_{i=1}^n (x_{i2} - \bar{x}_2)^2}} = \frac{S_{12}}{\sqrt{S_{11} \cdot S_{22}}} \\
&= r_{12}
\end{aligned}$$

Note that the covariance(x_1, x_2) is defined by:

$$S_{12} := \frac{1}{n-1} \sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$$

And r_{12} the correlation between x_1 and x_2, and $r_{12} \in [-1, 1]$
We have the following situations:

- $\theta = 0 \iff r_{12} = 1$
- $\theta = \pi \iff r_{12} = -1$
- $\theta = \pi/2 \iff r_{12} = 0$

- $0 < \theta < \pi/2 \iff r_{12} \in (0, 1)$

Resuming:

- Data: $\mathbb{X} \in \mathbb{R}^{n \times p}$
- Mean Vector:

$$\underline{\bar{x}} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \dots \\ \bar{x}_p \end{pmatrix} \in \mathbb{R}^p$$

- Covariance Matrix:

$$S = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ s_{21} & s_{22} & \dots & s_{2p} \\ & \dots & \dots & \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{bmatrix} \in \mathbb{R}^{p \times p}$$

- Correlation Matrix:

$$R = \begin{bmatrix} 1 & r_{12} & \dots & r_{1p} \\ r_{21} & 1 & \dots & r_{2p} \\ & \dots & \dots & \\ r_{p1} & r_{p2} & \dots & 1 \end{bmatrix} \in \mathbb{R}^{p \times p}$$

2 The problem

2.1 The data

The data received consists of

- A feature, regressor, or covariance matrix:

$$\mathbb{X} = \begin{bmatrix} x_{ij} \end{bmatrix} \in \mathbb{R}^{n \times p}$$

- A target:

$$y = (y_1, y_2, \dots, y_n)^T \in \mathbb{R}^n$$

2.2 Supervised Problems

There are problems where both $(x_1, \dots, x_p)^T = \underline{X}$ and y are observed on n stat units. Then, the association between $\underline{X}$ and y is directly observed in the training set.

Unsupervised problems are the problems where the target y is hidden. We can explore the data to find significant differences between data groups (its data distribution), estimate labels, and transform the problem into supervised learning.

The supervised problem's goal is to find:

$$f : \mathbb{R}^p \to \mathbb{R}$$

or

$$f : \mathbb{R}^p \to \{a, b, c, d, \dots\}$$

such that $f(\underline{X})$ is "optimal" for explaining the variation of y.

Definition 2.1: Optimality problem in supervised learning

Find $f : \mathbb{R}^p \to \mathbb{R} \quad st$

$$f = \arg\min_f E\left[(f(\underline{X}) - y)^2\right]$$

Proposition 2.2

The solution to the supervised learning problem is $E[y|\underline{X}])$

Proof. Note that:

$$\begin{aligned}
E\left[(f(\underline{X}) - y)^2\right] &= E\left[(f(\underline{X}) - E[y|\underline{X}] + E[y|\underline{X}] - y)^2\right] \\
&= E\left[(f(\underline{X}) - E[y|\underline{X}])^2\right] + E\left[(E[y|\underline{X}] - y)^2\right] + 2E\left[(f(\underline{X}) - E[y|\underline{X}])(E[y|\underline{X}] - y)\right]
\end{aligned}$$

Since:

$$\begin{aligned}
E\left[(f(\underline{X}) - E[y|\underline{X}])(E[y|\underline{X}] - y)\right] &= E\left[E\left[(f(\underline{X}) - E[y|\underline{X}])(E[y|\underline{X}] - y)|\underline{X}\right]\right] \qquad \text{(Conditional E theorem)} \\
&= E\left[(f(\underline{X}) - E[y|\underline{X}])|X\right] E\left[(E[y|\underline{X}] - y)|X\right] \\
&= E\left[(f(\underline{X}) - E[y|\underline{X}])\right] \underbrace{(E[y|\underline{X}] - E[y|\underline{X}])}_{0} \\
&= 0
\end{aligned}$$

Then,

$$E\left[(f(\underline{X}) - y)^2\right] = E\left[(f(\underline{X}) - E[y|\underline{X}])^2\right] + E\left[(E[y|\underline{X}] - y)^2\right]$$

As the second term is uncontrollable, the minimum argument is the one that makes the first term 0:

$$\arg\min_{f:\mathbb{R}^p \to \mathbb{R}} E\left[(f(\underline{X}) - y)^2\right] = E[y|\underline{X}]$$

□

This is actually the projection of y in the integrable $\sigma(\underline{X})$, the sigma-algebra of $\underline{X} \in L^2$.
Now, we can express y as:

$$y = E[y|\underline{X}] + \varepsilon$$

There always exists ε unless it is a deterministic problem, i.e. all the information to determine y is in X (but we are not interested in them)
Now, a general model is:

$$y = f(\underline{X}) + \varepsilon \qquad st \quad \varepsilon \perp \sigma(\underline{X}) \quad \text{i.e.} \quad E[\varepsilon|\underline{X}] = 0$$

Being $f(\underline{x}) = E(y|\underline{x})$ the regression function.

Assume the variance is finite:

$$Var(\varepsilon) = E[\epsilon^2] = \sigma^2 < \infty \quad \text{(since } E[\epsilon] = 0)$$

Use a training set $(\mathbb{X}, y)$ to estimate f using $\hat{f} : \mathbb{R}^p \to \mathbb{R}$, with $\hat{f}$ known once training data have been observed.

$$\mathcal{F} : data \to \hat{f}$$

$\mathcal{F}$ is the estimator, while $\hat{f}$ is the estimate.

2.3 K-nearest neighbors

Definition 2.3: K-nearest neighbors regression

Let be $k \geq 1$. Define:

$$N_k(\underline{x}) = \{k \quad \underline{x}_i \in \mathbb{R} \text{ closest to } \underline{x}\}$$

$$E[y|x] = \frac{1}{k} \sum_{i \in N_k(\underline{x})} y_i$$

Definition 2.4: K-nearest neighbors classification

$$\hat{f}(\underline{x}) = \begin{cases} 1 & \text{if there are more 1 than 0 in } N_k(\underline{x}) \\ 0 & \text{otherwise} \end{cases}$$

This is called vote by majority.

2.4 Curse of Dimensionality

Given a space or radius 1 in $\mathbb{R}^N$, the percentage of the space that a ball of radius r achieves is bigger as N is bigger:
For example, take r=0.1:

- **N=1**:
$$area = \frac{0.1}{1} = 0.1$$
- **N=2**:
$$area = \frac{0.1^2\pi}{1^2\pi} = 0.316$$
- **N=3**:
$$area = \frac{0.1^3\pi \cdot 4/3}{1^3\pi \cdot 4/3} = 0.464$$

  ...

- **N=100**:

$$area = 0.1^{1/100} = 0.977$$

Therefore, the neighborhood methods don't work in high dimensionality.

A solution could be to reduce the problem's dimensionality. One possible approach is to embed the space in a smaller one (e.g., PCA (see section 5), ICA, among others).

Another way is finding $\hat{f}$ among using a fewer number of parameters $\{f_\theta \quad st \quad \theta \in \Theta\}$.

2.5 Generalization Error

$\hat{f}$ estimate of f when data are known.
If we have $x_0 \notin \mathbb{X}$, we would predict y_0 with $\hat{f}(\underline{x}_0)$, and:

$$y_0 = f(\underline{x}_0) + \varepsilon_0$$

Then, the generalization error is

$$\begin{aligned} E_{data}\left[(y_0 - \hat{f}(x_0))^2\right] &= E_{data}\left[(f(\underline{x}_0) + \varepsilon_0 - \hat{f}(\underline{x}_0))^2\right] \\ &= E_{data}\left[(f(\underline{x}_0) - \hat{f}(\underline{x}_0))^2\right] + E_{data}\left[\varepsilon_0^2\right] + 2E_{data}\left[\varepsilon_0(f(\underline{x}_0) - \hat{f}(\underline{x}_0))\right] \\ &= \underbrace{(f(\underline{x}_0) - \hat{f}(\underline{x}_0))^2}_{\text{Reducible}} + \underbrace{Var(\epsilon_0)}_{\text{Irreducible}} \end{aligned}$$

Now,

$$\begin{aligned} Err(\hat{f}) = E\left[E_{data}\left[(y_0 - \hat{f}(x_0))^2\right]\right] &= E\left[(f(\underline{x}_0) - \hat{f}(\underline{x}_0))^2\right] + Var(\epsilon_0) \\ &= E\left[\left(f(\underline{x}_0) - E\left[\hat{f}(\underline{x}_0)\right] + E\left[\hat{f}(\underline{x}_0)\right] - \hat{f}(\underline{x}_0)\right)^2\right] + Var(\epsilon_0) \\ &= \underbrace{\left(f(x_0) - E\left[\hat{f}(x_0)\right]\right)^2}_{bias^2(\hat{f})} + \underbrace{Var(\hat{f}(\underline{x}_0))}_{Var(\hat{f})} + Var(\epsilon_0) \end{aligned}$$

There is a trade-off between bias and variance when we change the complexity of the function we choose:

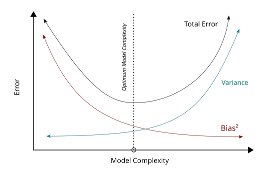

Figure 1: Representation Variance-Bias trade-off

3 Estimators for Mean and Variance

3.1 Probability Distribution

We assume that our data $\underline{x}_1, \underline{x}_2, \ldots, \underline{x}_n$ are iid$\sim$ $\underline{X} \in \mathbb{R}^p$, where iid means independently and identically distributed. In other words, $\underline{x}_i$ are realizations of X random variable.
Plus, it is defined a probability distribution ν:

$$\nu_{\underline{x}} : \mathbb{B} \to [0,1]$$
$$\nu_{\underline{x}}(\mathbb{B}) = P[\underline{X} \in \underline{\mathbb{B}}]$$

Special case:

$$\nu_{\underline{x}}(\mathbb{B}) = \int_{\mathbb{B}} f(\underline{x}) d\underline{x}$$

where f is the density of $\nu_{\underline{x}}$ or of $\underline{\text{x}}$.
Summaries of $\nu_{\underline{x}}$:

- Mean:

$$\underline{\mu} = E[\underline{X}] = \Big[\mu_1, \ldots, \mu_p\Big]^T$$

  Where $\mu_j = E[X_j] \quad j = 1, \ldots, p$

- The Covariance Matrix:

$$\Sigma = E[(\underline{X} - \underline{\mu})(\underline{X} - \underline{\mu})^T]$$

  i.e. $\Sigma = [\sigma_{ij}]_{p \times p}$ matrix
  Where $\sigma_{ij} = cov(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$

- The Correlation Matrix:

$$\rho = v^{-1/2} \Sigma v^{-1/2} = Cov(\underline{X}_{std})$$

  where:

$$V = \begin{bmatrix} \sigma_{11} & \ldots & 0 \\ 0 & \ldots & 0 \\ \ldots & \ldots & \ldots \\ 0 & \ldots & \sigma_{pp} \end{bmatrix}$$

$$V^{-1/2} = \begin{bmatrix} 1/\sqrt{\sigma_{11}} & \ldots & 0 \\ 0 & \ldots & 0 \\ \ldots & \ldots & \ldots \\ 0 & \ldots & 1/\sqrt{\sigma_{pp}} \end{bmatrix}$$

3.2 Linear combination of components of $\underline{\mathbf{X}}$

Let be p=2,

$$\underline{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \qquad \underline{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Consider the linear combination:

$$c_1X_1 + c_2X_2$$

The mean is:

$$E[c_1X_1 + c_2X_2] = c_1E[X_1] + c_2E[X_2] = c_1\mu_1 + c_2\mu_2$$

$$E[\underline{c}^T\underline{X}] = \underline{c}^T\underline{\mu}$$

The Variance is:

$$\begin{aligned} Var(c_1X_1 + c_2X_2) &= c_1^2Var(X_1) + c_2^2Var(X_2) + 2c_1c_2Cov(X_1, X_2) \\ &= c_1^2\sigma_{11} + c_2^2\sigma_{22} + 2c_1c_2\sigma_{12} \\ &= \underline{c}^T\Sigma\underline{c} \qquad , \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \end{aligned}$$

Anyway in general, for $x \in \mathbb{R}^p$: If $\underline{c} \in \mathbb{R}^p$

- $E[\underline{c}^T\underline{X}] = \underline{c}^T\underline{\mu}$
- $Var(c_1X_1 + c_2X_2) = \underline{c}^T\Sigma\underline{c}$

If $C \in \mathbb{R}^{k\times p}$

- $E[C\underline{X}] = C\underline{\mu}$
- $Cov(C\underline{X}) = C\Sigma C^T$

3.3 Estimators for μ and Σ

$\underline{\mu}$ and Σ are often unknown parameters, so we use data to estimate them.

- **Estimator for $\underline{\mu}$**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n}\underline{X}_i$$

- **Estimator for Σ**
  - In the case $\underline{\mu}$ is unknown

$$S = \frac{1}{n-1}\sum_{i=1}^{n}(\underline{X}_i - \underline{\bar{X}})(\underline{X}_i - \underline{\bar{X}})^T$$

  - In the case $\underline{\mu}$ is known

$$S_n = \frac{1}{n}\sum_{i=1}^{n}(\underline{X}_i - \underline{\mu})(\underline{X}_i - \underline{\mu})^T$$

The degrees of freedom are less in the case of $\underline{\mu}$ unknown

Proposition 3.1

The estimators are unbiased:

- $\bar{X}$ is unbiased for $\underline{\mu}$
$$E[\bar{X}] = \underline{\mu}$$
- S is unbiased for Σ
$$E[S] = \Sigma$$

Proof.

$$E[\underline{\bar{X}}] = E\left[\begin{bmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \dots \\ \bar{X}_p \end{bmatrix}\right] = \begin{bmatrix} E[\bar{X}_1] \\ E[\bar{X}_2] \\ \dots \\ E[\bar{X}_p] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_p \end{bmatrix}$$

On the other hand,

$$\begin{aligned} Cov(\underline{\bar{X}}) &= E[(\underline{\bar{X}} - \underline{\mu})(\underline{\bar{X}} - \underline{\mu})^T] \\ &= E\left[\left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \underline{\mu})\right)\left(\frac{1}{n}\sum_{j=1}^{n}(X_j - \underline{\mu})\right)^T\right] \\ &= \frac{1}{n^2}\sum_i\sum_j E\left[(X_i - \underline{\mu})(X_j - \underline{\mu})^T\right] \end{aligned}$$

Note that as X_i and X_j are independent for $i \neq j$, the expectation over the multiplication is the multiplication over the expectations. And $E(X_i - \underline{\mu}) = 0$, so

$$Cov(\underline{\bar{X}}) = \frac{1}{n^2}\sum_i E\left[(X_i - \underline{\mu})^2\right] = \frac{1}{n^2}n\Sigma = \frac{1}{n}\Sigma$$

On the other hand, the rest is left as an exercise:

$$E\left[\sum_{i=1}^{n}(\underline{X_i} - \underline{\bar{X}})(\underline{X_i} - \underline{\bar{X}})^T\right] = (n-1)\Sigma$$

□

Being unbiased is not enough for being a good estimator. We also need low variability

3.4 Column Perspective of S

Let analyze the estimator

$$S = \frac{1}{n-1}\sum_{i=1}^{n}(\underline{X}_i - \underline{\bar{X}})(\underline{X}_i - \underline{\bar{X}})^T$$

In a column perspective, i.e. $\mathbb{X} = [\underline{c}_1, \dots, \underline{c}_p]$, in $\mathbb{R}^n$

$$\underline{d}_j = \underline{c}_j - \frac{11^T}{1^T1}\underline{c}_j = \left[I - \frac{11^T}{1^T1}\right]\underline{c}_j$$

Define:

$$\mathbb{d} = [\underline{d}_1, \dots, \underline{d}_p] = \left(I - \frac{11^T}{1^T1}\right)\mathbb{X}$$

Then:

$$\begin{aligned} S = \begin{bmatrix} S_{11} & \dots & S_{1p} \\ S_{21} & \dots & S_{2p} \\ \dots & \dots & \dots \\ S_{p1} & \dots & S_{pp} \end{bmatrix} &= \frac{1}{n-1}\begin{bmatrix} \underline{d}_1^T\underline{d}_1 & \dots & \underline{d}_1^T\underline{d}_p \\ \underline{d}_2^T\underline{d}_1 & \dots & \underline{d}_2^T\underline{d}_p \\ \dots & \dots & \dots \\ \underline{d}_p^T\underline{d}_1 & \dots & \underline{d}_p^T\underline{d}_p \end{bmatrix} = \frac{1}{n-1}\mathbb{d}^T\mathbb{d} \\ &= \frac{1}{n-1}\mathbb{X}^T\left(I - \frac{11^T}{1^T1}\right)\left(I - \frac{11^T}{1^T1}\right)\mathbb{X} \end{aligned}$$

Having a column perspective of S

3.5 Variability of data

There are different ways to measure the variability in the data:

- Generalized variance: Measures the variability in each column considering their correlation
  - Sample: Det(S)
  - Population: Det(Σ)
- Total variance: Measures the variability in each column independently of their correlation
  - Sample: trace(S)
  - Population: trace(Σ)

Example.

Let be p=2:

$$S = \frac{1}{n-1}\begin{bmatrix} \underline{d}_1^T\underline{d}_1 & \underline{d}_1^T\underline{d}_2 \\ \underline{d}_1^T\underline{d}_2 & \underline{d}_2^T\underline{d}_2 \end{bmatrix} = \frac{1}{n-1}\begin{bmatrix} \|d_1\|^2 & \cos\theta_{12}\|d_1\|\|d_2\| \\ \cos\theta_{12}\|d_1\|\|d_2\| & \|d_2\|^2 \end{bmatrix}$$

Then,

$$\begin{aligned} Det(S) &= \frac{1}{(n-1)^2}\left(\|d_1\|^2\|d_2\|^2 - (\cos^2\theta_{12}\|d_1\|^2\|d_2\|^2)\right) \\ &= \frac{1}{(n-1)^2}\left(\sin^2\theta_{12}\|d_1\|^2\|d_2\|^2)\right) = \frac{1}{(n-1)^2}\left(\sin\theta_{12}\|d_1\|\|d_2\|\right)^2 \end{aligned}$$

> Therefore, note that:
> $$Det(S) \propto (\|d_1\| \sin\theta_{12} \|d_2\|)^2$$
> That's the square of the parallelogram area with base $\|d_2\|$ and height $\|d_1\|$.
>
> On the other side, total variance is calculated as:
> $$\text{trace}(S) = \|d_1\|^2 + \|d_2\|^2$$

Anyway, it can be generalized for a general p:

- General Variance $\propto Volume^2(\text{parallelotope}(\underline{d}_1, \dots, \underline{d}_p))$
- Total Variance: $\|\underline{d}_1\|^2 + \cdots + \|\underline{d}_p\|^2$

3.6 Dependency of variations

Proposition 3.2

Det(S)=0 $\iff$ $\underline{d}_1, \dots, \underline{d}_p$ are linear dependent

Proof. $\leftarrow$
$$\exists \underline{\alpha} \neq 0 \quad st \quad \alpha_1 \underline{d}_1 + \cdots + \alpha_p \underline{d}_p = 0$$
So $\mathbb{d}\underline{c} = \underline{0}$ Since $S = \frac{1}{n-1}\mathbb{d}^T\mathbb{d}$
$$S\underline{c} = \frac{1}{n-1}\mathbb{d}^T\mathbb{d}\underline{c} = \underline{0}$$
So columns of S are linearly dependent, and det(S) = 0.
$\rightarrow$ Assume $\det S = 0$
Then $\exists \underline{c} \neq 0 \quad st \quad S\underline{c} = \underline{0}$ So
$$\begin{aligned} 0 = \underline{c}^T S \underline{c} &= \frac{1}{n-1}\underline{c}^T \mathbb{d}^T \mathbb{d}\underline{c} \\ &= \frac{1}{n-1}\|\mathbb{d}\underline{c}\|^2 \end{aligned}$$
$$\implies \|\mathbb{d}\underline{c}\| = 0 \implies \mathbb{d}\underline{c} = 0$$
So $\underline{d}_1 + \cdots + \alpha_p \underline{d}_p$ are lin. dep. □

This is an important result since If $\underline{d}_1, \dots, \underline{d}_p$ are linear dependent,
$$\exists \underline{\alpha} \in \mathbb{R}^p \quad st \quad \alpha_1 \underline{d}_1 + \cdots + \alpha_p \underline{d}_p = 0 \quad \wedge \quad \underline{\alpha} \neq 0$$
Without loss of generality (WLOG), assume $\underline{\alpha}_1 \neq 0$.
$$\left[\frac{11^T}{1^T 1}\right] \underline{c}_1 = -\frac{\alpha_2}{\alpha_1}\underline{d}_2 - \frac{\alpha_3}{\alpha_1}\underline{d}_3 - \cdots - \frac{\alpha_p}{\alpha_1} d_p$$

Then, we can write c_1 as a linear combination of the other columns:

$$\underline{c}_1 = \frac{1}{\left[\frac{11^T}{1^T 1}\right]} \left[-\frac{\alpha_2}{\alpha_1}\underline{d}_2 - \frac{\alpha_3}{\alpha_1}\underline{d}_3 - \cdots - \frac{\alpha_p}{\alpha_1}\underline{d}_p \right]$$

Proposition 3.3

Let $\mathbb{X} \in \mathbb{R}^{n\times p}$ data.
If $p \geq n \quad \Longrightarrow \quad \det S = 0$

Remark.

> I.e., having more columns than rows in the data will necessarily imply an overfitting

Proof. Let be $\underline{d}_1, \ldots, \underline{d}_p \in \mathbb{R}^n$

$$\underline{d}_j = \underline{c}_j - \frac{11^T}{1^T 1}\underline{c}_j \in \mathcal{L}^{\perp}(1), \quad \dim(\mathcal{L}^{\perp}(1)) = n - 1$$

So $\underline{d}_1, \ldots, \underline{d}_p \in \mathbb{R}^n \in \mathcal{L}^{\perp}(1)$, but the max number of linear independent vectors in $\mathcal{L}^{\perp}(1)$ is n-1.
Therefore the vectors are linearly dependent □

4 Important concepts

4.1 Positive Definite

Definition 4.1: Positive semidefined

S is called positive semidefined if

$$\underline{c}^T S \underline{c} \geq 0 \quad \forall \underline{c} \in \mathbb{R}^p$$

Proposition 4.2

If S is positive semidefined $\det(S) > 0$ then S is positive defined

Remark.

> If $\underline{X} \sim \underline{\mu}, \Sigma$
> $$\underline{c}^T \Sigma \underline{c} = Var(\underline{c}^T \underline{X}) \geq 0$$

Proof. $S = \frac{1}{n-1}\mathbb{d}^T\mathbb{d}$

Note that, indeed, S is semidef. pos:

$$\forall c \neq 0 \quad \underline{c}^T S \underline{c} = \frac{1}{n-1} \underline{c}^T \mathbb{d}^T \mathbb{d} \underline{c} = \frac{1}{n-1} \|\underline{\mathbb{d}c}\| \geq 0$$

Now, by contradiction, assume $\exists \underline{c} \neq 0 \quad st \quad \underline{c}^T S \underline{c} = 0$

$$\implies \|\underline{\mathbb{d}c}\| = 0 \implies \underline{\mathbb{d}c} = 0 \implies \underline{d}_1, \ldots, \underline{d}_p \text{ are lim dependent } \implies \det(S) = 0$$

$$\implies \det(S) \neq 0 \implies \text{S is pos. def.}$$

This is a contradiction, since $\det S > 0$
Therefore, S is positive definited $\square$

Note that $S \in \mathbb{R}^{p \times p}$ is symmetric.
Then, it can be represented as

$$S = \sum_{i=1}^{p} \lambda_i \underline{e}_i \underline{c}_i^T \qquad \text{(spectral decomposition)}$$

where

$$\underline{e}_1, \ldots, \underline{c}_p \in \mathbb{R}^p$$

are pairwise independent, so: $\underline{e}_i^T \underline{e}_j$ is 1 if i=j and 0 if $i \neq j$.
Then, $\{\underline{e}_1, \ldots, \underline{c}_p\}$ is an orthonormal base for $\mathbb{R}^p$.
And $\forall i = 1, \ldots, p \quad (\lambda_i, \underline{e}_i)$ are the (eigenvalue, eigenvector) pairs Plus, S is semipositive def, so $\implies \lambda_i \geq 0 \quad i = 1, \ldots, p$.
Moreover, note that:

- General Variation:

$$\det(S) = \prod_{i=1}^{p} \lambda_i$$

  Then, if $\det(S) > 0 \implies \lambda_i > 0 \quad \forall i$

- Total variation:

$$tz(S) = \sum_{i=1}^{p} \lambda_i$$

Convention: From now on:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$$

which are the values associated with $\underline{e}_1, \ldots, \underline{e}_p$ respectively.

4.2 Mahalanobis

Consider $B \in \mathbb{R}^{p \times p}$ a positive def. matrix and $\underline{x} \in \mathbb{R}^p$. Q is a quadratic form:

$$Q(\underline{x}) = \underline{x}^T B \underline{x}$$

Now consider:

$$\mathcal{E}_r(\underline{0}) = \{\underline{x} \in \mathbb{R}^p \quad st \quad \underline{x}^T B \underline{x} \leq r\}$$

$$B = \sum_{i=1}^{p} \lambda_i \underline{e}_i \underline{e}_i^T$$

In p=2, this is an ellipse, defined by the matrix B, and with lengths λ_i in the direction of $\underline{e}_i$

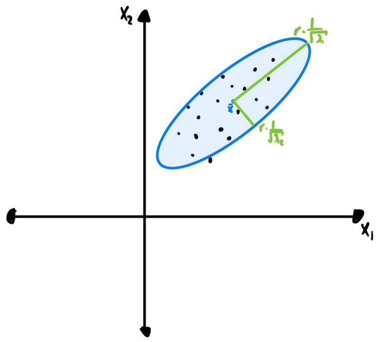

Figure 2: Mahalanobis distance example for a fixed r.

Then the volume:

$$Vol(\mathcal{E}_r(\underline{0})) = k_p r^p \frac{1}{\prod_{i=1}^{p} \sqrt{\lambda_i}}$$

We introduce a new distance, the Mahalanobis (statistical) distance. Let be $\underline{x}, \underline{y} \in \mathbb{R}^p$, and S such that $\det S > 0$

$$\begin{aligned}
d_{S^{-1}}(\underline{x}, \underline{y}) &= \sqrt{(\underline{x} - \underline{y})^T S^{-1} (\underline{x} - \underline{y})} \\
&= \sqrt{(\underline{x} - \underline{y})^T S^{-1/2} S^{-1/2} (\underline{x} - \underline{y})} \\
&= \sqrt{(S^{-1/2}\underline{x} - S^{-1/2}\underline{y})^T (S^{-1/2}\underline{x} - S^{-1/2}\underline{y})} \\
&= d_2(S^{-1/2}\underline{x}, S^{-1/2}\underline{y}) \qquad \text{(Euclidean Distance)}
\end{aligned}$$

where:

$$\begin{aligned}
S &= \sum \lambda_i \underline{e}_i \underline{e}_j^T \\
\sqrt{S} &= \sum \sqrt{\lambda}_i \underline{e}_i \underline{e}_j^T \\
S^{-1} &= \sum \frac{1}{\lambda_i} \underline{e}_i \underline{e}_j^T \\
S^k &= \sum (\lambda_i)^k \underline{e}_i \underline{e}_j^T
\end{aligned}$$

Remark.

> In statistical analysis, euclidean distance is not as important as Mahalanobis distance, which considers the distance in terms of standard deviations

Example.

Let be p=2. Considers the case of covariance 0.

$$S = \begin{bmatrix} s_{11} & 0 \\ 0 & s_{22} \end{bmatrix}, \qquad s_{11} > s_{22}$$

Take:

$$\mathcal{E}_r(\underline{\bar{x}}) = \{\underline{x} \in \mathbb{R}^p \quad st \quad (\underline{x} - \underline{\bar{x}})^T S^{-1} (\underline{x} - \underline{\bar{x}}) \leq r^2\}$$

Note that in this case $\mathcal{E}$ is represented with the matrix power to -1, so:

$$Vol(\mathcal{E}_r(\bar{x})) \propto r^2 \sqrt{\lambda_1 \lambda_2} = r^2 \sqrt{\det(S)} = r^2 \sqrt{\text{Gen. var.}}$$

Take the Mahalanobis distance:

$$\begin{aligned} d_{S^{-1}}(\underline{x}, \underline{y}) &= \sqrt{(\underline{x} - \underline{y})^T S^{-1} (\underline{x} - \underline{y})} = \sqrt{\frac{(x_1 - y_1)^2}{S_{11}} + \frac{(x_2 - y_2)^2}{S_{22}}} \\ &= d_{eucl(\underline{x}_{std}, \underline{y}_{std})} \end{aligned}$$

4.3 Correlation with coordinate system

Let be general p and S:

$$\begin{aligned} S &= \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ s_{12} & s_{22} & \dots & s_{2p} \\ \dots & \dots & \dots & \dots \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{bmatrix} \\ &= \sum_i^p \lambda_i \underline{e}_i \underline{e}_i^T \\ &= P \Lambda P^T \end{aligned}$$

where:

- $P = \begin{bmatrix} \underline{e}_1, \dots, \underline{e}_p \end{bmatrix}$
- $\Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_p \end{bmatrix}$

We can represent our problem on two axes:

- Old system using $\underline{x}_1, \ldots, \underline{x}_p$:

$$\begin{aligned}\underline{\tilde{W}} &= \underline{W} \\ \mathbb{X} &= \begin{pmatrix}\underline{x}_1 & \underline{x}_2 & \ldots & \underline{x}_n\end{pmatrix}^T \\ S &= \frac{1}{n-1}\mathbb{X}^T(I - \frac{11^T}{1^T1})\mathbb{X}\end{aligned}$$

- New system using $\underline{e}_1, \ldots, \underline{e}_p$:

$$\begin{aligned}\underline{\tilde{W}} &= \begin{bmatrix}\underline{e}_1^T\underline{W} \\ \underline{e}_2^T\underline{W} \\ \ldots \\ \underline{e}_p^T\underline{W}\end{bmatrix} = P^TW \\ \tilde{\mathbb{X}} &= \begin{bmatrix}(P^T\underline{x}_1)^T \\ (P^T\underline{x}_2)^T \\ \ldots \\ (P^T\underline{x}_p)^T\end{bmatrix} = \mathbb{X}P \\ S &= \frac{1}{n-1}\tilde{\mathbb{X}}^T(I - \frac{11^T}{1^T1})\tilde{\mathbb{X}} = \frac{1}{n-1}P^T\mathbb{X}^T(I - \frac{11^T}{1^T1})\mathbb{X}P \\ &= P^TSP = P^TP\Lambda P^TP = \Lambda\end{aligned}$$

  In a similar way, $\tilde{\mathbb{X}} = \mathbb{X}\mathbb{P}$, and $\tilde{S} = PP^T\Lambda P^TP = \Lambda$

Note that in both:

$$\det S = \prod_i \lambda_i = \det \tilde{S}$$

Remark.

> The correlation matrix is dependent on the coordinate/reference system, so analyses based on correlation must take into account that correlation is only a measure over a particular reference system

5 Principal Component Analysis (PCA)

5.1 Introduction

Given $\mathbf{x} \in \mathbb{R}^p$ with $\mathbf{X} \sim (\boldsymbol{\mu}, \Sigma)$ and a vector $\mathbf{a} \in \mathbb{R}^p$, we define:

$$\text{Var}(\mathbf{a}^T\mathbf{X}) = \mathbf{a}^T\Sigma\mathbf{a}$$

We aim to find $\mathbf{a}$ that maximizes $\text{Var}(\mathbf{a}^T\mathbf{X})$, subject to $\|\mathbf{a}\| = 1$ to avoid arbitrary scaling.

Definition 5.1: Principal Component Analysis (PCA)

Find $\mathbf{a}_1, \ldots, \mathbf{a}_p$ such that:

- $\|\mathbf{a}_i\| = 1 \quad \forall i = 1, \ldots, p$
- $\text{Var}(\mathbf{a}_1^T \mathbf{X}) = \sup_{\mathbf{a} \in \mathbb{R}^p, \|\mathbf{a}\|=1} \frac{\mathbf{a}^T \Sigma \mathbf{a}}{\mathbf{a}^T \mathbf{a}} = \lambda_1$
- For $i > 1$, $\mathbf{a}_i$ maximizes $\text{Var}(\mathbf{a}_i^T \mathbf{X})$ subject to orthogonality constraints:

$$\text{Cov}(\mathbf{a}_i^T \mathbf{X}, \mathbf{a}_j^T \mathbf{X}) = 0, \quad \forall j < i$$

Definition 5.2: i-th principal component

Taking the a's from above, the i-th principal component (or i-th score) is:

$$y_i = \underline{a}_i^T \underline{x}$$

where $\underline{a}_i$ is called the i-th loading

Since the covariance matrix is symmetric, it has an eigenvalue decomposition. The eigenvectors correspond to the principal directions of the data, and the eigenvalues indicate the variance along these directions.

Theorem 5.3

If Σ has a Spectral decomposition: $\Sigma = \sum_{i=1}^{p} \lambda_i \underline{e}_i \underline{e}_i^T$ Then:

1. The i-th principal component is the i-th eigenvalue

$$y_i = \underline{e}_i^T \underline{x} \quad or \quad y_i = \underline{e}_i^T (\underline{x} - \underline{\mu})$$

2. The covariance between components satisfies:

$$Cov(y_i, y_j) = \begin{cases} \lambda_i & i = j \\ 0 & i \neq j \end{cases}$$

Lemma 5.4: Order of eigenvalues

Let $B \in \mathbb{R}^{p\times p}$ a positive semidefinite matrix, and $B = \sum_i^p \lambda_i \underline{e}_i \underline{e}_i^T = P\Lambda P^T$ Then:

- $$\sup_{\underline{x}\neq 0} \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} = \lambda_1 \qquad \arg\max_{\underline{x}\neq 0} \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} = \underline{e}_1$$

- $$\sup_{\underline{x}\neq 0,\ \underline{x}\perp \underline{e}_1} \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} = \lambda_2 \qquad \arg\max_{\underline{x}\neq 0,\ \underline{x}\perp \underline{e}_1} \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} = \underline{e}_2$$
  $$\dots$$

- $$\sup_{\underline{x}\neq 0,\ \underline{x}\perp \underline{e}_1,\dots,\underline{e}_{p-1}} \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} = \lambda_p = \inf_{\underline{x}\neq 0} \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} \qquad \arg\max_{\underline{x}\neq 0,\ \underline{x}\perp \underline{e}_1,\dots,\underline{e}_{p-1}} \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} = \underline{e}_p$$

Proof. Take a general $\underline{x}$

$$\begin{aligned}
\lambda_p \leq \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} &= \frac{\underline{x}^T P\Lambda P^T \underline{x}}{\underline{x}^T P P^T \underline{x}} \\
&= \frac{\underline{y}^T \Lambda \underline{y}}{\underline{y}^T \underline{y}} \qquad (\underline{y} = P^T \underline{x}) \\
&= \frac{\sum_i^p \lambda_i y_i^2}{\sum_i^p y_i^2} \leq \frac{\lambda_1 \sum_i^p y_i^2}{\sum_i^p y_i^2} = \lambda_1
\end{aligned}$$

Then,

$$\lambda_p \leq \frac{\underline{x}^T B \underline{x}}{\underline{x}^T \underline{x}} \leq \lambda_1 \qquad \forall \underline{x} \in \mathbb{R}^p$$

Taking $\underline{x} = \underline{e}_1$, $\frac{\underline{e}_1^T B \underline{e}_1}{\underline{e}_1^T \underline{e}_1} = \frac{\lambda_1 \underline{e}_1^T \underline{e}_1}{\underline{e}_1^T \underline{e}_1} = \lambda_1$
Taking $\underline{x} = \underline{e}_p$, $\frac{\underline{e}_p^T B \underline{e}_p}{\underline{e}_p^T \underline{e}_p} = \lambda_p$
Now, note that we want to find the next bigger value, but perpendicular to $\underline{e}_1$

$$\underline{y} = P^T \underline{x} = \begin{bmatrix} \underline{e}_1^T \underline{x} \\ \underline{e}_2^T \underline{x} \\ \dots \\ \underline{e}_p^T \underline{x} \end{bmatrix} = \begin{bmatrix} 0 \\ \underline{e}_2^T \underline{x} \\ \dots \\ \underline{e}_p^T \underline{x} \end{bmatrix}$$

Then, the next sup is λ_2, with argument $\underline{e}_2$, following the same argument as before. ☐

It is also useful to look at the smallest (last) eigenvalues/principal component. If they are too close to 0, it indicates that there are some linear dependences in the data. We may want to do changes on the dataset in order to remove these dependences.

5.2 PCA in Terms of Covariance and Correlation

The principal components can be computed as a transformation of the original variables, using the eigenvectors of Σ as a basis. $\underline{Y} = \underline{P}^T \underline{X}$, is the vector of principal, components.
Where $P = \left[\underline{e}_1, \underline{e}_2, \ldots, \underline{e}_p\right]$
Note the expected value is: $E[\underline{Y}] = \underline{P}^T \underline{\mu}$
The principal component of $\Sigma = P\Lambda P^T$ are uncorrelated, as we can see:

$$Cov(\underline{Y}) = P^T \Sigma P = P^T P \Lambda P^T P = \Lambda$$

Where $\Lambda = \begin{bmatrix} \lambda_1 & \ldots \\ \ldots & \ldots \\ \ldots & \lambda_p \end{bmatrix}$

Proposition 5.5

Let be σ_{kk} the standard deviation of x_k and λ_i the value of the variance in the component i:

$$Corr(y_i, x_k) = e_{ki} \frac{\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}$$

Proof. First note that:

$$\begin{aligned} Cov(y_i, x_k) &= Cov(\underline{e}_i^T \underline{x}, \underline{u}_k^T \underline{x}), \qquad \text{where } u_k = (0, \ldots, 0, \underbrace{1}_{k}, 0, \ldots, 0) \\ &= \underline{e}_i^T \Sigma \underline{u}_k = u_k^T \Sigma \underline{e}_i = \lambda_i \underline{u}_k^T \underline{e}_i = \lambda_i e_{ki} \end{aligned}$$

Then:

$$Corr(y_i, x_k) = \frac{Cov(y_i, x_k)}{\sqrt{\lambda_i \cdot \sigma_{kk}}} = \frac{\lambda_i e_{ki}}{\sqrt{\lambda_i \cdot \sigma_{kk}}} = e_{ki} \frac{\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}$$

□

Now, with the same $\underline{X} \sim \underline{\mu}, \Sigma$, take the normalized data

$$\underline{X}_{std} = \underline{Z} = \underline{V}^{-1/2}(\underline{X} - \underline{\mu})$$

where $V = \begin{bmatrix} \sigma_1 & \ldots & \\ \ldots & & \\ & & \sigma_p \end{bmatrix} = \begin{bmatrix} Var(X_1) & & \ldots \\ & \ldots & \\ & & Var(X_p) \end{bmatrix}$

$$cov(\underline{Z}) = V^{-1/2} \Sigma V^{-1/2} = \rho = \begin{bmatrix} 1 & \ldots & \rho_{ij} \\ \ldots & 1 & \ldots \\ \rho_{ji} & \ldots & 1 \end{bmatrix} = \sum_{i=1}^{p} \lambda_i \underline{e}_i \underline{e}_i^T = P^T \Lambda P$$

Note that this is not the spectral decomposition of Σ, but ρ. Actually, note that σ_i are the eigenvalues of ρ
Now, the vector of principal components for normalized data can be analyzed in terms of

correlation:

$$\underline{Y} = P^T \underline{Z} = P^T V^{-1/2}(\underline{x} - \underline{\mu})$$

Where:

$$E[\underline{y}] = \underline{0}$$

$$Cov(\underline{Y}) = P^T V^{-1/2} \Sigma V^{-1/2} P = \begin{bmatrix} \sigma_1 & \dots \\ \dots & \\ & \sigma_p \end{bmatrix}$$

Example.

> Take: $\underline{X} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \Sigma = \begin{bmatrix} 1 & 4 \\ 4 & 100 \end{bmatrix}, \underline{\mu} = \underline{0}$
> By the spectral decomposition: $\Sigma = \lambda_1 \underline{e}_1 \underline{e}_1^T + \lambda_2 \underline{e}_2 \underline{e}_2^T$. It can be shown:
>
> $$\lambda_1 = 100.16 \qquad \underline{e}_1 = \begin{pmatrix} 0.04 & 0.999 \end{pmatrix}^T$$
> $$\lambda_2 = 0.84 \qquad \underline{e}_2 = \begin{pmatrix} 0.999 & -0.04 \end{pmatrix}^T$$
>
> PC1: $0.999x_1 - 0.04x_2 = 0$
> $\implies x_2 = \frac{0.999}{0.04} x_1 = 25x_1$
> PC2: $0.04x_1 + 0.999x_2 = 0$
> Regression line:
>
> $$\frac{y - \mu_y}{\sqrt{\sigma_y^2}} = \rho \frac{x - \mu_x}{\sqrt{\sigma_x^2}}$$
>
> Let be $\rho = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 1 \end{bmatrix}$ We can find:
>
> $$\lambda_1 = 1.4 \qquad \underline{e}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)^T$$
> $$\lambda_2 = 0.6 \qquad \underline{e}_2 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)^T$$

Finally, we can use both Σ and ρ to analyze the variation of $\underline{Y}$, depending on normalization on $\underline{X}$.

Principal Components with Σ: For $\underline{Y} = P^T \underline{X}$

- General Variance($\underline{X}$) $= \prod_{i=1}^p \lambda_i =$ General Variance($\underline{Y}$)
- Total Variance($\underline{X}$) $= trace(\Sigma) = \sum_{i=1}^p \lambda_i = trace(\Lambda) =$ Total Variance($\underline{Y}$)

Principal Components with ρ: For $\underline{Y} = P^T \underline{Z}$

- General Variance($\underline{Z}$) = General Variance($\underline{Y}$)
- Total Variance($\underline{Z}$) = Total Variance($\underline{Y}$)

5.3 Total Variability explained

The total variability explained by each component is:

$$\frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j}$$

So the cumulative variability until the k PC is:

$$\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{j=1}^{p} \lambda_j}$$

We can then do an analysis over the total variability the model gain adding one more variable. In this way, we can decide in marginal terms whether decide add another dimension or not.

A k can be decided with the elbow method. The problem with this is that it is purely mathematics. A better approach can be choosing a threshold (or evaluate a trade-off) for how much variability we want to explain regarding our problem we are solving.

5.4 Correspondence Analysis

Principal Components on Contingency/frequency tables for categorical values.
Do something very similar to standard principal component, but taking the spectral decomposition on the frequencies table instead of the values itself

5.5 Different perspective on PCA

Fitting the first PC, we wanted the element that maximize the variance in the dataset. Anyway, we can obtain the same result fitting a line that minimizes the distance to the points in the dataset.

Note that this is not the regression, since the regresion minimizes the y distance of the points. We want, instead, the line that considers the Euclidean distance to minimize the distance in all axes.

Let be $\mathbb{X} = \left[\underline{x}_1, \ldots, \underline{x}_n\right], \quad \underline{x}_i \in \mathbb{R}^p$.

We want to find the space $\mathcal{L}$ of dimension $k \ll p$ closest to the data.

$\mathcal{L}$ is spammed by $\underline{\eta}_1, \ldots, \underline{\eta}_k$ orthonormal basis. i.e., the problem is:

$$\text{Find } \underline{\eta}_1, \ldots, \underline{\eta}_k \quad st \quad \sum_{i=1}^{n} \|(\underline{x}_i - \underline{\bar{x}}) - \sum_{j=1}^{k} \underline{\eta}_j \underline{\eta}_j (\underline{x}_i - \underline{\bar{x}})\|^2 \quad \text{is minimum}$$

Let be $\underline{v}_i = \underline{x}_i - \underline{\bar{x}}$:

$$
\begin{aligned}
\|v_i - \sum_{j=1}^{k} \underline{\eta}_j \underline{\eta}_j v_i\|^2 &= \left(v_i - \sum_{j=1}^{k} \underline{\eta}_j \underline{\eta}_j v_i\right)^T \left(v_i - \sum_{j=1}^{k} \underline{\eta}_j \underline{\eta}_j v_i\right) \\
&= \underline{v}_i^T \underline{v}_i - 2\sum_{i=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_j \underline{v}_i + (\sum_{j=1}^{k}\sum_{t=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_j^T \underline{\eta}_t \underline{\eta}_t^T \underline{v}_t) \\
&= \underline{v}_i^T \underline{v}_i - 2\sum_{i=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_j \underline{v}_i + (\sum_{j=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_t^T \underline{v}_t) \qquad (*) \\
&= \underline{v}_i^T \underline{v}_i - (\sum_{j=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_t^T \underline{v}_t)
\end{aligned}
$$

(*) Note that

$$
\underline{\eta}_j^T \underline{\eta}_t = \begin{cases} 0 & j \neq t \\ 1 & j = t \end{cases}
$$

Now:

$$
\sum_{i=1}^{n} \|(\underline{x}_i - \underline{\bar{x}}) - \sum_{j=1}^{k} \underline{\eta}_j \underline{\eta}_j (\underline{x}_i - \underline{\bar{x}})\|^2 = \sum_{i=1}^{n} \underline{v}_i^T \underline{v}_i - \sum_{i=1}^{n}\sum_{j=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_t^T \underline{v}_t
$$

We can reformulate it further:

$$
\text{Find } \underline{\eta}_1, \dots, \underline{\eta}_k \quad st \quad \sum_{i=1}^{n}\sum_{j=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_t^T \underline{v}_t \qquad \text{is maximum}
$$

And even more:

$$
\begin{aligned}
\sum_{i=1}^{n}\sum_{j=1}^{k} \underline{v}_i^T \underline{\eta}_j \underline{\eta}_t^T \underline{v}_t &= \sum_{j=1}^{k} \underline{\eta}_j^T \left(\sum_{i=1}^{n} (\underline{v}_i \underline{v}_i^T)\right) \underline{\eta}_j \\
&= \sum_{j=1}^{k} \underline{\eta}_j^T \left(\sum_{i=1}^{n} ((\underline{x}_i - \underline{\bar{x}})(\underline{x}_i - \underline{\bar{x}})^T\right) \underline{\eta}_j \\
&= (n-1)\sum_{j=1}^{k} \underline{\eta}_j^T S \underline{\eta}_j
\end{aligned}
$$

Getting:

$$
\max_{\substack{\underline{\eta}_1 \\ \|\underline{\eta}_1\|=1}} (n-1)\underline{\eta}_1^T S \underline{\eta}_1 = \lambda_1
$$

A problem that we know how to solve, and being argmax is $\underline{e}_1$ if this is the largest eigenvector of $S = \sum_{i=1}^{p} \lambda_i \underline{e}_i \underline{e}_i^T$. This works the same for the next k-1 maxima.

So far, we have proved that if $\underline{x}_1, \dots, \underline{x}_n \in \mathbb{R}^p$. The linear spam $\mathcal{L}$ of dim k that minimize the euclidean distance is $\mathcal{L} = span(\underline{e}_1, \dots, \underline{e}_k)$, the k first eigenvalues of S.

And actually, the approximation error is:

$$\sum_{i=1}^{n} \underline{v}_i^T \underline{v}_i - (n-1) \sum_{j=1}^{k} \underline{e}_j^T S \underline{e}_j^T = \sum_{i=1}^{n} \underline{v}_i^T \underline{v}_i - (n-1) \sum_{j=1}^{k} \lambda_j$$

On the other hand,

$$\begin{aligned}
\sum_{i=1}^{n} \underline{v}_i^T \underline{v}_i &= \operatorname{trace}\left(\sum_{i=1}^{n} \underline{v}_i^T \underline{v}_i\right) && \text{(The trace of a number is the number)} \\
&= \operatorname{trace}\left(\sum_{i=1}^{n} (\underline{x}_i \underline{\bar{x}})^T (\underline{x}_i \underline{\bar{x}})\right) \\
&= \sum_{i=1}^{n} \operatorname{trace}\left((\underline{x}_i \underline{\bar{x}})^T (\underline{x}_i \underline{\bar{x}})\right) && \text{(Linearity of trace)} \\
&= \sum_{i=1}^{n} \operatorname{trace}\left((\underline{x}_i \underline{\bar{x}})(\underline{x}_i \underline{\bar{x}})^T\right) && (\operatorname{trace}(AB) = \operatorname{trace}(BA)) \\
&= \operatorname{trace} \sum_{i=1}^{n} \left((\underline{x}_i \underline{\bar{x}})(\underline{x}_i \underline{\bar{x}})^T\right) && \text{(linearity)} \\
&= (n-1) \operatorname{trace}(S) = (n-1) \sum_{i=1}^{p} \lambda_i
\end{aligned}$$

Coming back to the aproximation error:

$$\sum_{i=1}^{n} \underline{v}_i^T \underline{v}_i - (n-1) \sum_{j=1}^{k} \lambda_j = (n-1) \sum_{j=1}^{p} \lambda_j - (n-1) \sum_{j=1}^{k} \lambda_j = (n-1) \sum_{j=k+1}^{p} \lambda_j$$

Finally, the approximation error is exactly aproximated by n-1 times the sum of the eigenvalues that are out of the k first eigenvectors chosen for the span.

Remark.

> Note that there are an infinite number of bases of find k that generate $\mathcal{L}$. For instance, we may find a basis that is sparse. This basis has a better interpretability with respect to the original columns of $\mathbb{X}$, plus it makes easier the calculations. An algorithm that finds this basis is called VARIMAX.
> Anyway, our basis allows us to interpret the k first element on the basis as the k direction with the most variability in the data.

Remark.

> Let be $\mathbb{X} \in \mathbb{R}^{n \times p}$ data. Then, the centering data is:
>
> $$\left(I - \frac{11^T}{1^T 1}\right) \mathbb{X}$$
>
> We can calculate the Singular Value Descomposition (SVD)of the centered data, UDV, being:

- U a $n \times n$ unitary matrix: $U^T U = U U^T = I_n$
- V $p \times p$ unitary
- D a rectangular diagonal matrix:

$$D = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & d_p \\ 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

So:

$$S = \frac{1}{n-1}\mathbb{X}^T \left(I - \frac{11^T}{1^T 1}\right)^T \left(I - \frac{11^T}{1^T 1}\right)\mathbb{X} = \frac{1}{n-1} V^T D^T U^T U D V$$
$$= \frac{1}{n-1} V^T D^T D V = \sum_{i=1}^{p} \lambda_i \underline{v}_i \underline{v}_i^T \qquad \lambda_i = \frac{d_i^2}{n-1}$$

Since $D^T D = \begin{bmatrix} d_1^2 & 0 & \dots & 0 \\ 0 & d_2^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & d_p^2 \end{bmatrix}$

Note that actually we found a spectral decomposition:

$$\lambda_i = \frac{d_i^2}{n-1} \qquad \underline{v}_i = \underline{e}_i$$

That's the connection between PCA and SVD of a centered dataframe.

6 Gaussian Distribution

6.1 Definition

We say that $\underline{X}$ distributes Gaussian, i.e.:

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

where

$$\underline{\mu} \in \mathbb{R}^p \qquad \text{and} \qquad \Sigma \in \mathbb{R}^{p \times p} \text{ positive defined matrix}$$

if density of $\underline{X}$ is:

$$\begin{aligned}\varphi(\underline{x}) &= \frac{1}{\sqrt{(q\pi)^p \det(\Sigma)}} \exp\left[-\frac{1}{2}(\underline{x}-\underline{\mu})^T\Sigma^{-1}(\underline{x}-\underline{\mu})\right] \\ &\propto \exp\left[-\frac{1}{2}d^2_{\Sigma^{-1}}(\underline{x},\underline{\mu})\right]\end{aligned}$$

Remark.

> If $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$:
> $$\implies E[\underline{X}] = \underline{\mu} \quad \wedge \quad Cov(\underline{X}) = \Sigma$$

6.2 Transformations

Theorem 6.1

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma) \iff \underline{a}^T\underline{X} \sim N_1(\underline{a}^T\underline{\mu}, \underline{a}^T\Sigma\underline{a}) \quad \forall \underline{a} \in \mathbb{R}^p$$

Proof. Any distribution is completely identified by a characteristic function. Uses the characteristic function of N_p. See (Johnson & Wichern, 2007). □

Proposition 6.2

Let be $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$, $A \in \mathbb{R}^{q\times p}$, then:

$$\underbrace{A\underline{X}}_{\in\mathbb{R}^p} \sim N_q(A\underline{X}, A\Sigma A^T)$$

Proof. By the theorem 6.2, it is enough to show that $\forall a \in \mathbb{R}^q$

$$\underline{a}^T(A\underline{X}) \sim N_1(\underline{a}^T(A\underline{\mu}), \underline{a}^T(A\Sigma A^T)\underline{a})$$

$$\begin{aligned}\underline{a}^T(A\underline{X}) &= \underline{a}^T A\underline{X} = (A^T\underline{a})^T\underline{X} \\ &= N_1\left((A^T\underline{a})^T\mu, (A^T\underline{a})^T\Sigma(A^T\underline{a})\right) \qquad \text{(Theorem 6.2, since Gaussian variable and } (A^T\underline{a}) \in \mathbb{R}^p \\ &= N_1\left((\underline{a}^T(A^T\mu), \underline{a}^T(A\Sigma A^T)\underline{a}\right)\end{aligned}$$

Since this is true for any $\underline{a}$

$$A\underline{X} \sim N(A\underline{\mu}, A\Sigma A^T)$$

□

Corollary 6.3

If $\underline{X} \sim N_p(\underline{\mu}, \Sigma), \underline{d} \in \mathbb{R}^p$, then:

$$\underline{X} + \underline{d} \sim N_p(\underline{\mu} + \underline{d}, \Sigma)$$

Proof. Exercise $\square$

Remark.

Let be $z_1, z_2, \dots, z_p$ iid $\sim N_1(0,1)$ So $\underline{Z} = \begin{bmatrix} z_1 \\ \dots \\ z_p \end{bmatrix}$ has the following density function:

$$\varphi_p(\underline{Z}) = \prod_{i=1}^{p} \varphi_1(z_i) = \prod_{i=1}^{p} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z_i^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\sum_{i=1}^{p} z_i^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\underline{z}^T I \underline{z}}$$

So $\underline{Z} \sim N_p(\underline{0}, I)$.
Note that this works only because we know that variables are iid.

Remark.

Consider $\underline{\mu} \in \mathbb{R}^p, \Sigma \in \mathbb{R}^{p \times p}$ positive definite. Then:

$$\underline{X} = \Sigma^{1/2}\underline{Z} + \underline{\mu}$$

Where:

$$\Sigma = \sum_{i=1}^{p} \lambda_i \underline{e}_i \underline{e}_i^T$$

$$\Sigma^{1/2} = \sum_{i=1}^{p} \sqrt{\lambda_i} \underline{e}_i \underline{e}_i^T$$

Indeed:

$$\Sigma^{1/2}\underline{Z} \sim N_p(\underline{0}, \Sigma^{1/2} I \Sigma^{1/2}) = N(\underline{0}, \Sigma)$$
$$\Sigma^{1/2}\underline{Z} + \underline{\mu} \sim N_p(\underline{\mu}, \Sigma)$$

Note that:

$$\Sigma^{-1/2} = \sum_{i=1}^{p} \frac{1}{\sqrt{\lambda_i}} \underline{e}_i \underline{e}_i^T$$

6.3 Distance mean to point

Proposition 6.4

Let be $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$.

$$P\left[d^2_{\Sigma^{-1}}(\underline{X}, \underline{\mu}) \leq \chi^2_{1-\alpha}(p)\right] = 1 - \alpha$$

i.e., the Mahalanobis distance (squared) between a point that distributes normal and the mean is with a $1 - \alpha$ probability less than $\chi^2_{1-\alpha}(p)$

Proof.

$$\begin{aligned} W &= (\underline{X} - \underline{\mu})^T \Sigma^{-1} (\underline{X} - \underline{\mu}) = d^2_{\Sigma^{-1}}(\underline{X}, \underline{\mu}) \\ &= (\underline{X} - \underline{\mu})^T \Sigma^{-1/2} \Sigma^{-1/2} (\underline{X} - \underline{\mu}) = \underline{Z}^T \underline{Z} = \sum_{i=1}^{p} z_i^2, \qquad Z \sim N_p(\underline{0}, I) \end{aligned}$$

We know that if $z_1 \sim N(0, 1)$:

$$z_1^2 \sim \chi_1(1)$$

Then:

$$\sum_{i=1}^{p} z_i^p \sim \chi^2_1(p)$$

Now, it is known by the χ distribution:

$$P\left[d^2_{\Sigma^{-1}}(\underline{X}, \underline{\mu}) \leq \chi^2_{1-\alpha}(p)\right] = 1 - \alpha$$

□

6.4 Marginal Distributions

Corollary 6.5

If $\underline{X} = \begin{bmatrix} X_1 \\ \dots \\ X_p \end{bmatrix}$ and $\underline{X} \sim N_p\left(\underline{\mu} = \begin{bmatrix} \mu_1 \\ \dots \\ \mu_p \end{bmatrix}, \Sigma = [\sigma_{ij}]\right)$

$$\implies \forall i = 1, \dots, p \quad X_i \sim N_1(\mu_i, \sigma_{ii})$$

Proof. For $i = 1, \dots, p$. Let $\underline{u}_i = (0, \dots, 0, \underbrace{1}_{i}, 0, \dots, 0)$. $\underline{u}_i^T \underline{X} = X_i$.

By theorem 6.2:

$$X_i \sim N_1(\underline{u}_i^T \mu, \underline{u}_i^T \Sigma \underline{u}_i) = N_1(\mu_i, \sigma_{ii})$$

□

Example.

Let be p=2, and $\underline{X} \sim N(\underline{0}, I)$ and different W as in figure 3

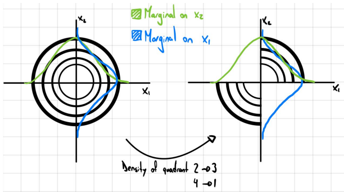

Figure 3: Same Marginals Different Joint

Note that both have marginals N(0,1) and N(0,1) on x_1, x_2 axes, respectively. Anyway, as we don't know their relationship, we cannot be sure of the joint distribution.

Proposition 6.6

Given $\underline{X}_1 \in \mathbb{R}^q$ and $\underline{X}_2 \in \mathbb{R}^{p-q}$ let be $\underline{X} \in \mathbb{R}^p$:

$$\underline{X} = \begin{pmatrix} \underline{X}_1 \\ \underline{X}_2 \end{pmatrix} \sim N_p\left(\begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$$

Where $\Sigma_{11} \in \mathbb{R}^{q \times q}$, $\Sigma_{12} \in \mathbb{R}^{q \times (p-q)}$, $\Sigma_{21} \in \mathbb{R}^{(p-q) \times q}$, and $\Sigma_{22} \in \mathbb{R}^{(p-q) \times (p-q)}$

$$\implies \underline{X}_1 \sim N_q(\underline{\mu}_1, \Sigma_{11})$$

Proof. Take $A = \begin{bmatrix} I_{q \times q} & \underline{0} \end{bmatrix}$ □

6.5 Independence

Definition 6.7: $\perp\!\!\!\perp$

We say that two random variables are independent if their joint density function is equal to the multiplication of their marginals. i.e., for x_1, x_2:

$$\varphi_X(\underline{x}) = \varphi_{X_1}(\underline{x}) \cdot \varphi_{X_2}(\underline{x})$$

Some notations use $\perp$ as $\perp\!\!\!\perp$.

Proposition 6.8

Let be $\underline{X} = \begin{pmatrix} \underline{X}_1 \\ \underline{X}_2 \end{pmatrix}$.

$$\implies \underline{X}_1 \perp\!\!\!\perp \underline{X}_2 \iff \Sigma_{12} = \Sigma_{21}^T = \{0\}$$

Proof. Show that $\forall \underline{t} = \begin{pmatrix} \underline{t}_1 \\ \underline{t}_2 \end{pmatrix} \in \mathbb{R}^p$:

$$\varphi_X(\underline{t}) = \varphi_{X_1}(\underline{t}) \cdot \varphi_{X_2}(\underline{t})$$

□

6.6 Conditional Distribution

Theorem 6.9

Let be $\begin{pmatrix} \underline{X}_1 \\ \underline{X}_2 \end{pmatrix} \sim N_p\left(\begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right)$

$$\implies X_1|X_2 = x_2 \sim N_q(\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

Proof. Let be

$$A = \begin{bmatrix} I_{q\times q} & -\Sigma_{12}\Sigma_{22}^{-1} \\ \underline{0}_{(p-q)\times q} & I_{(p-q)\times(p-q)} \end{bmatrix}$$

Consider

$$A\begin{pmatrix} \underline{X}_1 - \underline{\mu}_1 \\ \underline{X}_2 - \underline{\mu}_2 \end{pmatrix} \sim N_p\left(\begin{pmatrix} \underline{0} \\ \underline{0} \end{pmatrix}, A\Sigma A^T\right), \qquad A\Sigma A^T = \begin{pmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} & \underline{0} \\ \underline{0} & \Sigma_{22} \end{pmatrix}$$

Plus:

$$A\begin{pmatrix} \underline{X}_1 - \underline{\mu}_1 \\ \underline{X}_2 - \underline{\mu}_2 \end{pmatrix} = \begin{pmatrix} \underline{X}_1 - \underline{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\underline{X}_2 - \underline{\mu}_2) \\ \underline{X}_2 - \underline{\mu}_2 \end{pmatrix}$$

Hence:

$$\underline{X}_1 - \underline{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\underline{X}_2 - \underline{\mu} - 2) \perp\!\!\!\perp \underline{X}_2 - \underline{\mu}_2$$

And:

$$\underline{X}_1 - \underline{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\underline{X}_2 - \underline{\mu} - 2) \sim N_q(\underline{0}, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

Therefore the conditional does not change with respect to the marginal:

$$\underline{X}_1 - \underline{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\underline{X}_2 - \underline{\mu} - 2)|\underline{X}_2 = x_2 \sim N_q(\underline{0}, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

And:

$$\underline{X}_1|\underline{X}_2 = x_2 \sim N_q(\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{X}_2 - \underline{\mu} - 2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

$\square$

Remark.

Let be $X_1 \sim N_q(\underline{\mu}_1, \Sigma_{11})$ with no information about $\underline{X}_2$

$$X_1|X_2 = x_2 \sim N_q(\underbrace{\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2)}_{E[\underline{X}_1|\underline{X}_2=\underline{x}_2]}, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

Note that $E[\underline{X}_1|\underline{X}_2 = \underline{x}_2]$ is the linear regression, it is exactly the best predictor we can find given we know X_2.
Plus, note that in the case of knowing information about X_2, the variance, and therefore the uncertainty decreases.

Specifically, if $\underline{X}_1 \perp\!\!\!\perp \underline{X}_2 \implies \Sigma_{12} = 0$:

$$\implies \underline{X}_1|\underline{X}_2 = \underline{x}_2 \sim N_q(\underline{\mu}_1, \Sigma_{11})$$

Example.

For p=2:

$$\underline{X} = \begin{pmatrix} y \\ x \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_y \\ \mu_x \end{pmatrix}, \begin{bmatrix} \sigma_{yy} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{xx} \end{bmatrix}\right)$$

Then $y \sim N_1(\mu_y, \sigma_{yy})$, and by corollary 6.4:

$$Y|X = x \sim N_1\left(\mu_y + \frac{\sigma_{xy}}{\sigma_{xx}}(x - \mu_x), \sigma_{yy} - \frac{\sigma_{xy}^2}{\sigma_{xx}}\right)$$

Now: $\rho_{xy} = \frac{\sigma_{xy}}{\sqrt{\sigma_{xx}\sigma_{yy}}}$, and:

$$Y|X = x \sim N_1(\underbrace{\mu_y + \rho_{xy}\frac{\sqrt{\sigma_{xy}}}{\sqrt{\sigma_{xx}}}(x - \mu_x)}_{E[Y|X=x]}, \sigma_{yy}(1 - \rho_{xy}^2))$$

Note that:

$$y = \mu_y + \rho_{xy}\frac{\sqrt{\sigma_{xy}}}{\sqrt{\sigma_{xx}}}(x - \mu_x)$$

Is the regression line $y = \beta_0 + \beta_1 x$, where:

$$\beta_1 = \rho_{xy}\frac{\sqrt{\sigma_{xy}}}{\sqrt{\sigma_{xx}}} \qquad \beta_0 = \mu_y - \beta_1\mu_x$$

and:

$$\frac{y - \mu_y}{\sqrt{\sigma_{yy}}} = \rho_{xy}\frac{x - \mu_x}{\sqrt{\sigma_{xx}}}$$

Therefore, with the conditional expected value, we can find the linear regression of points that distribute Normal Now, set:

$$Z_x = \frac{x-\mu_x}{\sqrt{\sigma_{xx}}} \qquad Z_y = \frac{y-\mu_y}{\sqrt{\sigma_{yy}}}$$

Note that now, $z_y = \rho_{xy} z_x$, having a regression passing over the 0.
Now, take, as an example, x=3. Then,

$$y = \mu_y + 3\rho_{xy}\sqrt{\sigma_{yy}}$$

Now note that $x = \mu_x + 3\sqrt{\sigma_{xx}}$:
As $\rho \leq 1$, the y will be much closer to its mean than x, this is called **the regression effect**.

Anyway, trying to interpret the regression effect could commit a regression fallacy. This is trying to describe something that happens in nature as something purely analytical.

Indeed, we can note that the prediction of y based on x is different than x based on y.

6.7 Maximum Likelihood Estimator (MLE)

Unbiased estimators for a general distribution are

- Estimator for $\underline{\mu}$:

$$\underline{\bar{X}_n} = \frac{1}{n}\sum_{i=1}^{n} \underline{X}_i$$

- Estimator for Σ:

$$S = \frac{1}{n-1}\sum_{i=1}^{n}(\underline{X}_i - \underline{\bar{X}})(\underline{X}_i - \underline{\bar{X}})^T$$

Definition 6.10: Likelihood

$$L(\theta|\underline{X}) = \varphi(\underline{X}|\theta)$$

Definition 6.11: log likelihood

$$\ell(\theta|\underline{X}) = \log L(\theta|\underline{X})$$

Definition 6.12: Maximum Likelihood Estimator

$$\hat{\theta} = \arg\max_{\theta\in\Theta} L(\theta)$$

Remark.

Note that by monotony of log function, the maximum parameter of ℓ is the same as maximum of L

Example.

Let be $X_i \sim Bernoulli(p)$:

$$x_1, \dots, x_5 \quad iid \sim Bern(p), \text{ and } P[x_i = 1] = p$$

Given data: $x_1 = 1, x_2 = 0, x_3 = 0, x_4 = 1, x_5 = 0$

$$P[x_1 = 1, x_2 = 0, x_3 = 0, x_4 = 1, x_5 = 0] = p(1-p)(1-p)p(1-p) = p^2(1-p)^3$$

Then, the likelihood $L(p|data) = p^2(1-p)^3$
In this case, $\ell(p|data) = 2\log p + 3\log(1-p)$
In the case of the example, it can be shown that the maximum likelihood is $\hat{p} = \frac{1}{5}\sum_{i=1}^{5} x_i = 2/5$

Example.

The likelihood of a Gaussian given the data is:

$$L((\mu, \sigma)|x_i) = \prod_{i=1}^{n} \frac{1}{\sqrt{(2\pi)^p \det \Sigma}} e^{-\frac{1}{2}(\underline{x}_i - \underline{\mu})^T \Sigma^{-1}(\underline{x}_i - \underline{\mu})}$$

Now, the loglikehood is

$$\ell((\mu, \sigma)|x_i) == \sum_{i=1}^{n} \log(\frac{1}{\sqrt{(2\pi)^p}}) - n \log \sqrt{\det \Sigma} - \frac{1}{2}\sum_{i=1}^{n}(\underline{x}_i - \underline{\mu})^T \Sigma^{-1}(\underline{x}_i - \underline{\mu})$$

So:

$$(\underline{\hat{\mu}}, \hat{\Sigma}) = \arg \max_{\substack{\underline{\mu} \in \mathbb{R}^p \\ \Sigma \text{ is sdp}}} \ \ell(\underline{\mu}, \Sigma)$$

The result of this is:

$$\underline{\hat{\mu}} = \frac{1}{n}\sum_{i=1}^{n} \underline{X}_i = \underline{\bar{X}}_n$$
$$\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n}(\underline{x}_i - \hat{\mu})(\underline{x}_i - \hat{\mu})^T = \frac{n-1}{n} S$$

$\hat{\mu}$ is unbiased, but $\hat{\Sigma}$ biased.

6.8 Invariance Property of MLE

Given $\theta \in \mathbb{R}^p$:

$$T(\underline{x}_1, \dots, \underline{x}_n) \text{ the MLE of } \theta$$

The standard notation for T is $\hat{\theta} = \widehat{\theta(\underline{x}_1, \dots, \underline{x}_n)}$.
Now, given a new parameter, a function of θ: $\eta = h(\theta)$. The maximum likelihood estimator or η is:

$$\hat{\eta} = h(T(x_1, \dots, x_n)) = h(\hat{\theta})$$

Summarizing:

$$\hat{h(\theta)} = h(\hat{\theta})$$

7 Distribution of estimators

7.1 Distribution of $\underline{\bar{X}}_n$

Now, we will study the distribution of $\underline{\bar{X}}_n$ when $\underline{X}_1, \dots, \underline{X}_n \quad iid \sim N_p\left(\underline{\mu}, \Sigma\right)$:

Proposition 7.1

$$\underline{\bar{X}}_n \sim N_p\left(\underline{\mu}, \frac{1}{n}\Sigma\right)$$

Proof. Let be $\underline{\tilde{X}} = \begin{pmatrix} \underline{x}_1 \\ \dots \\ \underline{x}_n \end{pmatrix} \in \mathbb{R}^{np}$ such that

$$\underline{\tilde{X}} \sim N_{np}\left(\begin{pmatrix} \mu \\ \dots \\ \mu \end{pmatrix}, \begin{bmatrix} \Sigma & 0 & \dots & 0 \\ 0 & \Sigma & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \Sigma \end{bmatrix}\right)$$

Now, take:

$$A = \underbrace{\begin{bmatrix} I_{p\times p} & I_{p\times p} & \dots I_{p\times p} \end{bmatrix}}_{\text{n times}} \in \mathbb{R}^{p\times pn}$$

Now:

$$\frac{1}{n}A\underline{\tilde{X}} = \underline{\bar{X}}_n \sim N_p\left(\underline{\mu}, \frac{1}{n}\Sigma\right)$$

□

7.2 Wishart Distribution

Now, we will study the distribution of S when $\underline{X}_1, \dots, \underline{X}_n \quad iid \sim N_p\left(\underline{\mu}, \Sigma\right)$. We need to introduce a new distribution:

Definition 7.2: Wishart

Let be $\underline{Z}_1, \underline{Z}_2, \ldots, \underline{Z}_m$ iid $\sim N_p(\underline{0}, \Sigma)$. Then, the Wishart distribution is the distribution induced by:

$$\sum_{i=1}^{m} \underline{Z}_i Z_i^T \sim Wishart(\Sigma, m)$$

Proposition 7.3

Properties of Wishart Distribution:

1. Let be $A_1 \sim Wishart(\Sigma, m_1)$ and $A_2 \sim Wishart(\Sigma, m_2)$. If $A_1 \perp\!\!\!\perp A_2$:

$$\implies A_1 + A_2 \sim Wishart(\Sigma, m_1 + m_2)$$

2. Let be $C \in \mathbb{R}^{k \times p}$ matrix of constants. Let be $A \sim Wish(\Sigma, m)$

$$CAC^T \sim Wish(C\Sigma C^T, m)$$

3. Let be $\sigma^2 > 0$, $A \sim Wish(\Sigma, m)$

$$\implies \sigma^2 A \sim Wish(\sigma^2 \Sigma, m)$$

4. Let be $A \sim Wish(\Sigma, m)$, with $\Sigma = [\sigma^2]$

$$\implies A \sim \sigma^2 \chi^2(m)$$

Proof. **1:**
Let be $\underline{Z}_{11}, \underline{Z}_{12}, \ldots, \underline{Z}_{1m_1}$ iid $\sim N_p(\underline{0}, \Sigma)$ and$\underline{Z}_{21}, \underline{Z}_{22}, \ldots, \underline{Z}_{2m_2}$ iid $\sim N_p(\underline{0}, \Sigma)$.
Let be $A_1 = \sum_{i=1}^{m_1} \underline{Z}_{1i} \cdot \underline{Z}_{1i}^T$ and $A_2 = \sum_{i=1}^{m_2} \underline{Z}_{2i} \cdot \underline{Z}_{2i}^T$.
Define

$$\underline{W}_1 = Z_{11}, \ldots, \underline{W}_{m_1} = Z_{1m_1}, \underline{W}_{m_1+1} = Z_{21}, \underline{W}_{m_1+m_2} = Z_{2m_2} \qquad \text{iid } \sim N(\underline{0}, \Sigma)$$

Then,

$$A_1 + A_2 = \sum_{i=1}^{m_1+m_2} \underline{W}_i \underline{W}_i^T \sim Wish(\Sigma, m_1 + m_2)$$

2:
Let be $A = \sum_{i=1}^{m} \underline{Z}_i \underline{Z}_i^T$, with Z_i normal.

$$CAC^T = \sum_{i=1}^{m} C\underline{Z}_i \underline{Z}_i^T C^T$$

As $C\underline{Z}_i \sim N_k(\underline{0}, C\Sigma C^T)$

$$CAC^T \sim Wish(C\Sigma C^T, m)$$

3:

$$\sigma^2 A = \Sigma \sigma \underline{Z}_i \underline{Z}_i^T \sigma$$

$$\sigma \underline{Z}_i \sim N_p(\underline{0}, \sigma^2 \Sigma)$$

4:
Note that:

$$A = \sum_{i=1}^{m} Z_i^2, \text{ with } Z_1, \dots, Z_m \text{ iid } N_1(0, \sigma^2)$$

Now:

$$\frac{1}{\sigma^2} A = \frac{1}{\sigma^2} \sum_{i=1}^{m} Z_i^2 \sim \chi^2(m)$$

□

7.3 Distribution of S

Theorem 7.4

Let be $\underline{X}_1, \dots, \underline{X}_n$ iid $\sim N_p(\underline{\mu}, \Sigma)$, with μ unknown.

$$\Longrightarrow \sum_{i=1}^{n} (\underline{X}_i - \underline{\bar{X}})(\underline{X}_i - \underline{\bar{X}})^T \sim Wish(\Sigma, n-1)$$

Corollary 7.5

Let be $\underline{X}_1, \dots, \underline{X}_n$ iid $\sim N_p(\underline{\mu}, \Sigma)$, with μ unknown.

$$S \sim Wish\left(\frac{1}{n-1}\Sigma, n-1\right)$$

$$\hat{\Sigma}_{\text{MLE}} \sim Wish\left(\frac{1}{n}\Sigma, n-1\right)$$

Remark.

If $\underline{\mu}$ is known:

$$\sum_{i=1}^{n} (\underline{X}_i - \underline{\mu})(\underline{X}_i - \underline{\mu})^T \sim Wish(\Sigma, n)$$

Since the expression has one more degree of freedom (dof).

Theorem 7.6

Let be $\underline{X}_1, \dots, \underline{X}_n$ iid $\sim N_p(\underline{\mu}, \Sigma)$.

$$\underline{\bar{X}} \perp\!\!\!\perp S$$

Proof. Review Books $\square$

Remark.

> Summing up, if $\underline{X}_1, \dots, \underline{X}_n$ iid $\sim N_p(\underline{\mu}, \Sigma)$, with μ and Σ unknown:
>
> - $\underline{\bar{X}_n} \sim N_p\left(\underline{\mu}, \frac{1}{n}\Sigma\right)$
> - $(n-1)S \sim Wish(\Sigma, n-1)$
> - $\underline{\bar{X}} \perp\!\!\!\perp S$

8 Distribution of $\underline{\mu}$

8.1 Large n

8.1.1 Central Limit Theorem

Let be $\underline{X}_1, \dots, \underline{X}_n$ iid $\underline{\mu}, \Sigma$, with $\underline{X}_i \in \mathbb{R}^p$

$$\underline{\bar{X}}_n \xrightarrow{P} \mu \text{ as } n \to \infty$$

i.e.

$$P(\|\underline{\bar{X}}_n - \underline{\mu}\| > \epsilon) \to 0 \text{ as } n \to \infty \quad \forall \epsilon > 0$$

and

$$S \xrightarrow{P} \Sigma \text{ as } n \to \infty$$

Remark.

> $X \sim \mu, \Sigma$ means that X is a random variable whose distribution has mean μ and covariance Σ, but it can be a general distribution

Definition 8.1: Asymptotically Gaussian

A random variable X is said to be Asymptotically Gaussian, i.e. $X \sim AN\left(\mu, \sigma^2\right)$, if:

$$\implies X \overset{n \to \infty}{\to} \sim N(\mu, \sigma)$$

Theorem 8.2

Let be $\underline{X}_1, \ldots, \underline{X}_n$ iid $\underline{\mu}, \Sigma$, with $\underline{X}_i \in \mathbb{R}^p$

$$\sqrt{n}(\underline{\bar{X}} - \underline{\mu}) \sim AN(\underline{0}, \Sigma)$$

Remark.

> For practical purposes, if n is "large enough":
>
> $$\underline{\bar{X}} \sim N_p\left(\underline{\mu}, \frac{\Sigma}{n}\right)$$

8.1.2 Inference for $\underline{\mu}$

Let be "Large sample size", i.e., n huge. Let be $\underline{X}_1, \ldots, \underline{X}_n$ iid $\underline{\mu}, \Sigma$.
By CLT:

$$\sqrt{n}(\underline{\bar{X}} - \underline{\mu}) \sim N_p(\underline{0}, \Sigma)$$

Then:

$$(\sqrt{n}(\underline{\bar{X}} - \underline{\mu})^T)\Sigma^{-1}(\sqrt{n}(\underline{\bar{X}} - \underline{\mu})) = n(\underline{\bar{X}} - \underline{\mu})^T\Sigma^{-1}(\underline{\bar{X}} - \underline{\mu}) \sim \chi^2(p)$$

Anyway, using the approximation of S, we can state this with the estimator:

$$n(\underline{\bar{X}} - \underline{\mu})^T S^{-1}(\underline{\bar{X}} - \underline{\mu}) \sim \chi^2(p)$$

Note that this is a random quantity depending on a parameter, but its distribution does not depend on the parameters. This is called **Pivotal Statistic**. We know the distribution but we don't know the parameter necessarily.

Definition 8.3: $\chi^2_{1-\alpha}(p)$

We use $\chi^2_{1-\alpha}(p)$ as a notation to state the point where $\chi^2(p)$ has a $1-\alpha$ cumulated distribution

Remark.

> Some books/softwares work with $\chi^2_\alpha(p)$

8.1.3 Confidence Region

Now, fixing $\alpha \in (0, 1)$

$$P\left[n(\underline{\bar{X}} - \underline{\mu})^T S^{-1}(\underline{\bar{X}} - \underline{\mu}) \leq \chi^2_{1-\alpha}(p)\right] = 1 - \alpha$$

i.e. (see section 4.2, mahalanobis distance):

$$P\left[d^2_{(\frac{1}{n}S)^{-1}}(\underline{\bar{X}}, \underline{\mu}) \leq \chi^2_{1-\alpha}(p)\right] = 1-\alpha$$

now, consider the sets:

$$\mathcal{E}^\alpha(\underline{\mu}) = \left\{\underline{x} \in \mathbb{R}^p \quad st \quad d^2_{(\frac{1}{n}S)^{-1}}(\underline{\bar{X}}, \underline{\mu}) \leq \chi^2_{1-\alpha}(p)\right\}$$

$$\mathcal{E}^\alpha(\underline{\bar{x}}) = \left\{\underline{\eta} \in \mathbb{R}^p \quad st \quad d^2_{(\frac{1}{n}S)^{-1}}(\underline{\eta}, \underline{\bar{x}}) \leq \chi^2_{1-\alpha}(p)\right\}$$

This is an ellipse. Note that by distance symmetry:

$$\underline{\bar{x}} \in \mathcal{E}^\alpha_{(\frac{1}{n}S)^{-1}}(\underline{\mu}) \iff \underline{\mu} \in \mathcal{E}^\alpha_{(\frac{1}{n}S)^{-1}}(\underline{\bar{x}})$$

Note that $\underline{\bar{X}}$ is random, $\underline{\mu}$ is not.

Definition 8.4: Confidence region

The confidence region for $\underline{\mu}$ at level $1-\alpha, \quad \alpha \in (0,1)$ is

$$CR_{1-\alpha}(\underline{\mu}) = \mathcal{E}^\alpha_{(\frac{1}{n}S)^{-1}}(\underline{\bar{x}})$$

Remark.

> Note that there is a formula that relates the data directly with the confidence region.

8.1.4 Testing $\underline{\mu}$

We may want to test a hypothesis over μ:

$$H_0 : \underline{\mu} = \underline{\mu}_0 \in \mathbb{R}^p \qquad vs \qquad H_1 : \underline{\mu} \neq \underline{\mu}_0$$

Assume H_0 is true:

$$T_0^2 = \underbrace{n(\underline{\bar{X}} - \underline{\mu}_0)^T S^{-1}(\underline{\bar{X}} - \underline{\mu}_0)}_{d^2_{(\frac{1}{n}S)^{-1}}(\underline{\bar{X}}, \underline{\mu}_0)} \sim \chi^2(p)$$

Reject H_0 at level α if:

$$T_0^2 > \chi^2_{1-\alpha}(p)$$

The probability of the distribution to be higher than T_0^2 is called p-value.

Then, another decision rule could be to reject H_0 if p-value is small.

Remark.

> Very important regarding how to use p-value: Read (Wasserstein & Lazar, 2016)

Remark.

Note that assuming H_0 is true, we are looking for the probability that the observed value is higher than a certain confidence, in order to reject our assumption. This is a contradiction reasoning.

Remark.

$$T_0^2 > \chi^2_{1-\alpha}(p) \iff \underline{\mu}_0 \in \left(\mathcal{E}^{\alpha}_{(\frac{1}{n}S)^{-1}}(\underline{\bar{x}})\right)^C = \left(CR_{1-\alpha}(\underline{\mu})\right)^C$$

i.e., the confidence region is the values that we cannot reject at level α

8.2 Small n

8.2.1 F Distribution

In the case of small n, it is not that simple. We must introduce an assumption:

$$\underline{X}_1, \dots, \underline{X}_n \quad iid \sim N(\underline{\mu}, \Sigma)$$

Now, define:

Definition 8.5: F distribution

Let be $Y \sim \chi^2(n)$ and $W \sim \chi^2(m)$. Define the F(n,m) distribution as:

$$F = \frac{Y/n}{w/m}$$

Remark.

- $t = \frac{Z}{\sqrt{\frac{W}{m}}}$, where $Z \sim N(0,1)$ and $W \sim \chi^2(m)$. Then,

$$t^2 = \frac{Z^2}{\frac{W}{m}} \sim F(n,m), \qquad Z^2 \sim \chi^2(1) \quad and \quad W \sim \chi^2(m)$$

- $$F(n,m) \overset{m\to\infty}{\to} \frac{1}{n}\chi^2(n)$$

  Indeed, remember that $W \sim \chi^2(m)$. Plus, it can be written:

$$W = \sum_{i=1}^{m} Z_i^2, \qquad Z_i \sim N(0,1) \text{ independent}$$

  So:

$$\frac{W}{m} = \frac{1}{m}\sum_{i=1}^{m} Z_i^2 \overset{m\to\infty}{\to} E[Z_i^2] = \chi^2(m)$$

We will use this in section 8.2.3

8.2.2 Hotellings Theorem

Theorem 8.6: Hotelling Theorem

Let be $X \sim N_p(\underline{\mu}, \Sigma)$ and $W \sim Wish(\frac{1}{m}\Sigma, m)$, where Σ is the same for both distributions. If $\underline{X} \perp\!\!\!\perp \underline{W}$:

$$\Longrightarrow \frac{m-p+1}{mp}(\underline{X}-\underline{\mu})^T W^{-1}(\underline{X}-\underline{\mu}) \sim F(p, m-p+1)$$

Proof. We won't prove it □

Corollary 8.7

Let be $\underline{X}_1, \ldots, \underline{X}_n \quad iid \sim N(\underline{\mu}, \Sigma)$

$$\Longrightarrow n(\underline{\bar{X}}-\underline{\mu})^T S^{-1}(\underline{\bar{X}}-\underline{\mu}) \sim \frac{(n-1)p}{n-p} F(p, n-p)$$

Proof. Let be $\underline{\bar{X}} \sim N_p(\underline{\mu}, \frac{1}{n}\Sigma)$. We know that:

$$\sqrt{n}(\underline{\bar{X}}-\underline{\mu}) \sim N_p(\underline{0}, \Sigma) \quad \perp\!\!\!\perp \quad (n-1)S \sim Wish(\Sigma, n-1)$$

Now, applying Hotellings theorem with m=n-1.

$$\Longrightarrow n(\underline{\bar{X}}-\underline{\mu})^T S^{-1}(\underline{\bar{X}}-\underline{\mu}) \sim \frac{(n-1)p}{n-p} F(p, n-p)$$

□

Remark.

Note this expression is a pivotal

Remark.

Note that this implies that if $n \to \infty$ (case large n):

$$\frac{(n-1)p}{n-p} F(p, n-p) \to \chi^2(p)$$

8.2.3 Confidence Region for $\underline{\mu}$

Fixing $\alpha \in (0,1)$

$$P\left[n(\underline{\bar{X}}-\underline{\mu})^T S^{-1}(\underline{\bar{X}}-\underline{\mu}) \leq \frac{(n-1)p}{n-p}F_{1-\alpha}(p,n-p)\right] = 1-\alpha$$

i.e.

$$P\left[d^2_{(\frac{1}{n}S)^{-1}}(\underline{\bar{X}},\underline{\mu}) \leq \frac{(n-1)p}{n-p}F_{1-\alpha}(p,n-p)\right] = 1-\alpha$$

Now, we consider:

$$CR_{1-\alpha}(\underline{\mu}) = \left\{\underline{x} \in \mathbb{R}^p \quad st \quad d^2_{(\frac{1}{n}S)^{-1}}(\underline{\bar{X}},\underline{\mu}) \leq \frac{(n-1)p}{n-p}F_{1-\alpha}(p,n-p)\right\}$$

Remark.

Now, note that this is consistent with what we obtained in the large n case:

$$\underbrace{\frac{(n-1)p}{n-p}}_{\to p}\underbrace{F_{1-\alpha}(p,n-p)}_{\frac{1}{p}\chi^2_{1-\alpha}(p)} \to \chi^2_{1-\alpha}(p)$$

Example.

Let be $\underline{X}_1,\dots,\underline{X}_{10}$ iid $\sim N_2(\underline{\mu},\Sigma)$ (Note that n=10), where:

$$\underline{\bar{X}} = \underline{0}, \qquad S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Let be $\alpha = 0.1$ Then:

$$\begin{aligned} CR_{1-\alpha}(\underline{\mu}) &= \left\{\underline{x} \in \mathbb{R}^2 \quad st \quad 10(0-\underline{\mu})^T S^{-1}(0-\underline{\mu}) \leq \frac{9\cdot 2}{8}F_{0.9}(2,8)\right\} \\ &= \left\{\underline{x} \in \mathbb{R}^2 \quad st \quad 10(x_1^2+x_2^2) \leq 6.997\right\} \\ &= \left\{\underline{x} \in \mathbb{R}^2 \quad st \quad x_1^2+x_2^2 \leq 0.6997\right\} \end{aligned}$$

So the points are whose are within the circle of radius $\sqrt{0.6997}$.
Assume now that the parameters are different:

$$\underline{\bar{X}} = \underline{0}, \qquad S = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

Then:

$$\begin{aligned}CR_{1-\alpha}(\underline{\mu}) &= \{\underline{x} \in \mathbb{R}^2 \quad st \quad \underline{x}^T S^{-1} \underline{x} \leq 0.6997\} \\ &= \left\{\underline{x} \in \mathbb{R}^2 \quad st \quad x_1^2 - x_1 x_2 + x_2^2 \leq \frac{3}{4} \cdot 0.6997\right\}\end{aligned}$$

These points are not in a circle but in an ellipse.

8.2.4 Testing $\underline{\mu}$

We may want to test a hypothesis over μ:

$$H_0 : \underline{\mu} = \underline{\mu}_0 \in \mathbb{R}^p \qquad vs \qquad H_1 : \underline{\mu} \neq \underline{\mu}_0$$

As well as in the case of large n, assume H_0 is true:

$$T_0^2 = \underbrace{n(\underline{\bar{X}} - \underline{\mu}_0)^T S^{-1} (\underline{\bar{X}} - \underline{\mu}_0)}_{d^2_{(\frac{1}{n}S)^{-1}}(\underline{\bar{X}}, \underline{\mu}_0)} \sim \frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)$$

Reject H_0 at level α if:

$$T_0^2 > \frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)$$

9 Distribution of $\underline{a}^T \underline{\mu}$

9.1 Estimator

Theorem 9.1

Given $\underline{a} \in \mathbb{R}^p$ and $\underline{\bar{X}} \sim N_p(\underline{\mu}, \frac{1}{n}\Sigma)$:

$$\underline{a}^T \underline{\bar{X}} \text{ is an unbiased estimator for } \underline{a}^T \underline{\mu}$$

Proof. We know that $\underline{a}^T \underline{\bar{X}} \sim N_1(\underline{a}^T \underline{\mu}, \frac{1}{n} \underline{a}^T \Sigma \underline{a})$
Then:

$$\frac{\underline{a}^T \underline{\bar{X}} - \underline{a}^T \underline{\mu}}{\sqrt{\underline{a}^T \Sigma \underline{a}}} \sqrt{n} \sim N(0, 1)$$

We also know that $(n-1)S \sim Wish(\Sigma, n-1)$.
And $(n-1)\underline{a}^T S \underline{a} \sim Wish(\underline{a}^T \Sigma \underline{a}, n-1) = (\underline{a}^T \Sigma \underline{a}) \chi^2(n-1)$
Then:

$$\frac{(n-1)\underline{a}^T S \underline{a}}{\underline{a}^T \Sigma \underline{a}} \sim \chi^2(n-1)$$

Plus:

$$\frac{\underline{a}^T \underline{\bar{X}} - \underline{a}^T \underline{\mu}}{\sqrt{\underline{a}^T \Sigma \underline{a}}} \sqrt{n} \perp\!\!\!\perp \frac{(n-1)\underline{a}^T S \underline{a}}{\underline{a}^T \Sigma \underline{a}}$$

Now, take:

$$\frac{\frac{\underline{a}^T\bar{\underline{X}}-\underline{a}^T\mu}{\sqrt{\underline{a}^T\Sigma\underline{a}}}\sqrt{n}\sim N(0,1)}{\sqrt{\frac{(n-1)\underline{a}^TS\underline{a}}{\underline{a}^T\Sigma\underline{a}}}\sqrt{\frac{1}{n-1}}\sim\frac{1}{n-1}\chi^2(n-1)}\sim t(n-1)$$

Simplifying:

$$\frac{\underline{a}^T\bar{\underline{X}}-\underline{a}^T\mu}{\sqrt{\underline{a}^TS\underline{a}}}\sqrt{n}\sim t(n-1)$$

Having a pivotal.

Now, take a $\alpha\in(0,1)$

$$P\left[-t_{1-\frac{\alpha}{2}}(n-1)\leq\frac{\underline{a}^T\bar{\underline{X}}-\underline{a}^T\mu}{\sqrt{\underline{a}^TS\underline{a}}}\sqrt{n}\leq t_{1-\frac{\alpha}{2}}(n-1)\right]=1-\alpha$$

In other words:

$$P\left[\underline{a}^T\mu\in\left[\underline{a}^T\bar{\underline{X}}\pm t_{1-\frac{\alpha}{2}}(n-1)\sqrt{\frac{\underline{a}^TS\underline{a}}{n}}\right]\right]=1-\alpha\qquad\forall\underline{a}\in\mathbb{R}^p$$

$\square$

Remark.

Note that this is not the same as:

$$P\left[\underline{a}^T\mu\in\left[\underline{a}^T\bar{\underline{X}}\pm t_{1-\frac{\alpha}{2}}(n-1)\sqrt{\frac{\underline{a}^TS\underline{a}}{n}}\right]\quad\forall\underline{a}\in\mathbb{R}^p\right]=1-\alpha$$

Note that the expression above is incorrect. We haven't found an interval for every a at the same time.

9.2 One-at-the-time Confidence Interval

So, the one-at-the-time confidence interval, i.e., without considering other directions, is:

$$CI_{1-\alpha}(\underline{a}_i^T\underline{\mu})=\left[\underline{a}^T\bar{\underline{X}}\pm t_{1-\frac{\alpha}{2}}(n-1)\sqrt{\frac{\underline{a}^TS\underline{a}}{n}}\right]$$

Example.

Specifically, taking $a_i=\begin{pmatrix}0&\cdots&\underbrace{1}_{i}&\cdots&0\end{pmatrix}$

$$CI_{1-\alpha}(\mu_i)=\left[\bar{X}_i\pm t_{1-\frac{\alpha}{2}}(n-1)\sqrt{\frac{S_{ii}}{n}}\right],\qquad i=1,\ldots,p$$

On the other hand, taking $a_i - a_j$:

$$CI_{1-\alpha}(\mu_i - \mu_j) = \left[\bar{X}_i - \bar{X}_j \pm t_{1-\frac{\alpha}{2}}(n-1)\sqrt{\frac{S_{ii} - 2S_{ij} + S_{jj}}{n}}\right], \qquad i = 1, \dots, p$$

Example.

Specifically, in the previous section example, the confidence interval is:

$$\begin{aligned} CI_{1-\alpha}(\mu_1) &= \left\{\bar{x}_1 \pm t_{1-\alpha/2}(10-1)\sqrt{\frac{S_{11}}{10}}\right\} \\ &= \{\bar{x}_1 \pm 0.5796\} \\ CI_{1-\alpha}(\mu_2) &= \{\bar{x}_2 \pm 0.5796\} \qquad \text{(Similarly)} \end{aligned}$$

Now, note that each separated interval is not taking into account the correlation between the components. Then, each confidence interval is narrower than each component of the CR, when we analyze every component together. This happens because the probability is divided in a wider space. i.e., the measure space is wider.

9.3 Test $\underline{a}^T \underline{\mu}$

We may want to test a hypothesis over $\underline{a}^T \mu$:

$$H_0 : \underline{a}^T \underline{\mu} \leq \delta_0 \qquad vs \qquad H_1 : \underline{a}^T \underline{\mu} \geq \delta_0$$

Then, the statistic is:

$$t_0 = \frac{\underline{a}^T \underline{\bar{X}} - \underline{a}^T \underline{\mu}}{\sqrt{\underline{a}^T S \underline{a}}} \sqrt{n}$$

Then, given a fixed $\alpha \in (0, 1)$:

$$\text{Reject } H_0 \text{ of } t_0 > t_{1-\alpha}(n-1)$$

On the other hand, we may want to test a different hypothesis:

$$H_0 : \underline{a}^T \underline{\mu} = \delta_0 \qquad vs \qquad H_1 : \underline{a}^T \underline{\mu} \neq \delta_0$$

Then, the statistic is:

$$|t_0| = \frac{|\underline{a}^T \underline{\bar{X}} - \underline{a}^T \underline{\mu}|}{\sqrt{\underline{a}^T S \underline{a}}} \sqrt{n}$$

Then, given a fixed $\alpha \in (0, 1)$:

$$\text{Reject } H_0 \text{ if } |t_0| > t_{1-\alpha/2}(n-1)$$

Example.

> Specifically, in the case of $\underline{a} = a_i - a_j$, we can obtain:
>
> $$H_0 : \mu_i - \mu_j \leq \delta_0 \qquad vs \qquad H_1 : \mu_i - \mu_j \geq \delta_0$$

9.4 Simultaneous CIs

Sometimes, we need to have the certainty that in every direction of μ, at the same time, the predictions will be correct simultaneously, taking into account the correlation between the directions. In this case, we have to use the simultaneous CIs.

Proposition 9.2

Let B positive definite $p \times p$, and $\underline{b}, \underline{d} \in \mathbb{R}^p$

$$\implies (\underline{b}^T \underline{d})^2 \leq (\underline{b}^T B \underline{b})(\underline{d}^T B^{-1} \underline{d})$$

This turns on an equality if $\underline{B} \in \mathcal{L}(B^{-1}\underline{d})$ (the linear space generated by the vector)

Proof.

$$\begin{aligned}(\underline{b}^T \underline{d})^2 &= (\underline{b}^T B^{1/2} B^{-1/2} \underline{d})^2 \\ &\leq (\underline{b}^T B \underline{b})(\underline{d}^T B^{-1} \underline{d}) \qquad \text{(Cauchy-schwarz)}\end{aligned}$$

This is an equality if $\underline{B} \in \mathcal{L}(B^{-1}\underline{d})$ (exercise based on extension of C-S) □

Lemma 9.3: Max Lemma

Let be $B \in \mathbb{R}^{p \times p}$ positive def., and $\underline{d} \in \mathbb{R}^p$

$$\implies \max_{\substack{\underline{x} \in \mathbb{R}^p \\ \underline{x} \neq \underline{0}}} \frac{(\underline{x}^T \underline{d})^2}{\underline{x}^T B \underline{x}} = \underline{d}^T B^{-1} \underline{d}$$

Proof. Let be $\underline{x} \in \mathbb{R}^p$. By the past proposition 9.4

$$(\underline{x}^T \underline{d})^2 \leq (\underline{x}^T B \underline{x})(d^T B^{-1} \underline{d})$$

If $x \neq 0$, as B is pos. def.:

$$\frac{(\underline{x}^T \underline{d})^2}{\underline{x}^T B \underline{x}} \leq \underline{d}^T B^{-1} \underline{d}$$

Again, with equality if $\underline{x} \in \mathcal{L}(B^{-1}\underline{d})$ □

Now, take the statistic introduced in section 9.3:

$$t_0 = \frac{\underline{a}^T \underline{\bar{X}} - \underline{a}^T \underline{\mu}}{\sqrt{\underline{a}^T S \underline{a}}} \sqrt{n} = \frac{\underline{a}^T (\underline{\bar{X}} - \underline{\mu})}{\sqrt{\underline{a}^T S \underline{a}}} \sqrt{n}$$

Take the square:

$$t_0^2 = \max_{\substack{\underline{a}\in\mathbb{R}^p \\ \underline{a}\neq\underline{0}}} \frac{(\underline{a}^T(\bar{\underline{X}}-\underline{\mu}))^2}{\underline{a}^T S \underline{a}} n$$

And by the lemma 9.4:

$$t_0^2 = \frac{(\underline{a}^T(\bar{\underline{X}}-\underline{\mu}))^2}{\underline{a}^T S \underline{a}} n = n \cdot (\bar{\underline{X}}-\underline{\mu})^T S^{-1} (\bar{\underline{X}}-\underline{\mu}) \sim \frac{(n-1)p}{n-p} F(p, n-p)$$

Then, we want to find a $c > 0$ st

$$P\left[-c \leq \frac{\underline{a}^T(\bar{\underline{X}}-\underline{\mu})}{\sqrt{\underline{a}^T S \underline{a}}}\sqrt{n} \leq c, \qquad \forall \underline{a}\in\mathbb{R}^p,\ \underline{a}\neq 0\right] = 1-\alpha$$

This probability can be rewritten:

$$\begin{aligned} P\left[\frac{(\underline{a}^T(\bar{\underline{X}}-\underline{\mu}))^2}{\underline{a}^T S \underline{a}} n \leq c^2, \qquad \forall \underline{a}\in\mathbb{R}^p,\ \underline{a}\neq 0\right] &= P\left[\max_{\substack{\underline{a}\in\mathbb{R}^p \\ \underline{a}\neq\underline{0}}} \frac{(\underline{a}^T(\bar{\underline{X}}-\underline{\mu}))^2}{\underline{a}^T S \underline{a}} n \leq c^2\right] \\ &= P\left[\frac{(\underline{a}^T(\bar{\underline{X}}-\underline{\mu}))^2}{\underline{a}^T S \underline{a}} n = n \cdot (\bar{\underline{X}}-\underline{\mu})^T S^{-1} (\bar{\underline{X}}-\underline{\mu}) \leq c^2\right] \end{aligned}$$

And as this distributes F:

$$c^2 = \frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)$$

Therefore,

$$P\left[-c \leq \frac{\underline{a}^T(\bar{\underline{X}}-\underline{\mu})}{\sqrt{\underline{a}^T S \underline{a}}}\sqrt{n} \leq \sqrt{\frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)}, \qquad \forall \underline{a}\in\mathbb{R}^p,\ \underline{a}\neq 0\right] = 1-\alpha$$

Then:

$$Sim\ CI_{1-\alpha}(\underline{a}^T\underline{\mu}) = \left[\underline{a}^T \bar{\underline{x}} \pm \sqrt{\frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)} \sqrt{\frac{\underline{a}^T S \underline{a}}{n}}\right]$$

Actually, Sim CI, for each direction $\underline{a}$, is the projection of the confidence region in that direction.

9.5 k-Simultaneous CIs

Given $\underline{a}_1, \ldots, \underline{a}_k \in \mathbb{R}^p$. Now, we are studying the k simultaneous confidence intervals for $\underline{a}_1^T \mu, \ldots, \underline{a}_k^T \mu$.

Note that by the Bonferroni ineq: $(P(A \cup B) \leq P(A) + P(B))$:

$$\begin{aligned}
P\left[\bigcap_{i=1}^{k} \{\underline{a}_i^T \underline{\mu} \in CI_{1-\alpha}(\underline{a}_i^T \underline{\mu})\}\right] &= 1 - P\left[\bigcup_{i=1}^{k} \{\underline{a}_i^T \underline{\mu} \notin CI_{1-\alpha}(\underline{a}_i^T \underline{\mu})\}\right] \\
&\geq 1 - \sum_{i=1}^{k} P\left[\underline{a}_i^T \underline{\mu} \notin CI_{1-\alpha}(\underline{a}_i^T \underline{\mu})\right] \\
&= 1 - \sum_{i=1}^{k} \alpha = 1 - k\alpha
\end{aligned}$$

And now, note that simply taking $1 - \frac{\alpha}{k}$ in each $\underline{a}$:

$$P\left[\bigcap_{i=1}^{k} \{\underline{a}_i^T \underline{\mu} \in CI_{1-\alpha}(\underline{a}_i^T \underline{\mu})\}\right] \geq 1 - k\frac{\alpha}{k} = 1 - \alpha$$

So, Bonferroni Sim CI's is:

$$BSim\ CI_{1-\alpha}(\underline{a}_i^T \underline{\mu}) = \left[\underline{a}^T \underline{\bar{X}} \pm t_{1-\frac{\alpha}{2k}}(n-1)\sqrt{\frac{\underline{a}^T S \underline{a}}{n}}\right]$$

The difference between the one-at-the-time CI and the Bonferroni is called the Bonferroni correction

9.6 Test k simultaneous $\underline{a}_i^T \underline{\mu}$

We may want to test a hypothesis over μ:

$$H_0 \begin{cases} H_{01} : \underline{a}_1^T \underline{\mu} = \delta_1 \\ and \\ H_{02} : \underline{a}_2^T \underline{\mu} = \delta_2 \\ and \\ \dots \\ and \\ H_{0k} : \underline{a}_k^T \underline{\mu} = \delta_k \end{cases} \quad vs \quad H_1 \begin{cases} H_{11} : \underline{a}_1^T \underline{\mu} \neq \delta_1 \\ or \\ H_{12} : \underline{a}_2^T \underline{\mu} \neq \delta_2 \\ or \\ \dots \\ or \\ H_{1k} : \underline{a}_k^T \underline{\mu} \neq \delta_k \end{cases}$$

Now, the decision rule will be reject H_0 if

$$\exists i \in \{1, \dots, k\} \quad st \quad \frac{\underline{a}^T \underline{\bar{X}} - \delta_i}{\sqrt{\underline{a}^T S \underline{a}}}\sqrt{n} > t_{1-\frac{\alpha}{2k}}(n-1)$$

i.e.:

$$P\left[\bigcup_{i=1}^{k}\left\{\frac{\underline{a}^T\underline{\bar{X}}-\delta_i}{\sqrt{\underline{a}^TS\underline{a}}}\sqrt{n}>t_{1-\frac{\alpha}{2k}}(n-1)\right\}\middle|\bigcap_i H_{0i}\text{ are true}\right]$$
$$\leq\sum_{i=1}^{k}P\left[\frac{\underline{a}^T\underline{\bar{X}}-\delta_i}{\sqrt{\underline{a}^TS\underline{a}}}\sqrt{n}>t_{1-\frac{\alpha}{2k}}(n-1)\middle|\bigcap_i H_{0i}\text{ are true}\right]=\sum_{i=1}^{k}\frac{\alpha}{k}=\alpha$$

Note that if k is too large, the statics will be non-informative at all. Indeed, we will never reject the hypotheses.

9.7 Large Scale Hypothesis Testing

9.7.1 Family-Wise Error Rate

To solve the large-scale hypothesis problem in Bonferroni formulation, Benjamin and Hochberg proposed a solution (Benjamini & Hochberg, 1995).
Suppose again a k-simultaneous hypothesis:

$$H_0\begin{cases}H_{01}:\underline{a}_1^T\underline{\mu}=\delta_1\\ and\\ H_{02}:\underline{a}_2^T\underline{\mu}=\delta_2\\ and\\ \dots\\ and\\ H_{0k}:\underline{a}_k^T\underline{\mu}=\delta_k\end{cases}\quad vs\quad H_1\begin{cases}H_{11}:\underline{a}_1^T\underline{\mu}\neq\delta_1\\ or\\ H_{12}:\underline{a}_2^T\underline{\mu}\neq\delta_2\\ or\\ \dots\\ or\\ H_{1k}:\underline{a}_k^T\underline{\mu}\neq\delta_k\end{cases}$$

Use a D strategy for testing these k hypotheses (e.g. Bonferroni Strategy). Then, define:

	Decision Following D		
Truth	No Reject H_0	Reject H_0	
H_0	U	V	$= K_0$
H_1	T	S	
		$= R$	

Where:

- U: Correct Omissions
- V: False Discover
- T: Missed Discover
- S: True Discover

However, in practice, we can acknowledge R, but no K_0, since we don't know what's the truth.

Now, define:

$$I_0 = \{i = \{1, \dots, k\} \quad st \quad H_{0i} \text{ is true}\}, \qquad K_0 = |I_0|$$

Note that:

$$P\left[\text{at least one } H_{0i} \text{ rejected} | \bigcap_{i \in I_0} H_{0i}\right] = P[V \geq 1] = \text{FWER} = P\left[\bigcup_{i \in I_0} \{\text{Reject } H_{0i} | H_{0i} \text{ is true}\}\right]$$

FWER stands for Family-Wise Error Rate. Specifically, note that in the case we use the Bonferroni strategy, $FWER \leq \alpha$:

$$FWER \leq \sum_{i \in I_0} \frac{\alpha}{K} = K_0 \frac{\alpha}{K} \leq K \frac{\alpha}{K} = \alpha$$

9.7.2 False Discovery Rate

Now, define:

$$Q = \begin{cases} 0 & if \ R = 0 \\ \frac{V}{R} & if \ R > 0 \end{cases}$$

Now, consider $\frac{V}{R}$

Definition 9.4: False Discovery Rate (FDR)

$$FDR = E[Q]$$

Remark.

Possible cases:

- $k_0 = k$ Then: $S = 0$ and $V = R$. So:

$$Q = \begin{cases} 0 & if \ V = R = 0 \\ 1 & if \ V = R > 0 \end{cases}$$

  Then:

$$FDR = E[Q] = P[V > 0] = P[V \geq 1] = FWER$$

- $k_0 < k$, then:
  - If $V = 0 \implies Q = 0$
  - If $V > 0 \implies Q = \frac{V}{R} \leq 1$

  Hence

$$Q \leq \mathbb{1}[V > 0], \qquad \mathbb{1} \text{ is the indicatrix}$$

  So:

$$FDR = E[Q] \leq E[\mathbb{1}[V > 0]] = P[V > 0] = P[V \geq 1] = FWER$$

Specifically, B&H introduced a strategy for controlling FDR (Benjamini & Hochberg, 1995).
Let p_i be the $p-value$ for the test $H_{0i} \quad vs \quad H_{1i}$.
Consider $p_1, \dots, p_k$ and order them:

$$\underbrace{p_{(1)}}_{H_{0(1)}\ vs\ H_{1(1)}} \leq \dots \leq \underbrace{p_{(k)}}_{H_{0(p)}\ vs\ H_{1(p)}}$$

Theorem 9.5

If p_i are independent. Let be:

$$m = \max\left\{ i \in \{1, \dots, k\} \quad st \quad p_{(i)} \leq i\frac{\alpha}{k} \right\}$$

Take the strategy D_{BH} of rejecting $H_{0(i)}$ if $i \leq m$

$$\implies D_{BH} \text{ controls FDR at level } \alpha$$

Remark.

> We are rejecting the hypotheses that have a p-value increasing more than linear wrt α/k

Theorem 9.6: (Benjamini & Yekutieli, 2001)

1. If $p_1, \dots, p_k$ are positive correlated

$$\implies D_{BH} \text{ controls FDR at level } \alpha$$

2. If $p_1, \dots, p_k$ are negatively correlated

$$\implies D^*_{BH} \text{ strategy controls FDR at level } \alpha$$

where D^*_{BH} uses m^*:

$$m^* = \max\left\{ i \in \{1, \dots, k\} \quad st \quad p_{(i)} \leq i\frac{\alpha}{k \cdot C(k)} \right\}, \qquad C(k) = \sum_{i=1}^{k} \frac{1}{k}$$

10 Comparing means of different Gaussian Distributions

10.1 The problem

Given n units of paired data with $X_1, \ldots, X_p$ features:

$$\left(\begin{pmatrix} \underline{X}_{11} \\ \underline{X}_{12} \end{pmatrix}, \begin{pmatrix} \underline{X}_{21} \\ \underline{X}_{22} \end{pmatrix}, \ldots, \begin{pmatrix} \underline{X}_{n1} \\ \underline{X}_{n2} \end{pmatrix} \right) \subset \mathbb{R}^{p \times p}$$

We will represent each unit as:

$$\underline{X}_{i1} = \begin{pmatrix} X_{i11} \\ X_{i12} \\ \cdots \\ X_{i1p} \end{pmatrix} \in \mathbb{R}^p \qquad \wedge \qquad \underline{X}_{i2} = \begin{pmatrix} X_{i21} \\ X_{i22} \\ \cdots \\ X_{i2p} \end{pmatrix} \in \mathbb{R}^p$$

Given overall independent data pairs, with means $\underline{\mu}_1$ and $\underline{\mu}_2$ over the first and second element of the pairs, respectively. The goal is to do inference on $\underline{\mu}_1 - \underline{\mu}_2$. *Note that this is independence between pairs, no between elements in the pair.*

10.2 Pair Difference

Consider the difference:

$$\underline{D}_i = \underline{X}_{i1} - \underline{X}_{i2}$$

Given $\underline{D}_1, \ldots, \underline{D}_n$ iid $\sim N_P(\underline{\delta}, \Sigma)$, with $\underline{\delta} = \underline{\mu}_1 - \underline{\mu}_2$.
Denote:

$$\bar{\underline{D}} = \frac{1}{n} \sum_{i}^{n} \underline{D}_i$$

$$S_D = \frac{1}{n-1} \sum_{i=1}^{n} (\underline{D}_i - \bar{\underline{D}})(\underline{D}_i - \bar{\underline{D}})^T$$

Note that, as seen in section 8.2.2:

$$n(\bar{\underline{D}} - \underline{\delta})^T S_D^{-1} (\bar{\underline{D}} - \underline{\delta}) \sim \frac{(n-1)p}{n-p} F(p, n-p)$$

And given an $\alpha \in (0, 1)$, the confidence interval is:

$$CR_{1-\alpha}(\underline{\delta}) = \left\{ \underline{\eta} \in \mathbb{R}^p \quad st \quad n(\bar{\underline{D}} - \underline{\eta})^T S_D^{-1} (\bar{\underline{D}} - \underline{\eta}) \leq \frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p) \right\}$$

So we can test:

$$H_0 : \underline{\delta} = \underline{\delta}_0 \qquad vs \qquad H_1 : \underline{\delta} \neq \underline{\delta}_0$$

We will reject at a level α if:

$$T_0^2 > \frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)$$

where:

$$T_0^2 = n(\underline{\bar{D}} - \underline{\eta})^T S_D^{-1} (\underline{\bar{D}} - \underline{\eta})$$

Plus, the simultaneous CI:

$$Sim\ CI_{1-\alpha} = (\underline{a}^T \underline{\delta}) = \left[\underline{a}^T \underline{\bar{D}} \pm \sqrt{\frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)} \sqrt{\frac{\underline{a}^T S_D^{-1} \underline{a}}{n}} \right] \qquad \forall \underline{a} \in \mathbb{R}^p$$

Example.

$$Sim\ CI_{1-\alpha} = (\underline{\delta}_j) = \left[\underline{\bar{D}}_i \pm \sqrt{\frac{(n-1)p}{n-p} F_{1-\alpha}(p, n-p)} \sqrt{\frac{S_{D\ jj}^{-1}}{n}} \right] \qquad \forall \underline{j} = 1, \ldots, p$$

Plus, the Bonferroni CI:

$$Bonf\ CI_{1-\alpha}(\delta_j) = \left[\bar{D}_i \pm t_{1-\frac{\alpha}{2p}}(n-1) \sqrt{\frac{S_{D\ jj}}{n}} \right]$$

Take now this example:

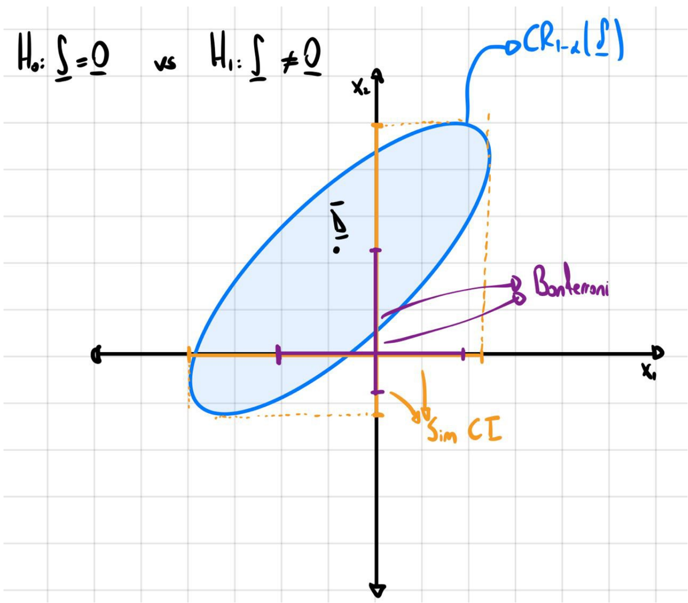

Figure 4: Example: Confidence intervals and regions

We can reject H_0, since $\underline{\delta}$ is outside the CR at a level α, then, if $\underline{\delta} = \begin{pmatrix} \delta_1 \\ \delta_2 \\ \dots \\ \delta_p \end{pmatrix} = \underline{\mu}_1 - \underline{\mu}_2$, we can conclude that $\underline{\mu}_1 \neq \underline{\mu}_2$.

Can we conclude whether $\underline{\delta}_1$, $\underline{\delta}_2$, or any of the p δ's differ from 0? As we can see in the Simultaneous Confidence Interval, which we have seen are the projections on the axis of the CR, there's not enough evidence to state this. We can say that the δs are different, but we cannot say which one is, the vector as a whole is different from 0.

We have shown only two δs, but it may happen that every δ SIM CI contains 0.

Now, we can think this happens because I am taking any possible linear combination. We then may think about Bonferroni, but in this case, each component contains 0 as well.

Note also that we can analyze if the mean is different from 0. We obviously know that $\underline{\bar{D}}$ is different from 0, but what we really need to find is if there is enough difference between $\underline{\bar{D}}$ and 0 to state that the population meaning is different from 0.

10.3 Repeated Measurements

Given the case $\underline{X}_1, \dots, \underline{X}_n$ iid $\sim N_q(\underline{\mu}, \Sigma)$:
Take the unit i. Suppose there are q observations: $X_{i1}, X_{i2}, \dots, X_{iq} \subset \mathbb{R}$, i.e. the same feature X observed in q instances:

$$\underline{X}_i = (X_{i1}, \dots, X_{iq}) \in \mathbb{R}^q$$

Now, we want to test:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_q \qquad vs \qquad H_1 : H_0^c$$

where $\mu_j = E[X_j] \quad j = 1, \dots, q$

Example.

> Possible interpretation: Is the variable changing over time/space?

Note that if $\underline{\mu} = \begin{pmatrix} \mu_1 \\ \dots \\ \mu_q \end{pmatrix}$, $H_0 : \underline{\mu} \in \mathcal{L}(\underline{1}) \qquad vs \qquad H_0 : \underline{\mu} \notin \mathcal{L}(\underline{1})$.
Now, consider :$\underline{c}_1, \dots, \underline{c}_{q-1} \in \mathcal{R}^q \quad st:$

- $\underline{c}_1, \dots, \underline{c}_{q-1}$ are linear independent
- $\underline{c}_i^T \underline{1} = 0 \quad i = 1, \dots, q-1 \qquad (i.e. \underline{c}_i \perp 1)$

i.e. $\mathcal{L}^{\perp}(\underline{1}) = span(\underline{c}_1, \dots, \underline{c}_{q-1})$.

Define the contrast matrix:

$$C = \begin{bmatrix} \underline{c}_1^T \\ \dots \\ \underline{c}_{q-1}^T \end{bmatrix} \in \mathbb{R}^{(q-1)\times q}$$

Now, the test turns on:

$$H_0 : C\underline{\mu} = \underline{0} \qquad vs \qquad H_1 : C\underline{\mu} \neq \underline{0}$$

Now, the unbiased estimator for $C\underline{\mu}$ is $C\underline{\bar{X}}$.
Plus, note we know that:

$$C\underline{\bar{X}} \sim N_{q-1}(C\underline{\mu}, \frac{1}{n}C\Sigma C^T)$$

So:

$$\sqrt{n}(C\underline{\bar{X}} - C\underline{\mu}) \sim N_{q-1}(0, C\Sigma C^T)$$

and

$$(n-1)CSC^T \sim Wishart(C\Sigma C^T, n-1)$$

And those are independent since $\underline{\bar{X}}$ and S are independent. Then, by Hotellings's theorem:

$$n(C\underline{\bar{X}} - C\underline{\mu})^T(CSC)^{-1}(C\underline{\bar{X}} - C\underline{\mu}) \sim \frac{(n-1)(q-1)}{n-q+1}F(q-1, n-q+1)$$

This is a pivotal, so we can use it for confidence regions:

$$T_0^2 = n(C\underline{\bar{X}} - 0)^T(CSC)^-1(C\underline{\bar{X}} - 0) = n(C\underline{\bar{X}})^T(CSC)^-1(C\underline{\bar{X}})$$

Reject at level $\alpha \in (0,1)$ if:

$$T_0^2 > \frac{(n-1)(q-1)}{n-q+1}F_{1-\alpha}(q-1, n-q+1)$$

Remark.

> Note that we can build C and $\tilde{C}$, as two different matrices, that both generate the complement of $\underline{1}$. In this case, it can be easily proven that since we can express $\tilde{C} = BC$, T_0^2 is the same

10.4 Hypotheses on Linear Spaces

As a generalization of past case, take $\underline{X}_1, \dots, \underline{X}_n$ iid $\sim N(\underline{\mu}, \Sigma)$, now we will test:

$$H_0 : \underline{\mu} \in \mathcal{L} \qquad vs \qquad H_1 : \underline{\mu} \notin \mathcal{L}$$

where $\mathcal{L}$ is a linear space of dimension k.
Take $\mathcal{L}^\perp = span(\underline{c}_1, \dots, \underline{c}_{p-k})$.

Define: $C = \begin{bmatrix} \underline{c}_1^T \\ \dots \\ \underline{c}_{p-k}^T \end{bmatrix}$ So the hypothesis is:

$$H_0 : C\underline{\mu} = \underline{0} \qquad vs \qquad H_1 : C\underline{\mu} = \underline{0}$$

So, as past case:

$$\sqrt{n}(C\bar{\underline{X}} - C\underline{\mu}) \sim N_{p-k}(0, C\Sigma C^T)$$

and

$$(n-1)CSC^T \sim Wishart(C\Sigma C^T, n-1)$$

And those are independent. And by Hotelling's theorem:

$$n(C\bar{\underline{X}} - C\underline{\mu})^T(CSC)^{-1}(C\bar{\underline{X}} - C\underline{\mu}) \sim \frac{(n-1)(p-k)}{n-p+k}F(p-k, n-p+k)$$

In general, it is a good idea to try to reframe a hypothesis test into a linear space hypothesis.

Example.

> In unit i, we have different measures across time:
>
> $$\underline{X}_i = \left(\begin{pmatrix} \underline{X}_{i11} \\ \underline{X}_{i21} \end{pmatrix}, \begin{pmatrix} \underline{X}_{i12} \\ \underline{X}_{i22} \end{pmatrix}, \dots, \begin{pmatrix} \underline{X}_{i1q} \\ \underline{X}_{i2q} \end{pmatrix}\right) = \begin{pmatrix} height \\ weight \end{pmatrix}$$
>
> We have: $\underline{\mu} = \begin{pmatrix} \underline{\mu}_{H1} \\ \dots \\ \underline{\mu}_{Hq} \\ \underline{\mu}_{W1} \\ \dots \\ \underline{\mu}_{Wq} \end{pmatrix} \in \mathbb{R}^{2q}$ So we can reframe H_0:
>
> $$H_0 : \begin{cases} \underline{\mu}_{H1} = \dots = \underline{\mu}_{Hq} \\ and \\ \underline{\mu}_{W1} = \dots = \underline{\mu}_{Wq} \end{cases} \qquad vs \qquad H_1 : H_0^c$$
>
> It can be proven that this can be reframed to the hypothesis with the linear space:
>
> $$\begin{pmatrix} 1 & \dots & 1 & 2 & \dots & 2 \end{pmatrix}$$

11 Multivariate Analysis of Variance (MANOVA)

11.1 The problem

Let be

$$\underline{X}_{11}, \dots, \underline{X}_{1n_1} \ iid \sim N_p(\underline{\mu}_1, \Sigma)$$
$$\underline{X}_{21}, \dots, \underline{X}_{2n_2} \ iid \sim N_p(\underline{\mu}_2, \Sigma)$$
$$\dots$$
$$\underline{X}_{g1}, \dots, \underline{X}_{gn_g} \ iid \sim N_p(\underline{\mu}_g, \Sigma)$$

such that they are independent.

Remark.

> This is not the same case presented in section 10. Note that here we don't have pairs, so the elements are not sorted from some perspective.

Note that the variance is always the same. We want to study the difference in the $\mu_1, \dots, \mu_g$. The goal of this is to, given different experiences, check if there is enough evidence to say that the population means are different, given $\underline{\bar{X}}_1, \dots \underline{\bar{X}}_g$. From another perspective, whether the variability between the group means is enough to say that they are different.

11.2 Hotelling's Test: Case $p \geq 1, g = 2$

Given $p \geq 1, g = 2$:

$$\underline{X}_{11}, \dots, \underline{X}_{1n_1} \ iid \sim N_p(\underline{\mu}_1, \Sigma)$$
$$\underline{X}_{21}, \dots, \underline{X}_{2n_2} \ iid \sim N_p(\underline{\mu}_2, \Sigma)$$

We want to do inference on $\underline{\mu}_1 - \underline{\mu}_2$ (Since $\neq 0$ indicates the means are different).

Note that:

$$\underline{\bar{X}_1} = \frac{1}{n_1} \sum_{i=1}^{n_1} \underline{X}_{i1} \sim N_p\left(\underline{\mu}_1, \frac{1}{n}\Sigma\right)$$

$$\underline{\bar{X}_2} = \frac{1}{n_2} \sum_{i=1}^{n_2} \underline{X}_{i2} \sim N_p\left(\underline{\mu}_2, \frac{1}{n}\Sigma\right)$$

$$\implies \underline{\bar{X}}_1 - \underline{\bar{X}}_2 \sim N_p\left(\underline{\mu}_1 - \underline{\mu}_2, \left(\frac{1}{n_1} + \frac{1}{n_2}\right)\Sigma\right)$$

Then:

$$\left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{-1/2} ((\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - (\underline{\mu}_1 - \underline{\mu}_2)) \sim N_p(\underline{0}, \Sigma)$$

Plus, given distributions of $\underline{\bar{X}}_1$ and $\underline{\bar{X}}_2$, S_1 and S_2 are estimators of Σ:

$$S_1 = \frac{1}{n_1 - 1}\sum_i^{n_1}(\underline{X}_{i1} - \underline{\bar{X}}_1)(\underline{X}_{i1} - \underline{\bar{X}}_1)^T \qquad (n_1 - 1)S_1 \sim Wish(\Sigma, n_1 - 1)$$

$$S_2 = \frac{1}{n_2 - 1}\sum_i^{n_2}(\underline{X}_{i2} - \underline{\bar{X}}_2)(\underline{X}_{i2} - \underline{\bar{X}}_2)^T \qquad (n_2 - 1)S_2 \sim Wish(\Sigma, n_2 - 1)$$

And, as the distributions are independent (since the populations are independent), by Wishart properties:

$$(n_1 - 1)S_1 + (n_2 - 1)S_2 \sim Wish(\Sigma, n_1 + n_2 - 2)$$

Now we can define:

$$S_{pooled} := \frac{(n_1 - 1)S_1 + (n_2 - 1)S_2}{n_1 + n_2 - 2}$$

So:

$$(n_1 + n_2 - 2)S_{pooled} \sim Wish(\Sigma, n_1 + n_2 - 2)$$

And $S_{pooled} \perp\!\!\!\perp \underline{\bar{X}}_1 - \underline{\bar{X}}_2$. So, by hotelling's theorem:

$$\left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{-1}((\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - (\underline{\mu}_1 - \underline{\mu}_2))^T (S_{pooled})^{-1}((\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - (\underline{\mu}_1 - \underline{\mu}_2)) \sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1}F(p, n_1 + n_2 - 1 - p)$$

Note that this is a pivotal.

Now, the test is:

$$H_0 : \underline{\mu}_1 - \underline{\mu}_2 = \underline{\delta}_0 \qquad vs \qquad \underline{\mu}_1 - \underline{\mu}_2 \neq \underline{\delta}_0$$

And:

$$T_0^2 = \left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{-1}((\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - \underline{\delta}_0)^T (S_{pooled})^{-1}((\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - \underline{\delta}_0)$$

And at a level $\alpha \in (0, 1)$, we will reject if:

$$T_0^2 > \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1}F_{1-\alpha}(p, n_1 + n_2 - 1 - p)$$

So the confidence region $CR_{1-\alpha}(\underline{\mu}_1 - \underline{\mu}_2)$ is $\underline{\eta} \in \mathbb{R}^p \quad st$:

$$\left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{-1}((\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - \underline{\eta})^T (S_{pooled})^{-1}((\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - \underline{\eta}) \leq \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1}F_{1-\alpha}(p, n_1 + n_2 - 1 - p)$$

We can find Sim CI and Bonf CI as well.

11.3 The variance fissure problem

Now, analyze a different problem, where variances are possibly different

$$\underline{X}_{11}, \ldots, \underline{X}_{1n_1} \; iid \sim N_p(\underline{\mu}_1, \Sigma_1)$$

$$\underline{X}_{21}, \ldots, \underline{X}_{2n_2} \; iid \sim N_p(\underline{\mu}_2, \Sigma_2)$$

Test:

$$H_0 : \Sigma_1 = \Sigma_2 \qquad vs \qquad H_1 : \Sigma_1 \neq \Sigma_2$$

This is a difficult problem.

There are tests generalizing level tests for p=1 (See Anderson's book).

There are non-parametric views of this problem, for example, permutation tests for the equality of the covariances (Cabassi, Pigoli, Secchi, & Carter, 2017).
Now, taking

$$\underline{\bar{X}}_1 \sim N_p(\underline{\mu}_1, \frac{1}{n_1}\Sigma_1)$$
$$\underline{\bar{X}}_2 \sim N_p(\underline{\mu}_2, \frac{1}{n_2}\Sigma_2)$$

$$\implies \underline{\bar{X}}_1 - \underline{\bar{X}}_2 \sim N_p(\underline{\mu}_1 - \underline{\mu}_2, \frac{1}{n_1}\Sigma_1 + \frac{1}{n_2}\Sigma_2)$$

Then:

$$\left[(\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - (\underline{\mu}_1 - \underline{\mu}_2)\right]^T \left(\frac{1}{n_1}\Sigma_1 + \frac{1}{n_2}\Sigma_2\right)^{-1} \left[(\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - (\underline{\mu}_1 - \underline{\mu}_2)\right] \sim \chi^2(p)$$

For large samples: n_1 and n_2 large:

$$S_1 \xrightarrow{p} \Sigma_1 \quad \text{as } n_1 \to \infty \qquad S_2 \xrightarrow{p} \Sigma_2 \quad \text{as } n_2 \to \infty$$

So:

$$\left[(\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - (\underline{\mu}_1 - \underline{\mu}_2)\right]^T \left(\frac{1}{n_1}S_1 + \frac{1}{n_2}S_2\right)^{-1} \left[(\underline{\bar{X}}_1 - \underline{\bar{X}}_2) - (\underline{\mu}_1 - \underline{\mu}_2)\right] \sim \chi^2(p)$$

And this is pivotal, so we can find the confidence region for $\underline{\mu}_1 - \underline{\mu}_2$, and the Sim CI and Bonf CI for $\underline{a}^T(\underline{\mu}_1 - \underline{\mu}_2)$.

11.4 ANOVA: Case $p = 1, g \geq 2$

Let be the ANOVA case (it is also called one-way anova, since there's only one independent variable):

$$X_{11}, \dots, X_{1n_1} \; iid \sim N_1(\mu_1, \sigma^2)$$
$$X_{21}, \dots, X_{2n_2} \; iid \sim N_1(\mu_2, \sigma^2)$$
$$\dots$$
$$X_{g1}, \dots, X_{gn_g} \; iid \sim N_1(\mu_g, \sigma^2)$$

such that they are independent.
Define $n = n_1 + n_2 + \dots + n_g$.
Given this case;

$$H_0 : \mu_1 = \dots = \mu_g \qquad vs \qquad H_1 : H_0^C$$

H_0 is rejected if we find a relevant difference on $\mu_i - \mu_j$.
Now, we perform a model reparameterization:

$$X_{ij} = \overbrace{\mu + \tau_i}^{\mu_i} + \varepsilon_{ij}$$

$$\varepsilon_{ij} \; iid \sim N(0, \sigma^2)$$

where

- μ is called the overall mean.
- τ_i is called the treatment of group i.

Now define an estimator of μ:

$$\hat{\mu}: \quad \bar{X} := \frac{1}{n} \sum_{i=1}^{g} \sum_{j=1}^{n_i} X_{ij}$$

And:

$$\begin{aligned} E[\bar{X}] &= \frac{1}{n} \sum_{i=1}^{g} \sum_{j=1}^{n_i} E[X_{ij}] = \frac{1}{n} \sum_{i} \sum_{j} \mu + \tau_i \\ &= \frac{1}{n} \sum_{i=1}^{g} n_i(\mu + \tau_i) = \frac{1}{n} \sum_{i} n_i \mu + \frac{1}{n} \sum_{i} n_i \tau_i = \mu + \frac{1}{n} \sum_{i=1}^{g} n_i \tau_i \end{aligned}$$

We add a condition to make the estimator unbiased:

$$\frac{1}{n} \sum_{i=1}^{g} n_i \tau_i = 0$$

Some softwares assume the following condition, since they assume the n_i are equal (be careful with this):

$$\sum \tau_i = 0$$

Now we construct an estimator of τ_i, i=1,..., g, first defining:

$$\bar{X}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij}$$

Then, the estimator $\hat{\tau}_i : \quad \bar{X}_i - \bar{X}$ is unbiased for every τ_i:

$$E[\bar{X}_i - \bar{X}] = \mu + \tau_i - \mu = \tau_i$$

To build hypothesis tests, we need the decomposition of variance tool.

11.5 Decomposition of Variance

Let be $\underline{X} = (X_{11}, \ldots, X_{1n_1}, X_{2n}, \ldots, X_{2n_2}, \ldots, X_{gn_g})^T \in \mathbb{R}^n$. Now, take:

$$\underline{u}_1 = \begin{pmatrix} 1 \\ \ldots (n_1 \ times) \\ 1 \\ 0 \\ \ldots \\ 0 \end{pmatrix}, \underline{u}_2 = \begin{pmatrix} 0 \\ \ldots (n_1 \ times) \\ 0 \\ 1 \\ \ldots (n_2 \ times) \\ 1 \\ 0 \\ \ldots \\ 0 \end{pmatrix}, \underline{u}_g = \begin{pmatrix} 0 \\ \ldots (n - n_g \ times) \\ 0 \\ 1 \\ \ldots (n_g \ times) \\ 1 \end{pmatrix}$$

The vectors satisfy the properties:

- $\underline{u}_1, \ldots, \underline{u}_g$ are linearly independent
- $\underline{u}_i^T \underline{u}_j = 0 \quad \forall i, j \quad st \quad i \neq j \qquad i.e.\ \underline{u}_1, \ldots, \underline{u}_g$ are orthogonal
- $\underline{1} \in span(\underline{u}_1, \ldots, \underline{u}_g)$, since $\underline{1} = \sum_{i=1}^{g} \underline{u}_i$

Now take the projections:

1. Over the span of u's:

$$\Pi_{\underline{u}_1, \ldots, \underline{u}_g}(\underline{X}) = \sum_{i=1}^{g} \frac{\underline{u}_i \underline{u}_i^T}{\underline{u}_i^T \underline{u}_i} \underline{X} = \sum_{i=1}^{g} \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij} \cdot \underline{u}_i = \sum_{i=1}^{g} \bar{X}_i \cdot \underline{u}_i$$

   Note that: $\underline{u}_i^T \underline{u}_i = n_i \quad \wedge \quad \underline{u}_i^T \underline{X} = \sum_{j=1}^{n_i} X_{ij}$

2. Over $\underline{1}$:

$$\Pi_{\underline{1}}(\underline{X}) = \frac{\underline{1}\underline{1}^T}{\underline{1}^T \underline{1}} \underline{X} = \left(\frac{1}{n} \sum_{i=1}^{g} \sum_{j=1}^{n_i} X_{ij} \right) \cdot \underline{1} = \bar{X} \cdot \underline{1} = \sum_{i=1}^{g} \bar{X} \cdot \underline{u}_i$$

3. Projection 1 over $\underline{1}$:

$$\begin{aligned} \Pi_{\underline{1}} \left(\sum_{i=1}^{g} \bar{X}_i \underline{u}_i \right) &= \frac{\underline{1}\underline{1}^T}{\underline{1}^T \underline{1}} \left(\sum \bar{X}_i \underline{u}_i \right) = \sum_{i=1}^{g} \bar{X}_i \frac{\underline{1}\underline{1}^T}{\underline{1}^T \underline{1}} \underline{u}_i \\ &= \left(\frac{1}{n} \sum_{i=1}^{g} \bar{X}_i n_i \right) \cdot \underline{1} = \frac{1}{n} \sum_{i=1}^{g} \sum_{j=1}^{n_i} X_{ij} \cdot \underline{1} = \bar{X} \cdot \underline{1} \end{aligned}$$

So we can decompose $\underline{X}$:

$$\underline{X} = \bar{X} \cdot \underline{1} + \sum_{i=1}^{g} (\bar{X}_i - \bar{X}) \underline{u}_i + (\underline{X} - \sum_{i=1}^{g} \bar{X}_i \underline{u}_i)$$

Being these 3 terms orthogonal. Then:

$$\|\underline{X}\|^2 = \underbrace{\|\bar{X} \cdot \underline{1}\|^2}_{1} + \underbrace{\left\| \sum_{i=1}^{g} (\bar{X}_i - \bar{X}) \underline{u}_i \right\|^2}_{2} + \underbrace{\left\| \underline{X} - \sum_{i=1}^{g} \bar{X}_i \underline{u}_i \right\|^2}_{3}$$

Then:

$$\sum_{i=1}^{g} \sum_{j=1}^{n_i} X_{ij}^2 = n\bar{X}^2 + \sum_{i=1}^{g} (\bar{X}_i - \bar{X})^2 n_i + \sum_{i=1}^{g} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

In words:

$$SS_{obs} = SS_{mean} + SS_{treat} + SS_{resid}$$

Remark.

- (1), (2), and (3) distribute Gaussian, because they are obtained from linear transformations of $\underline{X}$.
- (1), (2), (3) are independient

Note on the other hand that:

$$\|\underline{X} - \bar{X} \cdot \underline{1}\|^2 = \left\| \sum_{i=1}^{g} (\bar{X}_i - \bar{X}) \underline{u}_i \right\|^2 + \left\| \underline{X} - \sum_{i=1}^{g} \bar{X}_i \underline{u}_i \right\|^2$$

Then:

$$\sum_{i=1}^{g} \sum_{j=1}^{n_i} (X_{ij} - \bar{X})^2 = \sum_{i=1}^{g} (\bar{X}_i - \bar{X})^2 n_i + \sum_{i=1}^{g} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

In words:

$$Centered\ SS_{obs} = SS_{treat} + SS_{res}$$

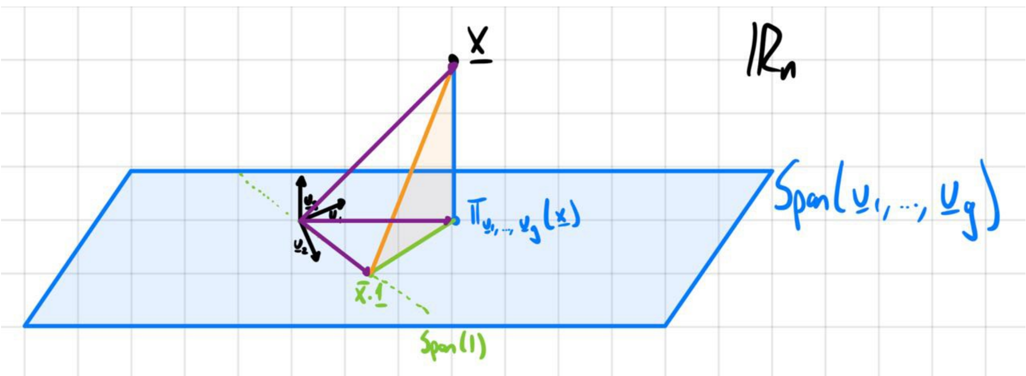

Figure 5: Example: $\underline{X}$ decomposition over projections on $\underline{u}_1, \ldots, \underline{u}_g$

Now, going back to the hypothesis test proposed in section 11.4:

$$H_0 : \mu_1 = \cdots = \mu_g \qquad vs \qquad H_1 : H_0^C$$

We reject H_0 if:

$$\frac{\sum_{i=1}^{g}(\bar{X}_i - \bar{X})^2 n_i}{\sum_{i=1}^{g}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2} \text{ is large}$$

Note that this indicates whether the difference in the means, and no the in-group variability gives big part of the variability.

Example.

> Given $P \in \mathbb{R}^{p\times p}$ orthogonal projection.
>
> $$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$
>
> $$\implies PX \perp\!\!\!\perp I \cdot P$$
>
> Hint: $\begin{pmatrix} P \\ I-P \end{pmatrix} \underline{X} \sim ?$

Remark.

> -
> $$\sum_{i=1}^{g}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2 = \sum_{i=1}^{g}(n_i - 1)S_i^2$$
>
>   being
>
> $$S_i^2 = \frac{1}{n_i - 1}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2 \text{ estimator of } \sigma^2$$
>
>   and:
>
> $$(n_i - 1)S_i^2 \sim \sigma^2\chi^2(n_i - 1)$$
>
> $$\implies \sum_{i=1}^{g}\sum_{j=1} n_i(X_{ij} - \bar{X}_i)^2 \sim \sigma\chi^2(n - g)$$
>
> - If H_0 is true:
>
> $$\sum_{i=1}^{g}\sum_{j=1}^{n_i}(X_{ij} - \bar{X})^2 = \sum_{i=1}^{g}(n-1)S^2 \sim \sigma^2\chi^2(n-1)$$
>
>   where:
>
> $$S^2 = \frac{1}{n-1}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}) \text{ estimator of } \sigma^2$$

When H_0 is true

$$F = \frac{\frac{\sum_{i=1}^{g}(\bar{X}_i - \bar{X})^2 n_i}{g-1}}{\frac{\sum_{i=1}^{g}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2}{n-g}} \sim F(g-1, n-g)$$

Reject H_0 at level $\alpha \in (0,1)$ if:

$$F > F_{1-\alpha}(g-1, n-g)$$

11.6 MANOVA: Case $p \geq 1$, $g \geq 2$

The multivariance is considered in MANOVA.

$$X_{ij} = \overbrace{\mu + \tau_i}^{\mu_i} + \varepsilon_{ij}$$

$$\varepsilon_{ij} \ iid \sim N_p(0, \Sigma)$$

st: $\sum_{i=1}^{g} n_i \underline{\tau}_i = \underline{0}$
For each component $k = 1, \ldots, q$ we can simply apply ANOVA in each component:

$$X_{ijk} = \overbrace{\mu_k + \tau_{ik}}^{\mu_i} + \varepsilon_{ijk}$$

$$\varepsilon_{ijk} \ iid \sim N_1(0, \sigma_{kk}^2)$$

st: $\sum_{i=1}^{g} n_i \underline{\tau}_i = \underline{0}$.
But in the real world, this condition is not satisfied.

Theorem 11.1: Covariance Decomposition:

$$\sum_{i=1}^{g} \sum_{j=1}^{n_i} (\underline{X}_{ij} - \bar{\underline{X}})(\underline{X}_{ij} - \bar{\underline{X}})^T = \underbrace{\sum_{i=1}^{g} n_i (\underline{X}_i - \bar{\underline{X}})(\underline{X}_i - \bar{\underline{X}})^T}_{B} + \underbrace{\sum_{i=1}^{g} \sum_{j=1}^{n_i} (\underline{X}_{ij} - \bar{\underline{X}}_i)(\underline{X}_{ij} - \bar{\underline{X}}_i)^T}_{W}$$

where:

$$\bar{\underline{X}} = \frac{1}{n} \sum_i \sum_j X_{ij} \qquad \underline{X}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} X_j$$

Proof. It can be proven that:

$$\underline{X}_{ij} - \bar{X} = (\underline{X}_{ij} - \bar{\underline{X}}_i) + (\underline{X}_{ij} - \bar{\underline{X}}_i)^T$$

□

B is called covariability between-group, and W is called covariability within-group.
We want to test:

$$H_0 : \underline{\mu}_1 = \cdots = \underline{\mu}_g \qquad vs \qquad H_1 : H_0^C$$

That is the same as:

$$H_0 : \underline{\tau}_1 = \cdots = \underline{\tau}_g \qquad vs \qquad H_1 : H_0^C$$

The test statistics could be:

- The Wilk's lambda:
$$\Lambda_W = \frac{\det W}{\det(W+B)}$$
And we will reject H_0 if Λ_W is small.

- The Pillai's lambda:
$$\Lambda_p = \text{trace}(B \cdot (B+W)^{-1})$$
And we will reject H_0 if Λ_p is small.

- Lawley-Hotelling'as lambda:
$$\Lambda_{CH} = \text{trace}\, BW^{-1}$$
And we will reject H_0 if Λ_{CH} is large.

The distribution of Λ_W when H_0 is true is known when:

- $p \geq 1$ and $g = 2, 3$
- $p = 2$ and $g \geq 1$

$\Lambda_W, \Lambda_p, \Lambda_{CH}$ are all based on eigenvalues $\lambda_1, \ldots, \lambda_s$ of BW^{-1} $(s = \min(g-1, p))$
The Barlett's approximation, when H_0 is true is:
$$-(n-1-\frac{p+g}{2}) \log \Lambda_W \sim \chi^2(p(g-1))$$
And we reject H_0 at level $\alpha \in (0,1)$ if:
$$-(n-1-\frac{p+g}{2}) \log \lambda_W > \chi^2_{1-\alpha}(p(g-1))$$
If H_0 is rejected: We would like to get the Bonferroni CI for
$$\tau_{i\ell} - \tau_{k\ell} \quad k, i = 1, \ldots, g; \ell = 1, \ldots, p$$
Now, we have to find an estimator for $\tau_{i\ell} - \tau_{k\ell}$. Note that:
$$(\bar{X}_{i\ell} - \bar{X}_\ell) - (\bar{X}_{k\ell} - \bar{X}_\ell) = \bar{X}_{i\ell} - \bar{X}_{k\ell} \sim N(\tau_{i\ell} - \tau_{k\ell}, \frac{\sigma_{\ell\ell}}{n_i} + \frac{\sigma_{\ell\ell}}{n_k})$$
where $\sigma_{\ell\ell}$ is the element (ℓ, ℓ) of Σ.
Now, on the other hand:
$$W = \sum_{i=1}^{g} \sum_{j=1}^{n_i} (\underline{X}_{ij} - \underline{\bar{X}}_i)(\underline{X}_{ij} - \underline{\bar{X}}_i)^T = \sum_{i=1}^{g} (n_i - 1) S_i$$
where S_i is the estimator of Σ in group i:
$$S_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\underline{X}_{ij} - \underline{\bar{X}}_i)(\underline{X}_{ij} - \underline{\bar{X}}_i)^T$$

Now, we called pooled estimator of Σ to (We divide W by its degrees of freedom):

$$\frac{1}{n-g}W$$

Therefore $\frac{1}{n-g}W_{\ell\ell}$ is an estimator for $\sigma_{\ell\ell}$. Now we can have the Bonferroni CI:

$$Bonf\ CI_{1-\alpha}(\tau_{i\ell}-\tau_{k\ell}) = \left[\bar{X}_{i\ell}-\bar{X}_{k\ell} \pm t_{1-\frac{\alpha}{2}\frac{2}{pg(g-1)}}(n-g)\sqrt{\frac{1}{n-g}W_{\ell\ell}}\right]$$

Note that $p\frac{g(g-1)}{2}$ are the possible combinations of k and i.

11.7 Two-way (M)ANOVA

In two-way MANOVA we have 2 factors. Factor 1 with g levels and factor 2 with b levels. Each combination of two factors i,j has n_{ij} factor of units. We will simplify the problem and we will study the balanced case (every combination of factors has the same number of units).

Example.

> Factor 1 can be liters of water and factor 2 fertilizer. For a fixed level of factor 1 and factor 2, we have n experiments.

Then, we have

$$X_{ijk} = \mu_{ij} + \epsilon_{ijk} \qquad\qquad \varepsilon_{ijk}\ iid \sim N(0,\sigma^2)$$

where:

$$i = 1,\dots,g \qquad j = 1,\dots,b \qquad k = 1,\dots n$$

We will reparametrize our model to the complete model, using:

$$\mu_{ij} = \mu + \tau_i + \beta_j + \gamma_{ij}$$

where:

- τ_i is the effect of factor 1 at level i.
- β_j is the effect of factor 2 at level j.
- γ_{ij} is the effect of interactions between the factors i and j.

Note that this parametrization has 1+g+b degrees of freedom more than the original one, so we have to add some constraints:

$$
\begin{aligned}
0 &= \sum_{i=1}^{g} \tau_i = \sum_{j=1}^{b} \beta_j \\
&= \sum_{i=1}^{g} \gamma_{ij} \quad \forall j = 1, \ldots, b \\
&= \sum_{j=1}^{b} \gamma_{ij} \quad \forall i = 1, \ldots, g
\end{aligned}
$$

Note that the first and second restrictions subtract 1 dof each one, the third one subtracts b restrictions, and the fourth one subtracts g-1 (since the third restriction completely defines one element of this restriction).
We will take the logic estimator for μ, τ_i, and β_j, respectively:

$$
\begin{aligned}
\bar{X} &= \frac{1}{gbn} \sum_{i=1}^{g} \sum_{j=1}^{b} \sum_{k=1}^{n} X_{ijk} \\
&= \bar{X}_{i\cdot} - \bar{X} \\
&= \bar{X}_{\cdot j} - \bar{X} \\
&= \bar{X}_{ij} - (\bar{X}_{i\cdot} - \bar{X}) - (\bar{X}_{\cdot j} - \bar{X}) - \bar{X} = \bar{X}_{ij} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} + \bar{X}
\end{aligned}
$$

Where:

$$
\begin{aligned}
\bar{X}_{i\cdot} &= \frac{1}{bn} \sum_{j=1}^{b} \sum_{k=1}^{n} X_{ijk} \\
\bar{X}_{\cdot j} &= \frac{1}{gn} \sum_{i=1}^{g} \sum_{k=1}^{n} X_{ijk} \\
\bar{X}_{ij} &= \frac{1}{n} \sum_{k=1}^{n} X_{ijk}
\end{aligned}
$$

11.8 Decomposition of Variance

The total variability, i.e. the centered sum of squares (SS_{center})

$$\sum_{i=1}^{g} \sum_{j=1}^{b} \sum_{k=1}^{n} (X_{ijk} - \bar{X})^2$$

Can be decomposed into:

$$SS_{center} = SS_{treat1} + SS_{treat2} + SS_{interact} + SS_{residual}$$

Where:

$$SS_{treat1} = \sum_{i=1}^{g} (\bar{X}_{i\cdot} - \bar{X})^2 nb \qquad \text{g-1 dof}$$

$$SS_{treat2} = \sum_{j=1}^{b} (\bar{X}_{\cdot j} - \bar{X})^2 ng \qquad \text{b-1 dof}$$

$$SS_{interact} = \sum_{i=1}^{g} \sum_{j=1}^{b} \sum_{k=1}^{n} (X_{ijk} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} - \bar{X})^2 n \qquad \text{(b-1)(b-1) dof}$$

$$SS_{residual} = \sum_{i=1}^{g} \sum_{j=1}^{b} \sum_{k=1}^{n} (X_{ijk} - \bar{X}_{ij})^2 \qquad \text{gb(b-1) dof}$$

$\frac{SS_{residual}}{gb(n-1)}$ is an estimator of σ^2. We can only estimate σ is $n > 1$.
We can get rid of γ_{ij}, reparametrizing as the additive model:

$$\mu_{ij} = \mu + \tau_i + \beta_j$$

In this model, the μ_{ij} will behave the same across one factor when the other is fixed, minus a constant.
In the complete model, the μ_{ij} will behave differently, this is the γ_{ij} effect.

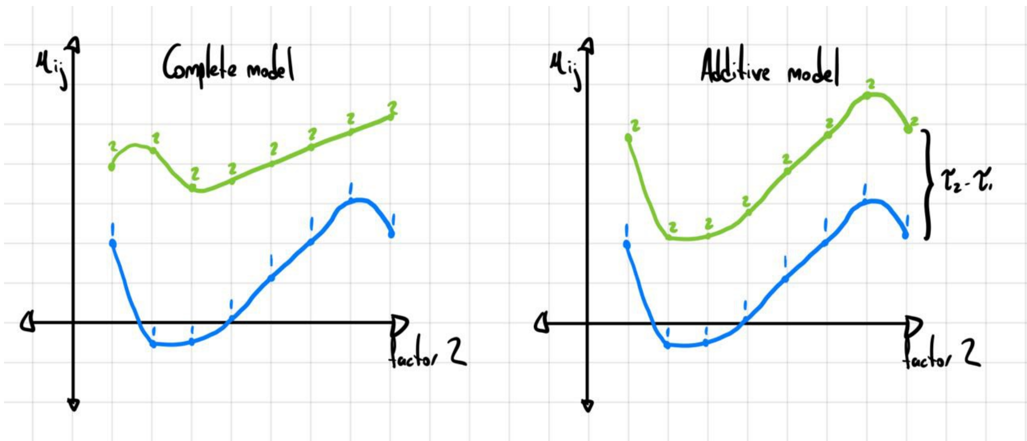

Figure 6: Example: Complete vs additive model, the relation between fixed levels of factor 1

We can build tests with this, some examples are:

- $$H_0 : \gamma_{ij} = 0 \quad \forall i, j \qquad vs \qquad H_1 : H_0^C$$

  The statistic used for this one will be:

  $$\frac{\frac{SS_{interact}}{(g-1)(b-1)}}{\frac{SS_{residual}}{gb(n-1)}} \sim F((g-1)(b-1),\ gb(n-1))$$

- $$H_0 : \tau_i = 0 \quad \forall i \qquad vs \qquad H_1 : H_0^C$$

  The statistic used for this one will be:

  $$\frac{\frac{SS_{treat1}}{(g-1)}}{\frac{SS_{residual}}{gb(n-1)}} \sim F((g-1),\ gb(n-1))$$

- There are other tests in the book

12 Supervised Classification

12.1 The problem

Each unit is represented by:

$$\mathbb{X}, L$$

where:

$$\underline{X} \in \mathbb{R}^p \qquad L \in \{1, 2, \dots, g\}$$

In the Supervised Classification, it will be a training dataset:

$$\begin{bmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{21} & X_{22} & \dots & X_{2p} \\ \dots & \dots & \dots & \dots \\ X_{n1} & X_{n2} & \dots & X_{np} \end{bmatrix} \begin{bmatrix} \ell_1 \\ \ell_2 \\ \dots \\ \ell_n \end{bmatrix}$$

The goal is to learn from data a function (called classifier):

$$\delta : \mathbb{R}^p \to \{1, 2, \dots, g\}$$

We say we look for $\delta(\underline{x})-$group membership. It is also called discriminant analysis, since we have to discriminate which group an element belongs to.

For building δ :

1. $f_i : \mathbb{R}^p \to [0, \infty$ density of dist of $\underline{X}$ when L=i.

   $$f_i(\underline{x})d\underline{x} = P[\underline{x} \in d\underline{x} | L = i] \qquad i = 1, \dots, g$$

   Note we have different density distributions for each i. We have to check whether they are different. e.g. with MANOVA.

2. Prior probability:

   $$P[L = i] = p_i \qquad i = 1, \dots, g$$

   where: $p_i \geq 0 \quad \wedge \quad \sum_{i=1}^{g} p_i = 1$.

3. Cost of misclassification:

$C(i|j)$ is the cost of attributing a unit to group i when in fact it belongs to group j. It must satisfy:

$$C(i|j) \geq 0 \qquad \wedge \qquad C(i|i) = 0 \qquad i,j = 1,\dots,g$$

Note that there's no symmetry in this function.

Both prior probability and misclassification costs come from the prior knowledge/environment. These are assumptions we need to set.

12.2 Optimal Classifier

Observe that:

$$\delta : \mathbb{R}^p \to \{1,\dots,g\} \iff \{R_1,\dots,R_g\} \text{ partition of } \mathbb{R}^p$$

i.e.:

$$R_i \subseteq \mathbb{R}^p, \qquad i = 1,\dots,g$$

where:

$$R_i \cap R_j = \emptyset \quad \forall i \neq j \qquad \wedge \qquad \bigcup_{i=1}^{g} R_i = \mathbb{R}^p$$

and note that:

$$R_i = \{\underline{x} \in \mathbb{R}^p \quad st \quad \delta(\underline{x}) = i\} \quad = \quad \delta^{-1}(\{i\}) \qquad i = 1,\dots,g$$

12.3 Optimality Criterion

The optimality criterion consists of minimizing the expected cost of misclassification:

$$ECM(\delta)$$

The subsets must be borel subsets since we have to integrate in the probability space.

Example.

Let be g=2. $\delta : \mathbb{R}^p \to \{1,2\} \iff \{R_1, R_2\}$

$$\begin{aligned} ECM(\delta) &= \underbrace{\int_{R_2} c(2|1) f_1(\underline{x}) p_1 d\underline{x}}_{\text{Expected cost of miss 1}} + \underbrace{\int_{R_1} c(1|2) f_2(\underline{x}) p_2 d\underline{x}}_{\text{Expected cost of miss 2}} \\ &= \int_{\mathbb{R}^p} c(2|1) f_1(\underline{x}) p_1 d\underline{x} - \int_{R_1} c(2|1) f_1(\underline{x}) p_1 d\underline{x} + \int_{R_1} c(1|2) f_2(\underline{x}) p_2 d\underline{x} \\ &= c(2|1) \cdot p_1 + \int_{R_1} \left[\underbrace{c(1|2) f_2(x) p_2 - c(2|1) f_1(x) p_1}_{g(\underline{x})} \right] \end{aligned}$$

We must choose R_1 such that ECM is minimum. Note that the R_1 that minimizes is

$$\begin{aligned} R_1 &= \{\underline{x} \quad st \quad g(\underline{x}) \leq 0\} \\ &= \{\underline{x} \quad st \quad c(1|2)f_2(x)p_2 \leq c(2|1)f_1(x)p_1\} \\ R_2 &= \{\underline{x} \quad st \quad c(1|2)f_2(x)p_2 > c(2|1)f_1(x)p_1\} \qquad \text{(Similarly)} \end{aligned}$$

Now, in the general case:

$$\begin{aligned} ECM(\delta) &= \sum_{k\neq 1}\int_k c(k|1)f_1(\underline{x})p_1 d\underline{x} + \cdots + \sum_{k\neq g}\int_k c(k|g)f_g(\underline{x})p_g d\underline{x} \\ &= \sum_{i=1}^{g}\sum_{k\neq i}\int_k c(k|i)f_i(\underline{x})p_i d\underline{x} \\ &= \int_{R_1}\sum_{k\neq 1} c(1|k)f_k(\underline{x})p_k d\underline{x} + \int_{R_2}\sum_{k\neq 2} c(2|k)f_k(\underline{x})p_k d\underline{x} + \cdots + \int_{R_g}\sum_{k\neq g} c(g|k)f_k(\underline{x})p_k d\underline{x} \\ &= \sum_{i=1}^{g}\int_{R_i}\sum_{k\neq i} c(i|k)f_k(\underline{x})p_k d\underline{x} \end{aligned}$$

Note that we have to build our optimal classifier. It can be proven that the optimal sets are:

$$\begin{aligned} &R_1 = \{\underline{x} \in \mathbb{R}^p \quad st \qquad \sum_{k\neq 1} c(1|k)f_k(\underline{x})p_k \leq \sum_{k\neq j} c(j|k)f_k(\underline{x})p_k \quad \forall j \neq 1\} \\ &\ldots \\ &R_g = \{\underline{x} \in \mathbb{R}^p \quad st \qquad \sum_{k\neq g} c(g|k)f_k(\underline{x})p_k \leq \sum_{k\neq j} c(j|k)f_k(\underline{x})p_k \quad \forall j \neq g\} \end{aligned}$$

Remark.

- Costs of misclassification can be defined up to a multiplicative constant. Plus, they are not in any constant, the important thing is that they reflect a relationship.
- $\delta(\underline{x}) = t \qquad t \in \{1, \ldots, g\}$

$$\sum_{k\neq t} c(t|k)f_k(\underline{x})p_k \leq \sum_{k\neq j} c(j|k)f_k(\underline{x})p_k$$
$$\frac{\sum_{k\neq t} c(t|k)f_k(\underline{x})p_k}{\sum_{i=1}^{g} f_i(\underline{x})p_i} \leq \frac{\sum_{k\neq j} c(j|k)f_k(\underline{x})p_k}{\sum_{i=1}^{g} f_i(\underline{x})p_i}$$

Note now that:

$$\begin{aligned} \frac{\sum_{k\neq t} f_k(\underline{x})p_k}{\sum_{i=1}^{g} f_i(\underline{x})p_i} &= \frac{P[\underline{X} \in d\underline{x}|L=k]P[L=k]}{\sum_{i=1}^{g} P[\underline{X} \in d\underline{x}|L=i]P[L=i]} \\ &= \frac{P[\underline{X} \in d\underline{x}|L=k]P[L=k]}{P[\underline{X} \in d\underline{x}]} = P[L=k|\underline{X} \in d\underline{x}] \quad \text{(Bayes theorem)} \end{aligned}$$

Therefore, the optimal classification is taking (rewritten):

$$R_i = \left\{ \underline{x} \in \mathbb{R}^p \quad st \quad \sum_{k \neq i} c(i|k) P[L = k | \underline{X} = \underline{x}] \leq \sum_{k \neq j} c(j|k) P[L = k | \underline{X} = \underline{x}] \quad i \neq j \right\}$$

12.4 Particular Cases

- Assume all the costs are the same:

$$c(i|j) = \begin{cases} c > 0 & i \neq j \\ 0 & i = j \end{cases} \qquad \forall i, j = 1, \ldots, g$$

Then, the optimum class is:

$$\begin{aligned} \delta(\underline{x}) = t &\iff \sum_{k \neq t} c \cdot P[L = k | \underline{X} = \underline{x}] \leq \sum_{k \neq j} c \cdot P[L = k | \underline{X} = \underline{x}] \quad \forall j \neq t \\ &\iff 1 - P[L = t | \underline{X} = \underline{x}] \leq 1 - P[L = j | \underline{X} = \underline{x}] \quad \forall j \neq t \\ &\iff P[L = t | \underline{X} = \underline{x}] \geq P[L = j | \underline{X} = \underline{x}] \quad \forall j \neq t \end{aligned}$$

Therefore we have the **Bayes Classifier**:

$$R_i = \{ \underline{x} \in \mathbb{R}^p \quad st \quad P[L = i | \underline{X} = \underline{x}] \geq P[L = j | \underline{X} = \underline{x}] \quad \forall j \neq i \}$$

NOTE THAT WE ARE NOT "NOT ASSUMING ANYTHING", WE ARE ASSUMING ALL THE PROBABILITIES ARE THE SAME.

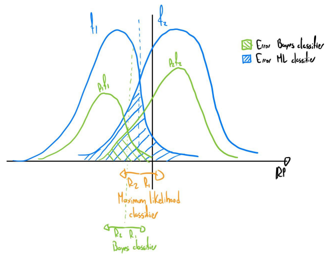

Figure 7: Bayes Classifier example

- **Uniform ignorance:** Assume now that:

$$c(i|j) = \begin{cases} c > 0 & i \neq j \\ 0 & i = j \end{cases} \qquad \forall i, j = 1, \ldots, g$$

And that all the prior probabilities are the same: $p_1 = \cdots = p_g = \frac{1}{g}$.

Then, by the past case:

$$\begin{aligned} R_i &= \{\underline{x} \in \mathbb{R}^p \quad st \quad P[L = t|\underline{X} = \underline{x}] \geq P[L = j|\underline{X} = \underline{x}] \quad \forall j \neq i\} \\ &= \{\underline{x} \in \mathbb{R}^p \quad st \quad f_t(\underline{x})p_t \geq f_j(\underline{x})p_j \quad \forall j \neq i\} \\ &= \{\underline{x} \in \mathbb{R}^p \quad st \quad f_t(\underline{x}) \geq f_j(\underline{x}) \quad \forall j \neq i\} \qquad \text{(equal priors)} \end{aligned}$$

This is the maximum likelihood class

- Now, assume:

$$c(i|j) = \begin{cases} c_j > 0 & i \neq j \\ 0 & i = j \end{cases} \qquad \forall i, j = 1, \dots, g$$

Then, the optimum class is:

$$\begin{aligned} \delta(\underline{x}) = t &\iff \sum_{k \neq t} c(t|k) \cdot P[L = k|\underline{X} = \underline{x}] \leq \sum_{k \neq j} c(j|k) \cdot P[L = k|\underline{X} = \underline{x}] \quad \forall j \neq t \\ &\iff \sum_{k \neq t} c_k \cdot f_k(\underline{x})p_k \leq \sum_{k \neq j} c_k \cdot f_k(\underline{x})p_k \quad \forall j \neq t \\ &\iff \sum_{k \neq t} f_k(\underline{x})\Pi_k \leq \sum_{k \neq j} f_k(\underline{x})\Pi_k \quad \forall j \neq t \qquad \Pi_k = \frac{c_k p_k}{\sum_{i=1}^{g} c_i p_i}, \sum \Pi_k = 1 \end{aligned}$$

The optimal classifier is the classifier in the base case, but the 'priors' (making a simile) are the true priors times the costs normalized: $\Pi_1, \dots \Pi_g$.
Specifically, softwares like R do not give the cost as an input, but the priors. In this case, we calculate the optimum introducing the costs by the means of the priors, without introducing the real priors of the problem

- Now, assume:

$$c(i|j) = \begin{cases} c_i > 0 & i \neq j \\ 0 & i = j \end{cases} \qquad \forall i, j = 1, \dots, g$$

Exercise.

- Now, assume:

$$c(i|j) = \begin{cases} c_i h_j & i \neq j \\ 0 & i = j \end{cases} \qquad \forall i, j = 1, \dots, g \qquad c_i > 0, h_j > 0$$

Exercise.

12.5 Quadratic Discriminant Analysis (QDA)

Let be $\underline{X}|L = i \sim N_p(\underline{\mu}_i, \Sigma_i), \qquad i = 1, \dots, g$. Consider Bayes Classifier:

$$
\begin{aligned}
\delta(\underline{x}) = t \iff & P[L = t|\underline{X} = \underline{x}] \geq P[L = j|\underline{X} = \underline{x}] \quad \forall j \neq t \\
\iff & f_t(\underline{x})p_t \geq f_j(\underline{x})p_j \\
\iff & p_t \cdot \frac{1}{\sqrt{(2p)^p|\Sigma_t|}} \exp\left(-\frac{1}{2}(\underline{x} - \underline{\mu}_t)^T \Sigma_t^{-1}(\underline{x} - \underline{\mu}_t)\right) \geq p_j \cdot \frac{1}{\sqrt{(2p)^p|\Sigma_t|}} \exp\left(-\frac{1}{2}(\underline{x} - \underline{\mu}_j)^T \Sigma_j^{-1}(\underline{x} - \underline{\mu}_j)\right) \\
\iff & p_t \cdot \frac{1}{\sqrt{\Sigma_t|}} \exp\left(-\frac{1}{2}(\underline{x} - \underline{\mu}_t)^T \Sigma_t^{-1}(\underline{x} - \underline{\mu}_t)\right) \geq p_j \cdot \frac{1}{\sqrt{|\Sigma_t|}} \exp\left(-\frac{1}{2}(\underline{x} - \underline{\mu}_j)^T \Sigma_j^{-1}(\underline{x} - \underline{\mu}_j)\right) \\
\iff & \log p_t - \frac{1}{2}\log|\Sigma_t| - \underbrace{\frac{1}{2}(\underline{x} - \underline{\mu}_t)^T \Sigma_t^{-1}(\underline{x} - \underline{\mu}_t)}_{d^2_{\Sigma_t^{-1}}(\underline{x}, \underline{\mu}_t)} \geq \log p_j - \frac{1}{2}\log|\Sigma_j| - \frac{1}{2}(\underline{x} - \underline{\mu}_j)^T \Sigma_j^{-1}(\underline{x} - \underline{\mu}_j)
\end{aligned}
$$

Now define the quadratic discriminant scores:

$$d_i^Q(\underline{x}) = \log p_i - \frac{1}{2}\log|\Sigma_i| - \frac{1}{2}(\underline{x} - \underline{\mu}_i)^T \Sigma_i^{-1}(\underline{x} - \underline{\mu}_i) \qquad i = 1, \dots, g$$

Now the bayes classifier is:

$$\delta(\underline{x}) = t \iff R_i\{\underline{x} \in \mathbb{R}^p \quad st \quad st \quad d_i^Q(\underline{x}) \geq d_j^Q(\underline{x}), \quad j \neq i\} \qquad \text{(QDA)}$$

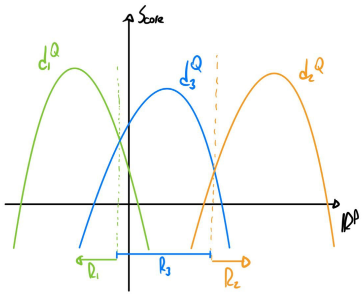

Figure 8: QDA curves: Example

Note that this gives quadratic split curves

12.6 Linear Discriminant Analysis (LDA)

Now, assume that we have the same variance for every group:

$$\underline{X}|L = i \sim N_p(\underline{\mu}_i, \Sigma), \quad i = 1, \dots, g$$

Then, the bayes classifier is:

$$\begin{aligned}
\delta(\underline{x}) = t \iff & \log p_t - \frac{1}{2}\log|\Sigma| - \frac{1}{2}(\underline{x}-\underline{\mu}_t)^T\Sigma^{-1}(\underline{x}-\underline{\mu}_t) \geq \log p_j - \frac{1}{2}\log|\Sigma| - \frac{1}{2}(\underline{x}-\underline{\mu}_j)^T\Sigma^{-1}(\underline{x}-\underline{\mu}_j) \\
\iff & \log p_t - \frac{1}{2}(\underline{x}-\underline{\mu}_t)^T\Sigma^{-1}(\underline{x}-\underline{\mu}_t) \geq \log p_j - \frac{1}{2}(\underline{x}-\underline{\mu}_j)^T\Sigma^{-1}(\underline{x}-\underline{\mu}_j) \\
\iff & \log p_t - \frac{1}{2}\underline{x}^T\Sigma^{-1}\underline{x} - \frac{1}{2}\underline{\mu}_i^T\Sigma^{-1}\underline{\mu}_i + \underline{\mu}_i^T\Sigma^{-1}\underline{x} \geq \log p_j - \frac{1}{2}\underline{x}^T\Sigma^{-1}\underline{x} - \frac{1}{2}\underline{\mu}_j^T\Sigma^{-1}\underline{\mu}_j + \underline{\mu}_j^T\Sigma^{-1}\underline{x} \\
\iff & \log p_t - \frac{1}{2}\underline{\mu}_i^T\Sigma^{-1}\underline{\mu}_i + \underline{\mu}_i^T\Sigma^{-1}\underline{x} \geq \log p_j - \frac{1}{2}\underline{\mu}_j^T\Sigma^{-1}\underline{\mu}_j + \underline{\mu}_j^T\Sigma^{-1}\underline{x}
\end{aligned}$$

Now, define the Linear Discriminant Score:

$$d_i(\underline{x}) = \log p_t - \frac{1}{2}\underline{\mu}_i^T\Sigma^{-1}\underline{\mu}_i + \underline{\mu}_i^T\Sigma^{-1}\underline{x}$$

Then, the Bayes classifier is:

$$\delta(\underline{x}) = t \iff R_i\{\underline{x} \in \mathbb{R}^p \quad st \quad st \quad d_i(\underline{x}) \geq d_j(\underline{x}), \quad j \neq i\} \qquad \text{(LDA)}$$

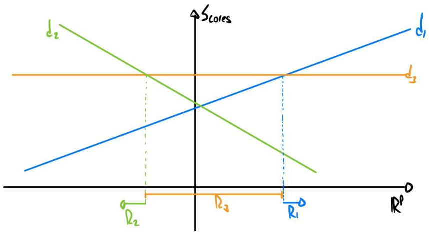

Figure 9: LDA curves: Example

Note that this gives linear split lines

Remark.

> The LDA classifier is robust to violations of normality assumptions. Indeed, we can recover the LDA classifier from deriving other classifiers that do not assume normality. A notable example is the Fisher Classifier (see 12.9)

Remark.

> Anyway, non-linearities can be inserted in the data (For example x^2), doing the curves linear in the data space, but possibly non-linear in the original one. This type of technique is called kernel technique.

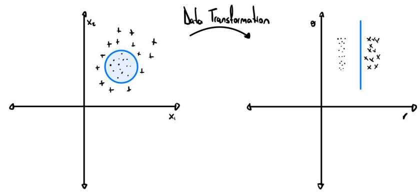

Figure 10: Kernel Example

12.7 Using Data

Training Set The training set is done by:

$$\mathbb{X} = \begin{bmatrix} \underline{x}_1^T \\ \dots \\ \underline{x}_n^T \end{bmatrix} \begin{bmatrix} \ell_1 \\ \dots \\ \ell_n \end{bmatrix}, \qquad \mathbb{X} \in \mathbb{R}^p, \quad \ell_i \in \{1, \dots, g\}$$

We want to use this data to "learn" f_i's, i.e. the distance of $\underline{X}$ to group $i = 1, \dots, g$.

Remark.

Be very careful with using the frequencies of the dataset as the prior probabilities.

In QDA:
Estimate $\underline{\mu}_1$ with:

$$\underline{\bar{X}_i} = \frac{1}{n_i} \sum_{j \in H_i} \underline{X}_i, \qquad H_i = \{j : e_j = i\}, \quad n_i = \#H_i$$

and Σ_i with:

$$S_i = \frac{1}{n_i - 1} \sum_{j \in H_i} (\underline{X}_j - \underline{\bar{X}}_i)(\underline{X}_j - \underline{\bar{X}}_i)^T$$

Then, simply plug th estimators in d_i^Q result.
In LDA:
Estimate Σ with:

$$\frac{1}{n-g} \sum_{i=1}^{g} (n_i - 1) S_i =: S_{pooled}$$

12.8 Naive Bayes Parametrization

Now, if p is large with respect to n_i's, then we can use the Naive Bayes parametrization of Σ_i (independence of the components) with the Gaussian Setting:

$$\Sigma_k = \begin{bmatrix} \sigma_{11}^{(k)} & \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots & \sigma_{pp}^{(k)} \end{bmatrix}, \qquad k = 1, \dots, g$$

$$d_k^Q(\underline{x}) = \log p_i - \frac{1}{2}\log\sum_{j=1}^{p}|\sigma_{jj}| - \frac{1}{2}\sum_{j=1}^{p}\frac{(x_j-\mu_{kj})^2}{\sigma_{jj}^{(k)}} \qquad i=1,\dots,g, \qquad \underline{X} = \begin{bmatrix} x_1 & x_2 & \dots & x_p \end{bmatrix}^T$$

12.9 Fisher Classifier

The Fisher Classifier looks for the direction where the groups can be discriminated the most. This is done from a general distribution, with groups that share the same covariance matrix:

$$\underline{X}|L=i \sim \underline{\mu}_i, \Sigma, \qquad i=1,\dots,g$$

Let be $\underline{a} \in \mathbb{R}^p$, then:

$$E[\underline{a}^T\underline{x}|L=i] = \underline{a}^T\underline{\mu}_i$$
$$Var[\underline{a}^T x|L=i] = \underline{a}^T\Sigma\underline{a}$$

Let be

$$B := \frac{1}{g-1}\sum_{i=1}^{g}(\underline{\mu}_i - \underline{\bar{\mu}})(\underline{\mu}_i - \underline{\bar{\mu}})^T$$
$$\underline{\mu} := \frac{1}{g}\sum_{i=1}^{g}\underline{\mu}_i$$

Now, the objective is to find the $\underline{a}$ that maximizes the variability between groups over the variability within groups:

$$\arg\max_{\underline{a}\in\mathbb{R}^p}\frac{\underline{a}^T B\underline{a}}{\underline{a}^T\Sigma\underline{a}} = \arg\max_{\underline{a}\in\mathbb{R}^p}\frac{1}{g-1}\frac{\sum_{i=1}^{g}(\underline{a}^T\underline{\mu}_i - \underline{a}^T\underline{\bar{\mu}})^2}{\underline{a}^T\Sigma\underline{a}}$$

Now, take $\Sigma^{1/2}\underline{a} = \underline{u}$, then:

$$\frac{\underline{a}^T B\underline{a}}{\underline{a}^T\Sigma\underline{a}} = \frac{\underline{u}^T\Sigma^{1/2}B\Sigma^{-1/2}\underline{u}}{\underline{u}^T\underline{u}}$$

And note that we can solve this as a spectral decomposition:

$$\arg\max_{\underline{u}\in\mathbb{R}^p}\frac{\underline{u}^T\Sigma^{1/2}B\Sigma^{-1/2}\underline{u}}{\underline{u}^T\underline{u}} = \underline{e}_1 \qquad \text{where } \Sigma^{-1/2}B\Sigma^{-1/2} = \sum_{i=1}^{S}\lambda_i\underline{e}_i\underline{e}_i^T \quad s=\min(g-1,p)$$

Then, as the eigenvectors are the vectors associated with the desired direction:

$$\arg\max_{\underline{a}\in\mathbb{R}^p}\frac{\underline{a}^T B\underline{a}}{\underline{a}^T\Sigma\underline{a}} = \Sigma^{-1/2}\underline{e}_1$$

Specifically, we call:

- $\underline{a}_1$: First discriminant direction
- $\underline{a}_1^T\underline{x}$: First discriminant score

And identically:

- $\underline{a}_1 = \Sigma^{-1/2}\underline{e}_1$: First discriminant direction
- ...
- $\underline{a}_s = \Sigma^{-1/2}\underline{e}_s$: s-th discriminant direction

Note that the discriminant directions are independent:

$$Cov(\underline{a}_i^T\underline{x}, \underline{a}_j^T\underline{x}) = \underline{a}_i^T\Sigma\underline{a}_j = \underline{e}_i^T\Sigma^{-1/2}\Sigma\Sigma^{-1/2}\underline{e}_j = \underline{e}_i^T\underline{e}_j^T = \mathbb{1}_{ij}, \qquad \mathbb{1}_{ij} = \text{Indicatrix}$$

Now, we can build classes based on discriminant scores. We want to estimate $\underline{\mu}_1, \ldots, \underline{\mu}_g$ by $\underline{\bar{X}}_1, \ldots \underline{\bar{X}}_g$.

Now, represent each $\underline{\bar{X}}_i$ with k the $k \leq s$ principal discriminant directions:

$$\underline{\bar{X}}_i = \begin{bmatrix} \underline{a}_1^T\underline{\bar{X}}_1 \\ \ldots \\ \underline{a}_k^T\underline{\bar{X}}_1 \end{bmatrix}$$

Then, the classes are defined as follows:

$$\delta(\underline{x}) = t \iff \sum_{j=1}^{k}(\underline{a}_j^T\underline{x} - \underline{a}_j^T\underline{\bar{X}})^2 \geq \sum_{j=1}^{k}(\underline{a}_j^T\underline{x} - \underline{a}_j^T\underline{\bar{X}}_h) \qquad \forall h \neq k$$

In this particular case, this gives the Voronoi Tesselation, assigning a point to its closest vertex in the dataset.

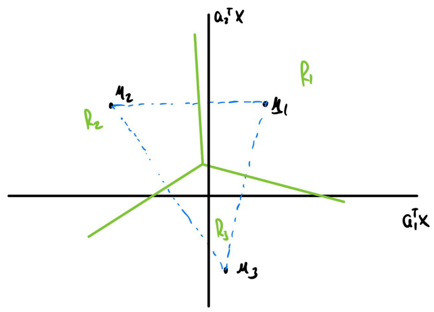

Figure 11: Voronoi Tesselation example

This is exactly the same as doing LDA if $k = s$ and $p_1 = \cdots = p_g = \frac{1}{g}$.

12.10 KNN classifier

Defining:

$$N_i = \left\{ \sum_{i=1}^{n} \mathbb{1}_{\ell_i}(y_i) \qquad x_i \in N_{\underline{x}} \right\}$$

KNN classifier:

$$\delta(\underline{x}) = t \iff N_t \leq N_j \quad \forall j$$

The bias increases as k increases, since the model oversimplifies the distribution by averaging over many neighbors, potentially ignoring local structures. The contrary occurs for the variance.

12.11 Evaluating a Classifier

Let be $\delta : \mathbb{R}^p \to \{1, \dots, g\}$ a classifier.

Definition 12.1: $AER(\delta)$

Define $AER(\delta)$ the actual error rate:

$$AER(\delta) = \sum_{i}^{g} \sum_{k \neq i} \int_{R_k} p_i f_i(\underline{x}) d\underline{x}$$

Remark.

> Sometimes, we refer to $AER(\delta)$ as $Accuracy(\delta)$.

Non, find a non-parametric estimate of AER:
Apply δ to the training set $\mathbb{X}$

$$\delta(\underline{x}_i) = \hat{\ell}_i, \qquad i = 1, \dots, n$$

And compare them with the true ℓ_i. It can be done in a table that compares true and assigned labels. Here, each element in the table n_{ij} indicates the elements that, belonging to the group i, were assigned to the group j.
Then, we can define a calculable error rate:

Definition 12.2: $A\hat{E}R(\delta)$

Define $A\hat{E}R(\delta)$ an estimator to $AER(\delta)$:

$$A\hat{E}R(\delta) = \frac{\#mistakes}{n} = 1 - \frac{\sum_{i=1}^{g} n_{ii}}{n} = APER(\delta) = e\bar{r}r$$

This last expression is called apparent error rate

Example.

> take g=2. so, $\delta : \mathbb{R}^p \to \{0, 1\}$.
> Then, the confusion matrix is:

	1	0	
1	n_{11}	n_{10}	$n_{\cdot 1}$
0	n_{01}	n_{00}	$n_{\cdot 1}$
	$n_{1\cdot}$	$n_{1\cdot}$	n

Table 1: Example of confussion matrix with g=2

Now:

$$\begin{aligned} APER(\delta) &= \frac{n_{10}+n_{01}}{n} = \frac{n_{1\cdot}}{n}\cdot\frac{n_{10}}{n_{1\cdot}} + \frac{n_{0\cdot}}{n}\cdot\frac{n_{01}}{n_{0\cdot}} \\ &= \hat{p_1}\int_{R_0}\hat{f_1(\underline{x})}d\underline{x} + \hat{p_0}\int_{R_1}\hat{f_0(\underline{x})}d\underline{x} \end{aligned}$$

With accuracy

$$\frac{n_{11}+n_{00}}{n}$$

Note that the actual AER is:

$$APER(\delta) = p_1\int_{R_0} f_1(\underline{x})d\underline{x} + p_0\int_{R_1} f_0(\underline{x})d\underline{x}$$

Now, different metrics are introduced:

- Precision:

  $$precision = \frac{n_{11}}{n_{\cdot 1}}$$

  Also called Positive Predicted Value (PPV):

  $$PPV = 1 - FDR$$

- Recall:

  $$\frac{n_{11}}{n_{1\cdot}}$$

  It is also called Sensitivity or Power:

  $$1 - P(\text{type II error})$$

- Specificity:

  $$\frac{n_{00}}{n_{0\cdot}}$$

  Note that this is

  $$1 - P(\text{type I error}) = 1 - \alpha$$

- F-Measure:

  $$2\cdot\frac{precision\cdot recall}{precision + recall} = \frac{2}{\frac{1}{precission}+\frac{1}{recall}}$$

  Note that this is the harmonic mean of precision and recall.

Ideal situation: Having a test set:

$$\tilde{\mathbb{X}} = \begin{bmatrix} \tilde{\underline{x}}_1^T \\ \dots \\ \tilde{\underline{x}}_n^T \end{bmatrix} \begin{bmatrix} \tilde{\ell}_1 \\ \dots \\ \tilde{\ell}_n \end{bmatrix}, \qquad \tilde{\mathbb{X}} \in \mathbb{R}^p, \quad \tilde{\ell}_i \in \{1, \dots, g\}$$

Now we could apply $\delta_{\mathbb{X}}$ to $\tilde{\mathbb{X}}$ and compare $\delta_{\mathbb{X}}(\tilde{\underline{x}}_i) = \hat{\ell}_i$ with $\tilde{\ell}_i$. We can then have a confusion matrix, and:

$$A\hat{E}R(\delta) = \frac{\sum_{i=1}^N \epsilon_i}{N} \qquad \epsilon_i = \begin{cases} 1 & \text{if } \tilde{\ell}_i = \hat{\ell}_i \\ 0 & \text{Otherwise} \end{cases}$$

Remark.

> The best way to build a test set is to choose it randomly (assuming the dataset distribution is right)

Remark.

> Data Augmentation, most of the time, is just to transform the data, not create new data.

12.12 Cross-validation

We see specifically one algorithm, called Leave-one-out (L1O).
For $i = 1, \dots, n$, take $(\underline{x}_i, \ell_i)$ out of the training set $\mathbb{X}$, to get:

$$\mathbb{X}_{-i}$$

Training the classificator on $\mathbb{X}_{-i}$, obtaining:

$$\delta_{-i} : \mathbb{R}^p \to \{1, \dots, g\}$$

Define $\delta_{-i}(\underline{x}_j) = \hat{\ell}_j$ and compare it with ℓ_j.
Then, take ε_j as before and calculate:

$$A\hat{E}R(\delta_{\mathbb{X}}) = \frac{\sum_j \varepsilon_j}{n}$$

The goal of this algorithm is to learn from the data, taking out the dependence of the data over a certain element in the dataset.

Anyway, taking just one out is not always sufficient. There also exists K-fold cross-validation. Set $k < n$. Now, permute the order of the data (rows), in the case the dataset has an order. Then, split $\mathbb{X}$ into k subsets. Now, for $J = \{C+1, \dots, C+k\}$, get $\mathbb{X}_{-J}$. Then, train a classifier on $\mathbb{X}_{-J}$, having:

$$\delta_{-J} : \mathbb{R}^p \to \{1, \dots, g\}$$

Now, apply δ_{-J} to subset J.
Finally, calculate:

$$\delta_{-J}(\underline{x}_i) = \hat{\ell}_i \qquad i \in J$$

And compare $\hat{\ell}_i$ with ℓ_i.

$$A\hat{E}R(\delta_{\mathbb{X}}) = \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i$$

Remark.

> We can repeat k-fold cross-validation B times by taking B different initial permutations of the data.

k is often 5 or 10, given a sufficiently large dataset.

Finally, after cross validation, we will have:

$$A\hat{E}R_1(\delta), \dots, A\hat{E}R_B(\delta)$$

We can take the mean and variance of these estimators, having a confidence interval:

$$CI_{1-\alpha}(E[A\hat{E}R(\delta)]) = \left[A\hat{E}R_1(\delta) \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{Var(A\hat{E}R_1(\delta))}{B}} \right]$$

Finally, the final classifier is the one trained with the whole data.

Remark.

> This works better in K-fold, since we are reducing the inter-variability, but increasing the intra-variability.

Remark.

> Be careful with doing Cross-Validation over a subset of selected components. All the processes should be taken into account in order to apply cross-validation. For example, we can first take the k-PCA of the data, and then apply cross-validation. This will "introduce" other data information in our data, making cross-validation effortless. See book Introduction to Statistical Learning.

12.13 Support Vector Machines (SVM)

(This is found in Elements of Statistical Learning Book).
Given data:

$$\mathbb{X} = \begin{bmatrix} \underline{x}_1^T \\ \dots \\ \underline{x}_n^T \end{bmatrix} \begin{bmatrix} \ell_1 \\ \dots \\ \ell_n \end{bmatrix}, \qquad \mathbb{X} \in \mathbb{R}^p, \quad \ell_i \in \{1, 2\}$$

The idea is to find a hyperplane h that separates the data of the two groups, i.e.:

$$\delta \iff \{R_1, R_2\}$$

Now, let be A and B subsets of $\mathbb{R}^p$. The first thing to do is to find out whether a separative hyperplane exists or not.

Definition 12.3: CH(A)

Define CH(A) as the convex hull identified by A

Theorem 12.4

Exists a hyperplane separating A and B if:

- $CH(A) \neq \emptyset \quad \wedge \quad CH(B) \neq \emptyset$
- $CH(A) \cap CH(B) = \emptyset$
- Either CH(A) or CH(B) is open

Proof. Geometric form of Hahn Banach Theorem □

Remark.

> 12.13 is equivalent to:
> CH(A) and CH(B) are closed and at list one of them is compact

Corollary 12.5

In case A and B are finite:
Needed only 12.13 and 12.13

Now, we have to characterize Hyperplanes in $\mathbb{R}^p$:

Definition 12.6: Hyperplanes

A hyperplane in $\mathbb{R}^p$ is an affine subsoace of $\mathbb{R}^p$ of dim p-1

Remark.

> A hyperplane is identified by $\underline{\beta} \in \mathbb{R}^p, \quad \underline{\beta} \perp \mathcal{L} \quad \wedge \quad \|\underline{\beta}\| = 1$

Define $\underline{x}_0 = \mathcal{L} \cap span(\underline{\beta})$.

$$\begin{aligned}
\underline{x} \in \mathcal{L} &\iff \Pi_{\underline{\beta}}(\underline{x}) = \underline{x}_0 \\
&\iff \frac{\underline{\beta}\underline{\beta}^T}{\underline{\beta}^T\underline{\beta}}\underline{x} = \underline{x}_0 \\
&\iff (\underline{\beta}^T\underline{x}) \cdot \underline{\beta} = \underline{x}_0 \\
&\iff \underline{\beta}^T\underline{x} = \underline{\beta}_0 \qquad \beta_0 := \|\underline{x}_0\| \\
&\iff \sum_{i=1}^{p} \beta_i x_i = \beta_0 \qquad \underline{x} = (x_1, \ldots, x_p)
\end{aligned}$$

Note now that $\underline{\beta}^T\underline{x} - \beta_0$ is a distance (with sign) between $\underline{x}$ and $\mathcal{L}$. Specifically, if:

1. $\underline{\beta}^T\underline{x} - \beta_0 = 0 \implies \underline{x} \in \mathcal{L}$
2. $\underline{\beta}^T\underline{x} - \beta_0 > 0 \implies \underline{x} \in \text{Plus(half space of } \mathbb{R}^p)$
3. $\underline{\beta}^T\underline{x} - \beta_0 < 0 \implies \underline{x} \in \text{Minus(half space of } \mathbb{R}^p)$

Now, assume that there is a separating hyperplane $\mathcal{L}$ for groups 1 and 2, identified by $\underline{\beta}$ and β_0. Let's notate 1 as plus and 2 as minus, so we define:

$$y_i \begin{cases} +1 & \text{if } \underline{x}_i \in Plus \\ -1 & \text{if } \underline{x}_i \in Minus \end{cases}$$

We want to find y_i such that:

$$y_i(\underline{\beta}^T\underline{x}_i - \beta_0) \geq 0$$

So every element will be well-classified. Further, we want to find a margin between the elements and the hyperplane:

$$M_1(\underline{\beta}, \underline{\beta}_0) = \min_i\{y_i(\underline{\beta}^T\underline{x}_i - \beta_0)\}$$

Now, the final objective is to find a hyperplane ($\underline{\beta}$ and β_0) such that $M_1(\underline{\beta}, \underline{\beta}_0)$ is maximum, i.e.:

$$\max_{\underline{\beta}, \underline{\beta}_0} M_1(\underline{\beta}, \underline{\beta}_0) = \max_{\underline{\beta}, \underline{\beta}_0} \min_i\{y_i(\underline{\beta}^T\underline{x}_i - \beta_0)\}$$

such to:

$$\begin{aligned}
&\|\underline{\beta}\| = 1 \\
&y_i(\underline{\beta}^T\underline{x}_i - \beta_0) \geq M \qquad \forall i = 1, \ldots, n
\end{aligned}$$

Remark.

> Specifically, this is called the hard version of SVM.

*The vectors that are close to the potential margin are called **support vectors**.*

12.13.1 Soft SVM

Anyway, there's no guarantee that there's always a separating hyperplane. We can relax the conditions of SVM, building a soft version of SVM.

In this version, it will be two cases where the elements won't satisfy the SVM conditions. The first is when an element is in the margin, it is at a distance $M(1-\epsilon)$, with $\epsilon \in (0,1)$. The second is when a point is at the other side of the hyperplane, where it does not belong, i.e., we say $(1-\epsilon)M$, with $\epsilon > 1$.

Now, the soft version of SVM is:

$$\max_{\underline{\beta},\underline{\beta}_0} M_1(\underline{\beta},\underline{\beta}_0) = \max_{\underline{\beta},\underline{\beta}_0} \min_i \{y_i(\underline{\beta}^T \underline{x}_i - \beta_0)\}$$

such to:

$$\begin{aligned}
&\|\underline{\beta}\| = 1 \\
&y_i(\underline{\beta}^T \underline{x}_i - \beta_0) \geq (1-\epsilon_i)M \qquad \forall i = 1, \dots, n \\
&\epsilon_i \geq 0 \qquad \wedge \qquad \sum_{i=1}^{n} \epsilon_i \leq C = \text{“Budget”}
\end{aligned}$$

It can be proven, that the solution is:

$$\begin{aligned}
\hat{\underline{\beta}} &= \sum_{i=1}^{n} \hat{\lambda}_i y_i \underline{x}_i \\
\hat{f}(\underline{x}) &= \hat{\underline{\beta}}^T \underline{x} + \hat{\beta}_0, \qquad \hat{\beta}_0 \in \mathbb{R} \\
C(\underline{x}) &= sign(\hat{f}(\underline{x}))
\end{aligned}$$

12.13.2 Non-linear Boundaries

If there's no obvious separating hyperplane, we can apply transformations to the data to use SVM.

Example.

> For example, instead of applying SVM with X_1, X_2, we can use:
>
> $$Z_1 = X_1, \quad Z_2 = X_2, \quad Z_3 = X_1^2, \quad Z_4 = X_2^2, \quad Z_5 = X_1 X_2$$
>
> Now, we can instead use the data $\mathbb{X}'$, where the columns are $Z_1, \dots, Z_5$ instead of X_1, X_2.

Kernel Track: For $i = 1, \dots, m$, let be $h_i : \mathbb{R}^p \to \mathbb{R}$ linear independent functions, and define:

$$\underline{h}(\underline{x}) = (h_1(\underline{x}), \dots, h_m(\underline{x}))^T \in \mathbb{R}^m$$

Transform the received dataset $\mathbb{X}$ into:

$$\tilde{\mathbb{X}} = \begin{bmatrix} \underline{h}_1^T(\underline{x}) \\ \dots \\ \underline{h}_n^T(\underline{x}) \end{bmatrix} \begin{bmatrix} \ell_1 \\ \dots \\ \ell_n \end{bmatrix}, \qquad \mathbb{X} \in \mathbb{R}^p, \quad \ell_i \in \{1, 2\}$$

And then separating the hyperplane using soft or hard SVM.
Now, take the optimal classifier:

$$C(\underline{x}) = sign(\hat{f}(\underline{h}(\underline{x}))$$

where:

$$\hat{\mathcal{F}}(\underline{x}) = \underline{f}(\underline{h}(\underline{x})) = \underline{\hat{\beta}}^T \underline{h}(\underline{x}) - \hat{\beta}_0 \qquad \wedge \qquad \underline{\hat{\beta}} = \sum_{i=1}^{n} \hat{\lambda}_i y_i \underline{h}(\underline{x}_i)$$

So:

$$\underline{f}(\underline{h}(\underline{x})) = \sum_{i=1}^{n} \hat{\lambda}_i y_i \underline{h}(\underline{x}_i)^T \underline{h}(\underline{x}) - \hat{\beta}_0$$

Remark.

> Only the inner product $\underline{h}^T(\underline{x}_i)\underline{h}(\underline{x})$ enter in the solution.
> Consider a function $\kappa : \mathbb{R}^p \times \mathbb{R}^p \to [0, \infty)$ (kernel):
>
> $$\kappa(\underline{x}, \underline{w}) = \underline{h}(\underline{x})^T \underline{h}(w), \qquad \underline{x}, \underline{w} \in \mathbb{R}^p$$
>
> $$\implies \hat{\mathcal{F}}(\underline{x}) = \sum_{i=1}^{n} \hat{\lambda}_i y_i \kappa(\underline{x}_i, \underline{x})$$

Examples of popular kernels:

- Polynomials of order d:
$$\kappa(\underline{x}, \underline{w}) = [1 + \underline{x}^T \underline{w}]^d$$
- Radial Basis with parameter γ:
$$\kappa(\underline{x}, \underline{w}) = \exp[-\gamma \|\underline{x} - \underline{w}\|^2]$$
- Hyperbolical tangent with parameters k_1, k_2:
$$\kappa(\underline{x}, \underline{w}) = \tanh(k_1^T \underline{x}^T \underline{w} + k_2)$$

Remark.

> Note that if we add more variables, by the curse of dimensionality, it will be easier to find a separating hyperplane, since the points will be more separated.

12.13.3 Multi-class SVM

It is possible to separate more than 2 classes in SVM, using more than 1 SVM.

- Doing 1v1 between every pair of groups.
- Doing 1 against all for every group.

13 Unsupervised Classification or Clustering

13.1 The problem

The main idea of unsupervised classification is to find the units belonging to the same group, being the elements of the group more similar between them than the units belonging to different groups.
In the unsupervised classification, it will be a training dataset:

$$\begin{bmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{21} & X_{22} & \dots & X_{2p} \\ \dots & \dots & \dots & \dots \\ X_{n1} & X_{n2} & \dots & X_{np} \end{bmatrix}$$

The goal is to learn from data a function that learns unseen groups:

$$\hat{\ell}_1, \dots, \hat{\ell}_n$$

This is called clustering analysis, and it is equivalent to learning a function:

$$\delta : \mathbb{R}^p \to \{1, 2, \dots, g\}$$

We don't know g here, so we will have to set it or estimate it.

13.2 Dissimilarity/Distance concept

In order to find this difference between units, it is used the distance or dissimilarity function $d : \mathbb{R}^p \times \mathbb{R}^p \to [0, \infty)$.
The functions used satisfy some of these properties:

1. $d(\underline{x}, \underline{y}) = 0 \iff \underline{x} = \underline{y} \qquad \forall \underline{x}, \underline{y} \in \mathbb{R}^p$

1'. $d(\underline{x}, \underline{x}) = 0 \qquad \forall \underline{x} \in \mathbb{R}^p$

2. Symmetry:
$$d(\underline{x}, \underline{y}) = d(\underline{y}, \underline{x}) \qquad \underline{x}, \underline{y} \in \mathbb{R}^p$$

3. Triangular Inequality:
$$d(\underline{x}, \underline{y}) \leq d(\underline{x}, \underline{z}) + d(\underline{z}, \underline{y})$$

4. Ultrametric Property:
$$d(\underline{x},\underline{y}) \leq \max\{d(\underline{x},\underline{z}), d(\underline{z},\underline{y})\}$$

There are three types of these functions:

- Metric: d is a metric if it satisfies (1), (2), (3).
- Pseudometric: d is a metric if it satisfies (1'), (2), (3).
- Ultra-metric: d is a metric if it satisfies (1), (2), (4).

Example.

> Let be $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$, define:
> $$d(f,g) = \int |f-g|^2$$
> Then, $d(f,g) = 0 \not\Rightarrow f = g$ But f satisfies 1'. It is actually a pseudometric

13.3 Steps for Clustering

1. Define the units and their representation.
2. Choosing a d.
3. Algorithm.

The most important step for clustering is choosing the distance, rather than the algorithm used

13.4 Distances examples

There are many distances that can be used, anyway, the most common ones are:

1. Euclidean:
$$d_2^2(\underline{x},\underline{y}) = \sum_{i=1}^{p}(x_i - y_i)^2 \qquad \underline{x},\underline{y} \in \mathbb{R}^p$$
   This distance does not take into account the scale differences, so it is often used over standardized variables.

2. Mahalanobious:
$$d(\underline{x},\underline{y}) = (\underline{x}-\underline{y})^T\Sigma^{-1}(\underline{x}-\underline{y})$$
   Where Σ is the covariance matrix.
   Since we don't know the covariance matrix or the groups, it is better to use one that assumes independence, as euclidean distance.

3. Distance p:

$$d_p(\underline{x}, \underline{y}) = \sqrt[p]{\sum_{i=1}^{p}(x_i - y_i)^p} \qquad p \geq 0$$

In p=1 it is called Manhattan distance, and p=2 it is the classical euclidean.

4. Canberra Distance: For $x \in (\mathbb{R}^+)^p$ and $y \in (\mathbb{R}^+)^p$:

$$d(\underline{x}, \underline{y}) = \sum_{i=1}^{p} \frac{|x_i - y_i|}{x_i + y_i}$$

13.5 Categorical Variables

The categorical variables can be represented as $\{0, 1\}^p$ points, since we can just use features that indicate if a characteristic is in a certain category or not.
Anyway, this is not the most efficient way if we have mutually exclusive variables. In this case, we can drop one column, and if all the rest are 0, it means the category is the remaining one.
For $\underline{x}, \underline{y} \in \{0, 1\}^p$, the euclidean distance is simply:

$$d_2^2(\underline{x}, \underline{y}) = \sum_{i=1}^{p}(x_i - y_i)^2 = \#discordances = n_{01} + n_{10}$$

	1	0	
1	n_{11}	n_{10}	$n_{\cdot 1}$
0	n_{01}	n_{00}	$n_{\cdot 1}$
	$n_{1\cdot}$	$n_{1\cdot}$	n

Table 2: Example of confussion matrix with g=2

In a confusion matrix, the number of discordances would be the sum of the values that are not on the diagonal.
A different distance could be:

$$d(\underline{x}, \underline{y}) = 1 - \frac{n_{11}}{n}$$

If, for example, we are only interested in classifying correctly the elements that are true.

Anyway, a lot of categorical distance examples can be found on (Johnson & Wichern, 2007).

13.6 Column-wise analysis

Now, we can see the data based on the columns instead of the units:

$$\mathbb{X} = \begin{bmatrix} \underline{x}_1^T \\ \dots \\ \underline{x}_n^T \end{bmatrix} = \begin{bmatrix} \underline{c}_1 & \dots & \underline{c}_p \end{bmatrix} \qquad \underline{c}_i \in \mathbb{R}^n$$

Let be $\underline{c}, \underline{b} \in \mathbb{R}^n$ columns, we can choose a distance:

$$d^2(\underline{c}, \underline{b}) = 2(1 - \rho) \qquad \rho = corr(\underline{c}, \underline{b})$$

Example.

> Let be $\underline{x}_1, \underline{x}_2, \underline{x}_3$ sequences of customer bills across time. A possible distance could be:
>
> $$2(1 - corr(\underline{x}_1, \underline{x}_3)$$
>
> That indicates how correlated (across time) the elements in two customers are.

13.7 Further Examples

Example.

> Let $\mathcal{U}$ be the space of sequences of symbols. Let be $\underline{u}, \underline{v} \in \mathcal{U}$.
> It can be defined $d : \mathcal{U} \times \mathcal{U} \to [0, \infty)$ as:
>
> $$d(\underline{u}, \underline{v}) = \frac{1}{2^k}$$
>
> where k is the first position where the symbols are different.
> This is an ultrametric.

Example.

> Let $\mathcal{U}$ be a set of texts (sequence of characters). Let be $\mathcal{O}$ the set of operations acting over $\underline{u} \in \mathcal{U}$. Let $\phi_1, \ldots, \phi_k \in \mathcal{O}$:
>
> $$\underline{u} \overset{\phi_1}{\to} \underline{u}_1 \overset{\phi_2}{\to} \underline{u}_2 \ldots \overset{\phi_k}{\to} \underline{v}$$
>
> if each of these operations have a cost $c_1, \ldots, c_k$, a possible distance could be:
>
> $$d(\underline{u}, \underline{v}) = \min_{\phi_i} \left(\sum_{i=1}^{k} c_i \right)$$
>
> This is called Edit Distance.

13.8 Linkage

Let be $U, V \subseteq \mathbb{R}^p$ and $d : \mathbb{R}^p \times \mathbb{R}^p \to [0, \infty)$. We are interested in what's the distance between U and V, i.e., $d(U, V)$. This is called linkage.
There are different kinds of linkages:

- Single Linkage:
$$d(U,V) = \min_{x,y}\{d(\underline{x},\underline{y}) \quad st \quad \underline{x} \in U, \quad \underline{y} \in V\}$$

- Complete Linkage:
$$d(U,V) = \max_{x,y}\{d(\underline{x},\underline{y}) \quad st \quad \underline{x} \in U, \quad \underline{y} \in V\}$$

- Average Linkage:
$$d(U,V) = \frac{1}{\#U\#V} \sum_{\substack{\underline{x}\in U \\ \underline{y}\in V}} d(\underline{x},\underline{y})$$

- Centroid Linkage: Let $\bar{x}$ and $\bar{y}$ the barycenter of U and V respectively, then:
$$d(U,V) = d(\bar{x},\bar{y})$$

13.9 Dissimilarity Matrix

The cluster algorithms are built not over the original data, but over the dissimilarity matrix:

$$D = [d_{ij}] = \begin{bmatrix} 0 & d_{12} & d_{13} & \dots & d_{1n} \\ d_{21} & 0 & d_{23} & \dots & d_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ d_{n1} & d_{n2} & d_{n3} & \dots & 0 \end{bmatrix}$$

Where $d_{ij} = d(\underline{x}_i, \underline{x}_j), \qquad i,j = 1, \dots, n.$
Therefore, it is not needed to have the exact points to build these algorithms, only the distances between the points.
This matrix is symmetric.

13.10 Hierarchical Agglomerative Algorithms

Hierarchical Agglomerative Algorithms are the algorithms that are built over the following procedure:

Algorithm 1 Hierarchical Agglomerative Algorithms Structure

Require: Dissimilarity Matrix: D
Ensure: Clusters: $U_1, \dots, U_k$
  Initialization of every unit as a cluster
  while not convergence **do**
    Aggregate the two most similar clusters
    Compute the new Dissimilarity Matrix
  end while

Finally, a dendrogram can be done, which consists of a graph that indicates where the elements joined.

Having the dendrogram, the Cophenetic Matrix $\tilde{D}$ can be determined, which consists of the distances where the elements were joined in a cluster.

Following, for every pair (i, j), we have two distances, d_{ij} distance and $\tilde{d}_{ij}$ cophenetic distance.

With these two values, we can contrast them and calculate their correlation, this is called the Cophenetic Correlation Coefficients:

$$CPCC = |corr(D, \tilde{D})|$$

The larger CPCC is, the better the algorithm "captured" the initial distance structure.

The last decision we must make is to decide at which level we will keep the clusters.

Example.

> Let be an algorithm using simple link. Let be:
>
> $$D = \begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 6 & 5 & 0 & & \\ 10 & 9 & 4 & 0 & \\ 9 & 8 & 5 & 3 & 0 \end{bmatrix}$$
>
> Where the rows are 1,2,3,4,5.
> Then 1 and 2 are grouped, and now:
>
> $$D = \begin{bmatrix} 0 & & & \\ 5 & 0 & & \\ 9 & 4 & 0 & \\ 8 & 5 & 3 & 0 \end{bmatrix}$$
>
> Where the rows are $\{1, 2\}, 3, 4, 5$.
> Then 4 and 5 are grouped, and now:
>
> $$D = \begin{bmatrix} 0 & & \\ 5 & 0 & \\ 8 & 4 & 0 \end{bmatrix}$$
>
> Where the rows are $\{1, 2\}, 3, \{4, 5\}$.
> Then 3 and {4,5} are grouped, and now:
>
> $$D = \begin{bmatrix} 0 & \\ 4 & 0 \end{bmatrix}$$
>
> Where the rows are $\{1, 2\}, \{3, 4, 5\}$.
> And finally, the two unique clusters are merged, and the cluster at the end is {1,2,3,4,5}.
> Then, we can do a dendrogram with the result, using the step-by-step clustering.

Finally, the cophenetic matrix can be calculated:

$$\tilde{D} = \begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 5 & 5 & 0 & & \\ 5 & 5 & 4 & 0 & \\ 5 & 5 & 4 & 3 & 0 \end{bmatrix}$$

Remark.

Often, some small random noise is added, to make sure that there are no two pairs with the same distance, so there is always a unique minimum. This is called Jittering.

13.11 Non-Hierarchical Algorithms

Let $\mathbb{X} \in \mathbb{R}^{n \times p}$ a training set. Let $d : \mathbb{R}^p \times \mathbb{R}^p \to [0, \infty)$ a dissimilarity function.

The idea is to find a partition of $\mathbb{X}$ in k subsets $C_1, \ldots, C_k$, where:

$$C_i \cap C_j = \emptyset \quad i \neq j \qquad \wedge \qquad \bigcup_i C_i = \mathbb{X}$$

Now,

Definition 13.1: Centroid of C_i

Define the centroid of a cluster $C_i \subset \mathbb{X}$ as:

$$\underline{\bar{x}}_i = \arg\min_{\underline{x} \in \mathbb{R}^p} \sum_{\underline{x}_j \in C_i} d^2(\underline{x}_j, \underline{x})$$

Solving this problem may be difficult, so we can simplify the problem to:

Definition 13.2: Medoid of C_i

$$\underline{\bar{x}}_i = \arg\min_{\underline{x} \in \mathbb{X}} \sum_{\underline{x}_j \in C_i} d^2(\underline{x}_j, \underline{x})$$

Example.

If d is the Euclidean distance, it can be proven that:

$$\underline{\bar{x}}_i = \frac{1}{|C_i|} \sum_{\underline{x}_j \in C_i} \underline{x}_j$$

Then, we would like to solve the general optimization problem:

Find $C_1, \dots, C_k \quad st \quad :$

$$\sum_{j=1}^{k} \sum_{\underline{x}_i \in C_j} d^2(\underline{x}_i, \bar{\underline{x}}_j) \text{ is minimum}$$

Remark.

> We may want to fit a supervised algorithm with the groups obtained in an unsupervised algorithm to evaluate the clusterization. If we are not able to find a supervised algorithm that can discriminate between the groups, it could mean that the clustering was not that good.

13.11.1 K-Means

Algorithm 2 K-Means Algorithm

Require: Dissimilarity Matrix: D
Require: Number of clusters: k
Ensure: Clusters: $U_1, \dots, U_k$
  Split $\mathbb{X}$ in K subsets $C_1, \dots, C_k$ or Choose $\bar{\underline{x}}_1, \dots, \bar{\underline{x}}_k$ initial centroids of the C_is
  while Centroids change in the step **do**
    Compute the centroids $C_1, \dots, C_k$
    Assign each $\underline{x} \in \mathbb{X}$ to the cluster identified by the closest centroid
  end while

Remark.

> The version that uses the medoids instead of centroids is called K-medoids.

Remark.

> The convergence of this algorithm was proved by Hartigan-Macqueen.

k can be chosen in multiple ways, one of them is to take:

$$W(k) = \sum_{j=1}^{k} \sum_{\underline{x}_i \in c_j} d^2(\underline{x}_i, \bar{\underline{x}}_j)$$

Then, the k can be chosen according to the marginal gain of increasing k, choosing the k before the marginal gain is not that much.
Another way is the Ward's method, an agglomerative method. It defines:

$$SS_j = \sum_{\underline{x} \in C_j} \|\underline{x}_i - \bar{\underline{x}}_j\|^2$$

and then:

$$SS = \sum_{i=1}^{k} SS_i$$

Then, again, k can be chosen in such a way that the increase of SS is minimum.

13.11.2 DBScan

The DBscan algorithm is a Density-Based algorithm that clusters points according to the density of elements in the space. Besides finding clustering, this algorithm also finds outliers, letting points without a high density without assigning to any group.

Definition 13.3: Neighborhood

Find $\epsilon > 0$, define the neighbors of $\underline{x}$ as:

$$N_\epsilon(\underline{x}) = \{\underline{y} \in \mathbb{R}^p \quad st \quad d(\underline{x}, \underline{y}) \leq \epsilon\}$$

Remark.

> Note that $|N_\epsilon(\underline{x})|$ is the size of the neighborhood, i.e., the number of units in $\mathbb{X}$ belonging to $N_\epsilon(\underline{x})$

Definition 13.4: Point Classes

Given a fixed integer $MinPts$, $\underline{x}_i \in \mathbb{X}$ is classified as:

- **Core Point:** If $|N_\epsilon(\underline{x}_i)| \geq MinPts$.
- **Border Point:** If $\underline{x}_i$ is not a core point, but belongs to $N_\epsilon(\underline{x}_j)$, and $\underline{x}_j$ is a core point.
- **Noise Point:** If it is either a core point nor a border point.

Definition 13.5: Directly Density Reachable Points

$\underline{x}_j \in \mathbb{X}$ is directly density reachable from $\underline{x}_i$ if:

- $\underline{x}_i$ is core.
- $\underline{x}_j \in N_\epsilon(\underline{x}_i)$.

Definition 13.6: Density Reachable Points

$\underline{x}_j \in \mathbb{X}$ is density reachable from $\underline{x}_i$ if there is a finite sequence $y_1, \ldots, y_q$ such that:

- $\underline{y}_1 = \underline{x}_i, \underline{y}_k = \underline{x}_i$
- $\underline{y}_{i+1}$ is directly density reachable from $\underline{y}_i \qquad \forall i = 1, \ldots, q-1$

Remark.

> $\underline{y}_1, \ldots, \underline{y}_{k-1}$ need to be core points.

Remark.

> (Directly) Density reachable is not a symmetric property.

Definition 13.7

Let be $\underline{x}_i, \underline{x}_j \in \mathbb{X}$. $\underline{x}_i$ and $\underline{x}_j$ are density connected if there is an $\underline{x} \in \mathbb{X}$ such that both $\underline{x}_i$ and $\underline{x}_j$ are density reachable from $\underline{x}$

What DBScan does is to identify subsets $C \subseteq \mathbb{X}$ such that:

- Let be $\underline{x}_i \in C$. If $\underline{x}_j$ is density reachable from $\underline{x}_i$, $\underline{x}_j \in C$
- If $\underline{x}_i, \underline{x}_j \in C$, then $\underline{x}_i$ and $\underline{x}_j$ are density connected

The obvious problem that appears is the curse of dimensionality. Even if ϵ is too large, points will be too far from others.

Remark.

> Some border points may be in two different clusters, so the clusters are not a partition.

13.11.3 Multidimensional Sealing (MDS)

Let δ be the euclidean distance in $\mathbb{R}^q$. The idea of multidimensional Sealing is to find an embedding of the data:

$$\tilde{\mathbb{X}} = \begin{bmatrix} \underline{y}_1^T \\ \ldots \\ \underline{y}_n^T \end{bmatrix} \in \mathbb{R}^q, \qquad q \ll p$$

such that:

$$d_{ij} \simeq \delta_{ij}$$

Where $d_{ij} = d(\underline{x}_i, \underline{x}_j)$ and $\delta_{ij} = \delta(\underline{y}_i, \underline{y}_j)$. In other words, the objective is find $y_1, \ldots, y_n$ st:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} (d_{ij} - \delta_{ij})^2 \text{ is minimum}$$

Remark.

> If d is euclidean, then $y_1, \ldots, y_n$ are represented by scores of principal components $PC_1, \ldots, PC_q$.

Another possible objective function is minimize the stress function:

$$\frac{\sum_{i=1}^{n} \sum_{j=1}^{n} (\theta \cdot d_{ij} - \delta_{ij})^2}{\sum_{i=1}^{n} \sum_{j=1}^{n} \delta_{ij}^2}$$

where $\theta : [0, \infty) \to [0, \infty)$ This time choosing both $y_i's$ and θ

Remark.

> This representation is not unique, we have to find the one that fits with our purposes, since there is an equivalent class of solutions.

14 Regression

14.1 Introduction

Regression is a supervised learning algorithm. Let be $\underline{X} \in \mathbb{R}^p$ vector of covariates and $y \in \mathbb{R}$ target. The regression goal is:

$$\arg \min_{f:\mathbb{R}^p \to \mathbb{R}} E[(y - f(\underline{x}))^2] = E[y|\underline{x}]$$

As the distribution is not known, we try to find estimates $\hat{f} : \mathbb{R}^p \to \mathbb{R}$ of $E[y|\underline{x}]$.
Recall that the available training set consists of:

$$\mathbb{X} = \underbrace{\begin{bmatrix} \underline{x}_1^T \\ \ldots \\ \underline{x}_n^T \end{bmatrix}}_{\underline{X}} \underbrace{\begin{bmatrix} y_1 \\ \ldots \\ y_n \end{bmatrix}}_{y}$$

To achieve this, there are two approaches:

- Data Driven: CART, Random Forest, Boosting.
- Model Based: Linear Regression, GLM, LMM, Kriging.

14.2 Classification and Regression Tree (CART)

Let be $\hat{f}$ step-wise function, constant over the **hyperrectangular** sets $R1, \dots, R_J$ portion of $\mathbb{R}^p$ subject to $R_i \cap \mathbb{X} \qquad \forall i = 1, \dots, J$.

Define now:

$$|R_i| = \text{Number of points of } \mathbb{X} \text{ in } R_i$$

$$\bar{y}_i = \frac{1}{|R_i|} \sum_{\underline{x}_j \in R_i} y_i = \text{`Average on } R_i\text{'}$$

Specifically, $\hat{f}$ is defined as follows:

$$\hat{f}(\underline{x}) = \sum_{i=1}^{J} \bar{y}_i \mathbb{I}[\underline{x} \in R_i]$$

The general problem to be optimized is:

$$\arg \min_{\{R_1, \dots, R_J\}} \sum_{i=1}^{J} \sum_{\underline{x}_j \in R_i} (y_j - \bar{y}_i)^2$$

Without overfitting the data.

CART is a greedy algorithm to solve this problem. CART splits a single rectangle R as follows:

1. R rectangle in $\mathbb{R}^p$:

$$R = (a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_p, b_p), \qquad a_i, b_i \in \bar{\mathbb{R}}$$

2. Given $\underline{x}_1, \dots, \underline{x}_m \in R \cap \mathbb{X}$:

$$\bar{y}_i = \frac{1}{|R|} \sum_{\underline{x}_j \in R} y_i$$

3. Average Evaluation Function:

$$\sum_{\underline{x}_j \in R} (y_j - \bar{y}_j)^2$$

4. Split options: R can be split along the direction X_1, and optimally find s_1 subject to:

$$\sum_{\underline{x}_j \in R} (y_j - \bar{y})^2 = \left[\sum_{x_{j1} \leq s_1} (y_j - \bar{y}_-)^2 + \sum_{x_{j1} > s_1} (y_j - \bar{y}_+)^2 \right] \text{ is max} \qquad (*)$$

where:

$$\bar{y}_- = \frac{1}{|\{\underline{x}_j \in R \quad st \quad x_{j1} \leq s_1\}|} \sum_{x_{i1} \leq s_1} y_i$$

$$\bar{y}_+ = \frac{1}{|\{\underline{x}_j \in R \quad st \quad x_{j1} > s_1\}|} \sum_{x_{i1} > s_1} y_i$$

The same can be done for directions $X_2, X_3, \dots, X_p$, having:

$$S_1^*, \dots, S_p^*$$

where in each case S_k^* is chosen such $(*)$ is maximized.
Then, we choose the direction that maximizes the Average Evaluation Function.

The CART algorithm proceeds by splitting rectangles recursively at each iteration.

The result corresponds to a tree where splits are made recursively.

Note that we can divide in the same direction more than once, for different rectangles.

A stop condition is given when all the rectangles have less or equal *MinPts* units.

Another possible stop condition is to add a penalization to the overall cost function:

$$\sum_{i=1}^{J} \sum_{\underline{x}_j \in R_i} (y_j - \bar{y}_i)^2 + \alpha J =: W(\alpha)$$

And minimize with respect to $R_1, \dots, R_J$ and J.

Now, we can prune the tree bottom up to minimize $W(\alpha)$. *α can be gotten from cross-validation.*

In the case of classification, given units $\underline{x}_1, \dots, \underline{x}_n$ with labels $1, \dots, k$, the function to maximize is the Gini index:

$$\sum_{\underline{x}_j \in R} \sum_{i=1}^{k} p_i(1 - p_i)$$

Where p_i is the frequency of label i in R.

Remark.

> More precise approach can be found in (Sohil, Sohail, & Shabbir, 2021).

14.3 Linear Models for Regression

Again, the goal is to approximate $E[y|\underline{X}]$.
We build the design matrix:

$$\mathbb{Z} = \begin{bmatrix} 1 & z_{11} & z_{12} & \dots & z_{1r} \\ 1 & z_{21} & z_{22} & \dots & z_{2r} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & z_{n1} & z_{n2} & \dots & z_{nr} \end{bmatrix} = f\left(\begin{bmatrix} \underline{x}_1^T \\ \dots \\ \underline{x}_n^T \end{bmatrix}\right)$$

Where columns are denoted as $Z_i = h_i(\underline{X}_1, \dots, \underline{X}_p)$, with h_i known.
Note that we can find:

$$\begin{aligned} E[y|X_1, \dots, X_n] &= E[y|Z_1, \dots, Z_r] \\ &= \beta_0 + \beta_1 Z_1 + \dots + \beta_r Z_r \end{aligned}$$

Having that:

$$\underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon} = [\beta_0 + \beta_1 Z_1 + \dots + \beta_r Z_r + \epsilon]_i$$

Where $\underline{\epsilon}$ is such that:

$$E[\underline{\epsilon}] = \underline{0} \qquad \wedge \qquad Var[\underline{\epsilon}] = \sigma^2 I \qquad \wedge \qquad \underline{\epsilon} \perp\!\!\!\perp X_1, \dots, X_p$$

And:

$$\underline{\beta} = \begin{pmatrix} \beta_0 & \beta_1 & \dots & \beta_r \end{pmatrix}^T \in \mathbb{R}^{r+1}$$

Remark.

> Note that we are not assuming Gaussianity.

14.4 Linear Models and ANOVA

Linear models are very flexible, indeed, we can use them for ANOVA.
The ANOVA problem is:

$$Indep \begin{cases} X_{11}, \dots, X_{1n_1} & iid \sim N(\mu_1, \sigma^2) \\ X_{21}, \dots, X_{2n_2} & iid \sim N(\mu_2, \sigma^2) \\ \dots \\ X_{g1}, \dots, X_{gn_g} & iid \sim N(\mu_g, \sigma^2) \end{cases}$$

Define: $\underline{Y} = \begin{pmatrix} X_{11} & \dots & X_{1n_1} & X_{21} & \dots & X_{gn_g} \end{pmatrix} \in \mathbb{R}^{n = n_1 + \dots + n_g}$.
Then, take $\mathbb{Z}$ the matrix with n_i ones in the positions related with its X_{ij}:

$$\mathbb{Z} = \begin{bmatrix} 1 & 1 & 0 & \dots & 0 \\ \dots & n_1 \dots & \dots & \dots & \dots \\ \dots & 1 & 0 & \dots & 0 \\ \dots & 0 & 1 & \dots & 0 \\ \dots & \dots & n_2 \dots & \dots & \dots \\ \dots & 0 & 1 & \dots & 0 \\ \dots & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \dots & 0 & 0 & \dots & 1 \\ \dots & \dots & \dots & \dots & n_g \dots \\ 1 & 0 & 0 & \dots & 1 \end{bmatrix}$$

Then, $\underline{Y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon}$.
Where $\underline{\beta} = \begin{pmatrix} \mu & r_1 & \dots & r_g \end{pmatrix}^T$ And note that as ANOVA:

$$x_{ij} = \mu + r_i + \epsilon_{ij}$$

And:

$$\underline{\epsilon} \sim N_n(\underline{0}, \sigma^2 I)$$

When we do ANOVA, we use exactly this model, but with a constraint:

$$\sum_i n_i r_i = 0 \iff r_g = \frac{-1}{n_g} \sum_{i=1}^{g-1} n_i r_i$$

So the problem change to:

$$\mathbb{Z} = \begin{bmatrix} 1 & 1 & 0 & \dots & 0 \\ \dots & n_1 \dots & \dots & \dots & \dots \\ \dots & 1 & 0 & \dots & 0 \\ \dots & -n_1/ng & 1 & \dots & 0 \\ \dots & \dots & n_2 \dots & \dots & \dots \\ \dots & -n_1/ng & 1 & \dots & 0 \\ \dots & -n_1/ng & -n_2/ng & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \dots & -n_1/ng & -n_2/ng & \dots & 1 \\ \dots & \dots & \dots & \dots & n_{g-1} \dots \\ 1 & -n_1/ng & -n_2/ng & \dots & 1 \\ 1 & -n_1/ng & -n_2/ng & \dots & -n_{g-1}/ng \end{bmatrix} \quad \wedge \quad \underline{\beta} = \begin{pmatrix} \mu \\ r_1 \\ r_2 \\ \dots \\ r_{g-1} \end{pmatrix}$$

And this is a full-rank matrix.
On the other side, we can consider other variables, this is called **ANCOVA**, for example:

$$X_{ij} = \mu + r_i + \beta W_i + \epsilon_{ij}$$

14.5 Ordinary Least Squares (OLS)

Now, we need to know how to estimate $\underline{\beta}$ and σ^2. This can be done with Ordinary Least Squares:

$$\hat{\beta} = \arg \min_{\underline{\beta} \in \mathbb{R}^{r+1}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 = \arg \min_{\underline{\beta}} \sum_{i=1}^{n} \sum (y_i - \underline{z}_i^T \underline{\beta})^2$$

Note that this is just the orthogonal projection of $\underline{Y}$ into $\mathcal{L}(\mathbb{Z})$, $\Pi_{\mathcal{L}(\mathbb{Z})}(\underline{Y})$.

Theorem 14.1

If $\mathbb{Z}$ is full rank, i.e., $rank(\mathbb{Z}) = r+1 \leq n$, then:

$$\hat{\underline{\beta}} = \arg\min_{\underline{\beta}\in\mathbb{R}^{r+1}} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 = \left(\mathbb{Z}^T\mathbb{Z}\right)^{-1}\mathbb{Z}^T\underline{y}$$

Proof. Taking $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{r+1} > 0$ eigenvalues of $\mathbb{Z}^T\mathbb{Z}$

$$\mathbb{Z}^T\mathbb{Z} = \sum_{i=1}^{r+1} \lambda_i \underline{e}_i \underline{e}_i^T$$

So:

$$\left(\mathbb{Z}^T\mathbb{Z}\right)^{-1} = \sum_{i=1}^{r+1} \frac{1}{\lambda_i} \underline{e}_i \underline{e}_i^T$$

For $i = 1, \dots, r+1$, define:

$$q_i := \frac{1}{\sqrt{\lambda_i}} \mathbb{Z}\underline{e}_i \in \mathcal{L}(\mathbb{Z})$$

So, by definition, and as λ_j is an eigenvalue of $\mathbb{Z}^T\mathbb{Z}$:

$$q_i^T q_j = \frac{1}{\sqrt{\lambda_i\lambda_j}} \underline{e}_i^T \mathbb{Z}^T\mathbb{Z}\underline{e}_j = \frac{\lambda_j}{\sqrt{\lambda_i\lambda_j}} \underline{e}_i^T \underline{e}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

So $\{q_1, \dots, q_{r+1}\}$ is an orthonormal basis for $\mathcal{L}(\mathbb{Z})$.
Therefore, we can express the projection as:

$$\begin{aligned}\Pi_{\mathcal{L}(\mathbb{Z})}\underline{y} &= \sum_{i=1}^{r+1} \Pi_{q_i}\underline{y} = \sum_{i=1}^{r+1} \frac{q_i q_i^T}{q_i^T q_i}\underline{y} = \sum_{i=1}^{r+1} \frac{1}{\lambda_i}\mathbb{Z}\underline{e}_i\underline{e}_i^T\mathbb{Z}^T\underline{y} \\ &= \mathbb{Z}\left(\sum_{i=1}^{r+1}\frac{1}{\lambda_i}\underline{e}_i\underline{e}_i^T\right)\mathbb{Z}^T\underline{y} = \mathbb{Z}\left(\mathbb{Z}^T\mathbb{Z}\right)^{-1}\mathbb{Z}^T\underline{y}\end{aligned}$$

Finally:

$$\hat{\underline{y}} = \Pi_{\mathcal{L}(\mathbb{Z})}\underline{y} = \mathbb{Z}\left(\mathbb{Z}^T\mathbb{Z}\right)^{-1}\mathbb{Z}^T\underline{y} = Hy$$

where $H := \mathbb{Z}\left(\mathbb{Z}^T\mathbb{Z}\right)^{-1}\mathbb{Z}^T$ is called hat operator.
So, by definition of the problem:

$$\hat{\underline{\beta}} = \left(\mathbb{Z}^T\mathbb{Z}\right)^{-1}\mathbb{Z}^T\underline{y}$$

□

14.6 Residual

Now, define

$$\hat{\underline{\epsilon}} = \underline{y} - \hat{\underline{y}}$$

is the residual, and $\underline{\hat{\epsilon}} \perp \underline{\hat{y}}$.
Observations:

- If rank($\mathbb{Z}$)=n=r+1:
$$\underline{\hat{y}} = \underline{y} \qquad \wedge \qquad \underline{\hat{\epsilon}} = \underline{0}$$
- The situation rank($\mathbb{Z}$)=$r+1 > n$ does not exist
- If $rank(\mathbb{Z}) = k < r+1 \leq n$:
$$\mathbb{Z}^T\mathbb{Z} = \sum_{i=1}^{r+1} \lambda_i \underline{e}_i \underline{e}_i^T$$
And then the generalized inverse of $\mathbb{Z}^T\mathbb{Z}$ is:
$$\left(\mathbb{Z}^T\mathbb{Z}\right)^- = \sum_{i=1}^{k} \frac{1}{\lambda_i} \underline{e}_i \underline{e}_i^T$$
And then:
$$\underline{\hat{\beta}} = \left(\mathbb{Z}^T\mathbb{Z}\right)^- \mathbb{Z}^T \underline{y}$$

Remark.

> $\underline{\hat{\epsilon}}$ is deterministic, it is not an error estimate.

14.7 Coefficient of determination (R^2)

As the estimate and residuals are perpendicular:

$$\|\underline{y}\|^2 = \|\underline{\hat{y}}\|^2 + \|\underline{\hat{\epsilon}}\|^2$$

And this is:

$$\underbrace{\sum_{i=1}^{n} y_i^2}_{\text{sum tot.}} = \underbrace{\sum_{i=1}^{n} \hat{y_i}^2}_{\text{sum regression}} + \underbrace{\sum_{i=1}^{n} \hat{\epsilon_i}^2}_{\text{sum res.}}$$

Now, note:

$$\Pi_{\mathcal{L}(1)}(\underline{y}) = \bar{y} \cdot 1$$

And since $\underline{1}$ is in the design matrix:

$$1^T H = (H^T 1)^T = (H1)^T = 1^T$$

$$\Pi_{\mathcal{L}(1)}(\underline{\hat{y}}) = \frac{11^T}{1^T1}\underline{\hat{y}} = \frac{11^T}{1^T1} H\underline{y} = \frac{11^T}{1^T1}\underline{y} = \bar{y} \cdot 1$$

And therefore, we can represent our variability around the mean:

$$\underbrace{\|\underline{y} - \bar{y} \cdot 1\|^2}_{\text{CSS tot.}} = \underbrace{\|\underline{\hat{y}} - \bar{y} \cdot 1\|^2}_{\text{CSS fitted}} + \underbrace{\|\underline{\hat{\epsilon}}\|^2}_{\text{CSS res.}}$$

Explicitly:

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n}\hat{\epsilon}_i^2$$

And R^2 indicates how much of the variability is explained by the values:

$$R^2 := \frac{\|\underline{\hat{y}} - \bar{y} \cdot 1\|^2}{\|\underline{y} - \bar{y} \cdot 1\|^2}$$

Or as it is usually calculated in the software:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \hat{\epsilon}_i^2}{\|\underline{y} - \bar{y} \cdot 1\|^2} = 1 - \sin^2(\theta)$$

Where $\sin(\theta)$ is the angle between $\mathcal{L}\mathbb{Z}$ and $\underline{y}$.
R can be:

- $R^2 \in [0, 1]$
- $R^2 = 1$: The fitting is perfect:

$$\theta = 0 \implies \underline{\hat{\epsilon}} = \underline{0}$$

- $R^2 = 0$: $\hat{y}_i = \bar{y} \qquad \forall i$

$$\theta = \frac{\pi}{2} \implies \underline{\hat{y}} = \bar{y} \cdot 1$$

Remark.

> It is essential to put 1 as part of the design matrix. If not, the CSS equality does not hold, since $1 \notin \mathcal{L}(\mathbb{Z})$

Remark.

> **Regression to the origin:**
> If we force $\beta_0 = 0$, because it is a need on our problem, we can do it, but the CSS equality won't hold, since our design matrix won't pass for $\mathcal{L}(1)$. However, the SS equality still holds, so we can build a "different R":
>
> $$\tilde{R}^2 = 1 - \frac{\|\underline{\hat{\epsilon}}\|^2}{\|\underline{y}\|^2}$$

Also, check the figure and analysis from section 11.5:

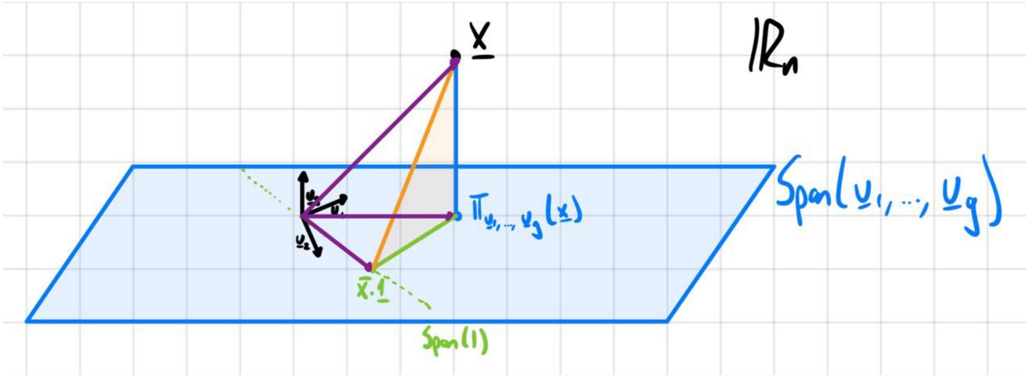

Figure 12: Example: $\underline{X}$ decomposition over projections on $\underline{u}_1, \ldots, \underline{u}_g$

14.8 Properties of $\hat{\underline{\beta}}$ and $\hat{\underline{\epsilon}}$

Theorem 14.2: Properties of $\hat{\underline{\beta}}$ and $\hat{\underline{\epsilon}}$

Assume rank($\mathbb{Z}$) = r+1:

1. $\hat{\underline{\beta}}$ is unbiased:
$$E[\hat{\underline{\beta}}] = \underline{\beta}$$
2. The variability depends on the design matrix in the following way:
$$Cov(\hat{\underline{\beta}}) = \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}$$
3. $E[\hat{\underline{\epsilon}}] = 0$
4. $Cov(\hat{\underline{\epsilon}}) = \sigma^2 \cdot (I - H)$
5. $E[\hat{\underline{\epsilon}}^T \hat{\underline{\epsilon}}] = E[\sum_i \hat{\epsilon}_i^2] = \sigma^2 \cdot (n - (r+1))$

Proof. 1.

$$E[\hat{\underline{\beta}}] = E[(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y}] = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\mathbb{Z}\underline{\beta} = \underline{\beta}$$

2.

$$\begin{aligned} Cov(\hat{\underline{\beta}}) &= Cov((\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y}) = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T Cov(\underline{y})\mathbb{Z}^T(\mathbb{Z}^T\mathbb{Z})^{-1} \\ &= \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1} = \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1} \end{aligned}$$

3.

$$\begin{aligned} E[\hat{\underline{\epsilon}}] &= E[\underline{y} - \hat{\underline{y}}] = \mathbb{Z}\underline{\beta} - E[H\underline{y}] \\ &= \mathbb{Z}\underline{\beta} - H\mathbb{Z}\underline{\beta} = \mathbb{Z}\underline{\beta} - \mathbb{Z}\underline{\beta} = 0 \end{aligned}$$

4.

$$
\begin{aligned}
Cov(\underline{\hat{\epsilon}}) &= Cov((I-H)\underline{y}) = (I-H)\sigma^2 I(I-H)^T \\
&= \sigma^2(I-H)(I-H)^T = \sigma^2(I-H) \qquad\qquad (I-H) = (I-H)^T \text{ in orthogonal proj}
\end{aligned}
$$

5. Noting that $(I_n - H)\underline{y} = (I_n - H)\mathbb{Z}\underline{\beta} + (I_n - H)\underline{\epsilon} = (I_n - H)\underline{\epsilon}$ for the second line:

$$
\begin{aligned}
E[\underline{\hat{\epsilon}}^T\underline{\hat{\epsilon}}] &= E[\text{trace}(\underline{\hat{\epsilon}}^T\underline{\hat{\epsilon}})] = E[\text{trace}(\underline{\hat{\epsilon}\hat{\epsilon}}^T)] = \text{trace}(E[\underline{\hat{\epsilon}\hat{\epsilon}}^T]) = \text{trace}(E[(I_n-H)\underline{yy}^T(I_n-H)^T]) \\
&= \text{trace}\left(E[(I_n-H)\underline{\epsilon\epsilon}^T(I_n-H)^T\right) \\
&= \text{trace}\left(((I_n-H)E[\underline{\epsilon\epsilon}^T](I_n-H)^T\right) = \text{trace}[\sigma^2(I_n-H)(I_n-H)^T] = \text{trace}[\epsilon^2(I_n-H)] \\
&= \sigma(\text{trace}(I_n) - \text{trace}(H)) = \sigma^2(n - \text{trace}(H)) = \sigma^2(n-(r+1))
\end{aligned}
$$

The last step, since:

$$
\text{trace}[H] = \text{trace}[(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T] = \text{trace}[\mathbb{Z}^T\mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}] = \text{trace}[I_{r+1}] = r+1
$$

□

Definition 14.3: S^2

$$
S^2 = \frac{\underline{\hat{\epsilon}}^T\underline{\hat{\epsilon}}}{n-(r+1)} = \frac{\|\underline{\hat{\epsilon}}\|^2}{n-(r+1)}
$$

Corollary 14.4

S^2 is unbiased for σ^2, i.e.:

$$
E[S^2] = \sigma^2
$$

Proof. Cor of point 5 in the theorem □

14.9 Gaussian Assumption

Let be a new assumption:

$$
\underline{\epsilon} \sim N_n(\underline{0}, \sigma^2 I)
$$

This implies:

$$
\underline{y} \sim N_n(\mathbb{Z}\underline{\beta}, \sigma^2 I)
$$

Theorem 14.5: Properties of $\underline{\hat{\beta}}$ and $\underline{\hat{\epsilon}}$ in Gaussianity

Assume $\underline{\epsilon} \sim N_n(\underline{0}, \sigma^2 I)$ and $rank(\mathbb{Z}) = r + 1 \leq n$:

1. $\underline{\hat{\beta}}$ and $\frac{\underline{\hat{\epsilon}}^T \underline{\hat{\epsilon}}}{n} = \hat{\sigma}^2$ are Maximum Likelihood estimators of $\underline{\beta}$ and σ^2 respectively.
2. $\underline{\hat{\beta}} \sim N_{r+1}(\underline{\beta}, \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1})$
3. $\underline{\hat{\epsilon}} \sim N_n(\underline{0}, \sigma^2(I - H))$
   Note that the Covariance is singular, has a support in the orthogonal space to $\mathcal{L}(\mathbb{Z})$.
4. $\underline{\hat{\epsilon}} \perp\!\!\!\perp \underline{\hat{\beta}}$
5. $\underline{\hat{\epsilon}}^T \underline{\hat{\epsilon}} \sim \sigma^2 \chi^2(n - (r + 1))$

Proof. 1. Write the likelihood and differenciate

2,3,4. :

$$\begin{pmatrix} \underline{\hat{\beta}} \\ \underline{\hat{\epsilon}} \end{pmatrix} = \underbrace{\begin{pmatrix} (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z} \\ I - H \end{pmatrix}}_{A} \underline{Y}$$

So:

$$\begin{pmatrix} \underline{\hat{\beta}} \\ \underline{\hat{\epsilon}} \end{pmatrix} \sim N_{r+1+n}\left(A\mathbb{Z}\underline{\beta}, \sigma^2 AA^T\right)$$

Where:

$$A\mathbb{Z}\underline{\beta} = \begin{pmatrix} \underline{\beta} \\ \underline{0} \end{pmatrix} \qquad \wedge \qquad AA^T = \begin{bmatrix} (\mathbb{Z}^T\mathbb{Z})^{-1} & \underline{0} \\ \underline{0} & I - H \end{bmatrix}$$

5. Note that

$$(\underline{\hat{\epsilon}} - \underline{0})^T [\sigma^2(I - H)]^{-1} (\underline{\hat{\epsilon}} - \underline{0}) \sim \chi^2(n)$$

So, it can be proved that if:

$$\underline{X} \sim N_n(\underline{0}, \Sigma), \qquad \Sigma = \sum \lambda_i \underline{e}_i \underline{e}_i^T \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > \lambda_{k+1} = 0 = \dots = \lambda_n$$

Then:

$$\underline{X}^T \Sigma^- \underline{X} \sim \chi^2(n - k)$$

And from here, the result is direct.

□

14.10 Confidence interval for $\underline{\beta}$

In the case of $\underline{z} \sim N_n(\underline{0}, \sigma^2 I)$, from theorem 14.9:

$$(\underline{\hat{\beta}} - \underline{\beta})^T (\sigma^2(\mathbb{Z}\mathbb{Z})^{-1})^{-1} (\underline{\hat{\beta}} - \underline{\beta}) \sim \chi^2(r + 1)$$

and it is independent to:

$$\frac{\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}}{\sigma^2} \sim \chi^2(n-(r+1))$$

so we can get a pivotal :

$$\frac{\frac{(\hat{\underline{\beta}}-\underline{\beta})^T\mathbb{Z}\mathbb{Z}(\hat{\underline{\beta}}-\underline{\beta})}{r+1}}{S^2} \sim F(r+1, n-(r+1))$$

and:

$$\frac{(\hat{\underline{\beta}}-\underline{\beta})^T\mathbb{Z}\mathbb{Z}(\hat{\underline{\beta}}-\underline{\beta})}{S^2} \sim (r+1)F(r+1, n-(r+1))$$

So, with a significance level $\alpha \in (0,1)$:

$$CR_{1-\alpha}(\underline{\beta}) = \{\underline{\eta} \in \mathbb{R}^{r+1} \quad st \quad (\underline{\eta}-\underline{\beta})^T\mathbb{Z}\mathbb{Z}(\underline{\eta}-\underline{\beta}) \leq (r+1)S^2F_{1-\alpha}(r+1, n-(r+1))\}$$

14.11 Confidence interval for $\underline{a}^T\underline{\beta}$

Given $\underline{a} \in \mathbb{R}^{r+1}$, from theorem 14.9:

$$\underline{a}^T\hat{\underline{\beta}} \sim N_1(\underline{a}^T\beta, \sigma^2\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a})$$

So:

$$\frac{\underline{a}^T\hat{\underline{\beta}} - \underline{a}^T\beta}{\sqrt{\sigma^2\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}}} \sim N_1(0,1)$$

That is independent to:

$$\frac{\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}}{\sigma^2} \sim \chi^2(n-(r+1))$$

So:

$$\frac{\frac{\underline{a}^T\hat{\underline{\beta}} - \underline{a}^T\beta}{\sqrt{\sigma^2\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}}}}{\sqrt{\frac{\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}}{\sigma^2(n-(r+1))}}} \sim \frac{N_1(0,1)}{\sqrt{\frac{\chi^2(n-(r+1))}{n-(r+1)}}} = t(n-(r+1))$$

And since $\sqrt{\frac{\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}}{(n-(r+1))}} = S^2$:

$$\frac{\underline{a}^T\hat{\underline{\beta}} - \underline{a}^T\beta}{\sqrt{S^2\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}}} \sim t(n-(r+1))$$

And given $\alpha \in (0,1)$ the one-at-the-time confidence interval is:

$$CI_{1-\alpha}(\underline{a}^T\underline{\beta}) = \left[\underline{a}^T\hat{\underline{\beta}} \pm t_{1-\alpha/2}(n-(r+1))\sqrt{S^2\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}}\right]$$

So the specific case:

$$CI_{1-\alpha}(\underline{\beta}_i) = \left[\hat{\underline{\beta}}_i \pm t_{1-\alpha/2}(n-(r+1))\sqrt{S^2 \cdot diag_i(\mathbb{Z}^T\mathbb{Z})^{-1}}\right]$$

So we can't reject $H_0 : \beta_i = 0$ if $0 \notin CI_{1-\alpha}(\beta_i)$, but one at a time. This is what the software usually gives.

Instead, if we want all at a time confidence interval, we can use Bonferroni's correction, using k as the number of βs we want in the CI.

On the other hand, we can find:

$$\max_{\underline{a}\in\mathbb{R}^{r+1}} \frac{(\underline{a}^T\hat{\underline{\beta}} - \underline{a}^T\beta)^2}{S^2\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}} = \frac{1}{S^2}(\hat{\underline{\beta}} - \underline{\beta})\mathbb{Z}^T\mathbb{Z}(\hat{\underline{\beta}} - \underline{\beta})^T \sim (r+1)F(r+1, n-(r+1)$$

And then:

$$Sim\ CI_{1-\alpha}(\underline{a}^T\underline{\beta}) = \left[\underline{a}^T\hat{\underline{\beta}} \pm \sqrt{S^2(r+1)F_{1-\alpha}(r+1, n-(r+1))}\sqrt{\underline{a}^T(\mathbb{Z}^T\mathbb{Z})^{-1}\underline{a}}\right]$$

14.12 Confidence interval for σ^2

As:

$$\frac{(n-(r+1))S^2}{\sigma^2} = \hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}} \sim \sigma^2\chi^2(n-(r+1))$$

So:

$$P\left[\chi^2_{\alpha/2}(n-(r+1)) \leq \frac{(n-(r+1))S^2}{\sigma^2} \leq \chi^2_{1-\alpha/2}(n-(r+1))\right] = 1-\alpha$$

And finally the confidence interval:

$$CI_{1-\alpha}[\sigma^2] = \left[\frac{(n-(r+1))S^2}{\chi^2_{1-\alpha/2}(n-(r+1))}, \frac{(n-(r+1))S^2}{\chi^2_{\alpha/2}(n-(r+1))}\right]$$

14.13 Confidence interval for $C\underline{\beta}$

Let $C \in \mathbb{R}^{p\times(t+1)}$ a matrix of known coefficients. In this case, we want to test:

$$H_0 : C\underline{\beta} = \kappa_0 \qquad vs \qquad H_1 : C\underline{\beta} \neq \kappa_0$$

Note that:

$$C\underline{\beta} = \begin{bmatrix} C_{11} & C_{12} & \dots & C_{1,r+1} \\ C_{21} & C_{22} & \dots & C_{2,r+1} \\ \dots & \dots & \dots & \dots \\ C_{p1} & C_{p2} & \dots & C_{p,r+1} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \dots \\ \beta_r \end{bmatrix} = \begin{bmatrix} C_{11}\beta_0 + C_{12}\beta_1 + \dots + C_{1,r+1}\beta_r \\ C_{21}\beta_0 + C_{22}\beta_1 + \dots + C_{2,r+1}\beta_r \\ \dots \\ C_{p1}\beta_0 + C_{p2}\beta_1 + \dots + C_{p,r+1}\beta_r \end{bmatrix}$$

So we are asking each one of these linear combinations to be equal to κ_0.
By normal properties:

$$C\hat{\underline{\beta}} \sim N_p(C\underline{\beta},\ \sigma^2C(\mathbb{Z}^T\mathbb{Z})^{-1}C)$$

Now:

$$(C\hat{\underline{\beta}} - C\underline{\beta})^T[\sigma^2C(\mathbb{Z}^T\mathbb{Z})^{-1}C^T](C\hat{\underline{\beta}} - C\underline{\beta}) \sim \chi^2(p)$$

And using $\frac{1}{\sigma^2}\hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}}$:

$$\frac{\frac{(C\hat{\underline{\beta}}-C\underline{\beta})^T[\sigma^2C(\mathbb{Z}^T\mathbb{Z})^{-1}C^T](C\hat{\underline{\beta}}-C\underline{\beta})}{p}}{S^2} \sim F(p, n-(r+1))$$

So, if $H_0 : C\underline{\beta} = \underline{0}$ is true, we reject H_0 at level $\alpha \in (0, 1)$ if:

$$\frac{1}{S^2}(C\hat{\underline{\beta}})^T[C(\mathbb{Z}^T\mathbb{Z})^{-1}C^T]C\hat{\underline{\beta}} \geq pF_{1-\alpha}(p, n-(r+1))$$

Example.

Let be the special case:

$$H_0 : \beta_r = \beta_{r-1} = ... = \beta_{r-(p-1)} = 0 \qquad vs \qquad H_1 : \exists\beta_i \neq 0, i \in \{r, r-1, \dots, r-(p-1)\}$$

In other words, we are testing if we can take out the last p-1 variables. Rewriting:

$$\mathbb{Z} = \begin{bmatrix} \mathbb{Z}_1 & \mathbb{Z}_2 \end{bmatrix}$$

And the hypothesis is equivalent to:

$$H_0 : \underline{y} = \mathbb{Z}_1\underline{\beta}_1 + \underline{\epsilon}_1 \qquad vs \qquad H_1 : \underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon}$$

And this is equivalent to take:

$$C = \begin{bmatrix} \underline{0} & I_p \end{bmatrix}$$

and the hypothesis being:

$$H_0 : C\underline{\beta} = \underline{0} \qquad vs \qquad H_1 : C\underline{\beta} \neq \underline{0}$$

By Pythagoras, it can be seen that this is equivalent to comparing the distances in the projections:

$$\hat{\underline{\epsilon}}_1^T\hat{\underline{\epsilon}}_1 - \hat{\underline{\epsilon}}^T\hat{\underline{\epsilon}} = SS_{res}(\mathbb{Z}_1) - SS_{res}(\mathbb{Z})$$

And reject if this amount is large.

Example.

Following the previous example, specifically, if:

$$H_0 : \beta_1 = \cdots = \beta_r = 0 \qquad vs \qquad H_1 : \exists\beta_i \neq 0$$

Then:

$$\mathbb{Z}_1 = \begin{bmatrix} 1 \\ 1 \\ \dots \\ 1 \end{bmatrix}$$

Then:

$$\hat{\underline{\epsilon}}_1^T\hat{\underline{\epsilon}}_1 = \sum_{i=1}^{n}\hat{\epsilon}_{1i}^2 = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

So, by decomposition of variance in the in *:

$$\underline{\hat{\epsilon}}_1^T \underline{\hat{\epsilon}}_1 - \underline{\hat{\epsilon}}^T \underline{\hat{\epsilon}} = \sum_{i=1}^{n} (y_i - \bar{y})^2 - \sum_{i=1}^{n} \hat{\epsilon}_{1i}^2 \stackrel{*}{=} \sum_{i=1}^{n} (\hat{y}_i - y)^2$$

And we reject the hypothesis:

$$\frac{\frac{\sum_{i=1}^{n} (\hat{y}_i - y)^2}{r}}{\frac{\underline{\hat{\epsilon}}^T \underline{\hat{\epsilon}}}{n-(r+1)}} \geq F_{1-\alpha}(r, n-(r+1))$$

This usually appears in the software and indicates whether one of the variables is significant or not. It does not indicate whether the fit of the model is good or not.

14.14 Prediction

Let be a new unit $\underline{z}_0$, we may want to predict y_0:

$$y_0 = \underline{z}_0^T \underline{\beta} + \epsilon_0$$

We can only estimate:

$$E[y_0|\underline{z}_0] = \underline{z}_0^T \underline{\beta}$$

And the obvious estimator is $\underline{z}_0^T \underline{\beta}^T$, where:

$$E[\underline{z}_0^T \underline{\hat{\beta}}] = \underline{z}_0^T \underline{\beta} \qquad \wedge \qquad Var[\underline{z}_0^T \underline{\hat{\beta}}] = \sigma^2 \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0$$

In particular, if $\underline{\epsilon} \sim N(\underline{0}, \sigma^2 I)$:

$$\underline{z}_0^T \underline{\hat{\beta}} = N\left(\underline{z}_0^T \underline{\beta}, \sigma^2 \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0\right)$$

In this case:

$$CI_{1-\alpha}(E[y_0|\underline{z}_0]) = CI_{1-\alpha}(\underline{z}_0^T \underline{\beta}) = \left[\underline{z}_0^T \underline{\hat{\beta}} \pm t_{1-\frac{\alpha}{2}}(n-(r+1)) \sqrt{S^2 \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0}\right]$$

With this, we can calculate the one-at-a-time interval of every point in the space (see figure 13).
Instead, if we want a curve where the model is contained with confidence $1-\alpha$, we need to use simultaneous CI (see figure 13):

$$Sim\, CI_{1-\alpha}(E[y_0|\underline{z}_0]) = CI_{1-\alpha}(\underline{z}_0^T \underline{\beta}) = \left[\underline{z}_0^T \underline{\hat{\beta}} \pm \sqrt{(r+1) F_{1-\alpha}(r+1, n-(r+1))} \sqrt{S^2 \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0}\right]$$

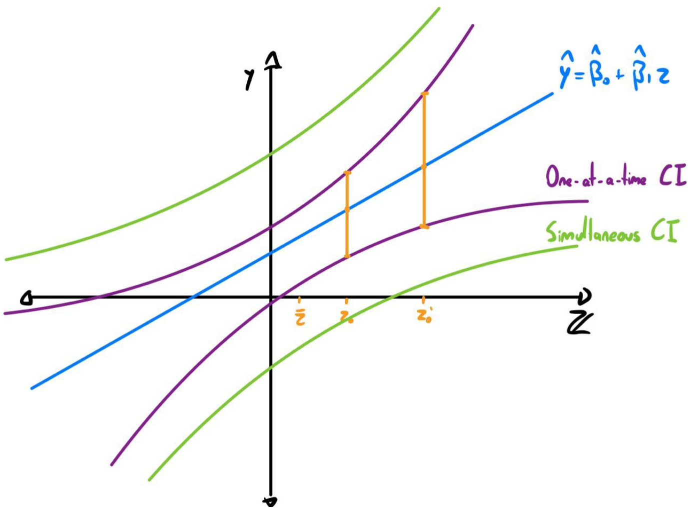

Figure 13: Example: Linear Regression, with one-at-a-time CI and Simultaneous CI

Note that in the previous cases, we have a confidence interval for the mean of y_0 ($E[y_0]$). If we want:

$$P[y_0 \in I_0 | \underline{z}_0] = 1 - \alpha$$

Note that:

$$y_0 \sim N(\underline{z}_0^T \underline{\beta}, \sigma^2)$$

And $y_0 \perp\!\!\!\perp \underline{z}_0^T \hat{\underline{\beta}}$, so:

$$y_0 - \underline{z}_0^T \hat{\underline{\beta}} \sim N(0, \sigma^2 \left[1 + \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0\right])$$

Then:

$$\frac{\frac{y_0 - \underline{z}_0^T \hat{\underline{\beta}}}{\sigma^2[1 + \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0]}}{\frac{S^2}{\sigma^2}} = \frac{y_0 - \underline{z}_0^T \hat{\underline{\beta}}}{S^2[1 + \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0]} \sim t(n - (r+1))$$

(Since the upper term $\sim N(0,1)$, and the $\frac{\hat{\underline{\epsilon}}^T \hat{\underline{\epsilon}}}{\sigma^2} \sim \chi^2(n - (r+1))$) So, the prediction interval for y_0 is:

$$CI_{1-\alpha}(y_0 | \underline{z}_0) = \left[\underline{z}_0^T \hat{\underline{\beta}} \pm t_{1-\frac{\alpha}{2}}(n - (r+1)) \sqrt{S^2 \left[1 + \underline{z}_0^T (\mathbb{Z}^T \mathbb{Z})^{-1} \underline{z}_0\right]}\right]$$

Again, note that this is one-at-a-time, if we want it to be simultaneous, we have to follow a similar process as above.

Theorem 14.6: Gauss-Markov Theorem

$\underline{z}_0^T \hat{\underline{\beta}}$ is the minimum variance estimator among the unbiased estimators of $\underline{z}_0^T \underline{\beta}$ that are linear functions of $\underline{y}$.

Proof. It won't be done □

14.15 Residual Analysis

The creation of a model consists of data, model, and assumptions, making an estimation procedure. This consists of estimates, confidence, statements, and finally conclusions. We have been centering on this second step.

Going back, we need to test the assumptions we have made so far, doing diagnostics and model selection.

The model consists of:

$$\underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon}$$

where $\underline{\epsilon} \in \mathbb{R}^n$.
The fitted model consists of:

$$\underline{\hat{y}} = \mathbb{Z}\underline{\hat{\beta}} + \underline{\hat{\epsilon}}$$

But $\underline{\hat{\epsilon}} \in \mathcal{L}^{\perp}(\mathbb{Z}) \in \mathbb{R}^{n-(r+1)}$.
Where (if $\epsilon \sim 0, \sigma I$) $E[\underline{\hat{\epsilon}}] = 0$ and $Var[\underline{\hat{\epsilon}}] = \sigma^2(I - H)$.
Actually, if $\underline{\epsilon} \sim N(\underline{0}, \sigma I)$:

$$\underline{\hat{\epsilon}} \sim N(\underline{0}, \sigma^2(I - H))$$

We have to check this Gaussianity, and correlation as well (Durbin-Watson test), we cannot assume it.
Indeed, if we graph $\underline{\hat{\epsilon}}$ against $\underline{\hat{y}}$, we should see a point cloud. In practice, we don't see this. We can see the following situations:

- Seeing heteroscedastic errors, where the variance increases as $\underline{\hat{y}}$ increases. To solve this. we can:
  - Transforming features of $\underline{y}$.
  - Model $\underline{\epsilon}$ where Cov($\underline{\epsilon}$)= $\sigma\Sigma$, with Σ different as identity.
  - Linear Mixed Models (see section 15).
- Finding outliers, points that are clearly outside the point cloud. In this case, we could eliminate them or find a reasonable treatment.
- Finding two different clusters. We should use a model that take into consideration this, or fit two different models.
- Finding polynomial (or general known functions) patterns. In this case, we could modify $\mathbb{Z}$, adding columns with the data powered by j (z^j).

14.16 Leverage

Now, note that $Var(\hat{\epsilon}_i) = \sigma(1 - h_{ii})$, where h_{ii} is called leverage:

$$h_{ii} = diag_i(H) = diag_i(\mathbb{Z}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T)$$

To analyze the residuals without considering the difference in variance in each variable, consider the studentized residuals:

$$-\frac{\hat{\epsilon}_i}{\sqrt{S^2(1-h_{ii})}}$$

And we can repeat the analysis using this type of residual. In this case, the leverage points, points have less leverage on the residuals.

14.17 Influencial cases

Let be the training set:

$$[Z][y] = \begin{bmatrix} \underline{z}_1^T \\ \underline{z}_2^T \\ \dots \\ \underline{z}_n^T \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}$$

The fitted model will be $\underline{\hat{\beta}} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y} \in \mathbb{R}^{r+1}$.
Now, let be the training set without the unit i:

$$[Z_{(-i)}][y_{(-i)}] = \begin{bmatrix} \underline{z}_1^T \\ \underline{z}_2^T \\ \dots \\ \underline{z}_{i-1} \\ \underline{z}_{i+1} \\ \dots \\ \underline{z}_n^T \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_{i-1} \\ y_{i+1} \\ \dots \\ y_n \end{bmatrix}$$

Now, the fitted model will be $\underline{\hat{\beta}}_{(-i)} = (\mathbb{Z}_{(-i)}^T\mathbb{Z}_{(-i)})^{-1}\mathbb{Z}_{(-i)}^T\underline{y}_{(-i)} \in \mathbb{R}^{r+1}$.
We can analyze the difference (or distance) between those estimators to measure the influence of the point. This is done with the Cook's distance:

$$D(\underline{\hat{\beta}}, \underline{\hat{\beta}}_{(-i)}) = D_i = \frac{\left(\underline{\hat{\beta}} - \underline{\hat{\beta}}_{(-i)}\right)^T \mathbb{Z}^T\mathbb{Z} \left(\underline{\hat{\beta}} - \underline{\hat{\beta}}_{(-i)}\right)^T}{S^2(r+1)}$$

Actually, remember that the confidence region of $\underline{\hat{\beta}}$ is:

$$CR_{1-\alpha}(\underline{\beta}) = \{\underline{\eta} \quad st \quad D(\underline{\eta}, \underline{\hat{\beta}} \leq F_{1-\alpha}(r+1, n-(r+1))\}$$

So we can see if $\underline{\hat{\beta}}_{(-i)}$ is in this region.

Remark.

> D_i can be rewritten:
>
> $$D_i = \left(\frac{\hat{\epsilon}_i}{\sqrt{S^2(1-h_{ii})}}\right)^2 \frac{h_{ii}}{1-h_{ii}} \cdot \frac{1}{r+1}$$

> The first term is the studentized residual, so if it is high, it indicates that it could be an outlier. The second term accounts for leverage, since as higher, the more it affects the regression

14.18 Collinearity

If $\mathbb{Z}$ columns are "close" to be linear dependent, this may arise a problem. Remember:

$$\underline{\hat{\beta}} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y} \qquad \wedge \qquad Cov(\underline{\hat{\beta}}) = \sigma^2(\mathbb{Z}^T\mathbb{Z})^{-1}$$

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{\sum_{i=1}^n (z_{ij} - \bar{z}_{ij})^2}\frac{1}{1-R_j^2}$$

R_j^2 is the R^2 coefficient of det when Z_j is regressed on $1, Z_1, \ldots, Z_{j-1}, Z_{j+1}, \ldots, Z_r$.

$$Var(\hat{\beta}_j)\downarrow \qquad \text{if } \sum_{i=1}^n (z_{ij} - \bar{z}_j)^2 \uparrow$$

$$Var(\hat{\beta}_j)\uparrow \qquad \text{if } R_j^2 \to 1$$

So we want the variable j to be slightly correlated with the other covariates.
Then, we define the variance inflation factor, which indicates how the variance of the element j increases due to colinearity:

$$VIF_j = \frac{1}{1-R_j^2}$$

Remark.

> A good tool to vanishes this factor is PCA.

14.19 Model Selection

Given $\mathbb{Z} \in \mathbb{R}^{n\times(r+1)}$, with r features, there are:

- 1 regressor with 0 features.
- r regressors with 1 feature.
- $\binom{r}{2}$ regressors with 2 features.
- And so on

In total, there are:

$$\sum_{i=0}^{r}\binom{r}{i} = r^r$$

We can, actually, find the best model extensively:
For every $k = 1, \ldots, r$

1. Fit all the models with k regressors.

2. Choose the best in terms of R^2, and set R_k^2

We can note that:

$$R_1^2 \le R_2^2 \le \cdots \le R_r^2$$

Since we cannot simply use this to choose between the best model among different k's. We can use other metrics, like:

- R_{adj}^2
- AIC
- BIC

14.19.1 Forward Selection

1. Start with the best model with 1 regressor.
2. Add a regressor and find the best model with the past regressors and the best added one.
3. Run an F test to evaluate whether the $\underline{\hat{\beta}} \neq 0$ or not.
4. If $\underline{\hat{\beta}} \neq 0$, add the regressor, if not, finish the loop.

14.20 Non-diagonal Covariance of $\underline{\epsilon}$

OLS assumes that $Cov(\underline{\epsilon}) = \sigma^2 I$.
Assume $W \in \mathbb{R}^{n\times n}$ is positive definite. Generaly, we want:

$$\arg\min_{\underline{\beta}} (y - \mathbb{Z}\underline{\beta})^T W^{-1} (y - \mathbb{Z}\underline{\beta})$$

Note that:

$$\begin{aligned}(y - \mathbb{Z}\underline{\beta})^T W^{-1}(y - \mathbb{Z}\underline{\beta}) &= (y - \mathbb{Z}\underline{\beta})^T W^{-1/2} W^{-1/2}(y - \mathbb{Z}\underline{\beta}) \\ &= (\underbrace{W^{-1/2}\underline{y}}_{y^*} - \underbrace{W^{-1/2}\mathbb{Z}}_{\mathbb{Z}^*}\underline{\beta})^T (W^{-1/2}y - W^{-1/2}\mathbb{Z}\underline{\beta}) \\ &= (y^* - \mathbb{Z}^*\underline{\beta})^T(y^* - \mathbb{Z}^*\underline{\beta}) = \|\underline{y}^* - \mathbb{Z}^*\underline{\beta}\|^2\end{aligned}$$

So with a general variance, the problem becomes a normal OLS, but using different y and $\mathbb{Z}$'s:

$$\underline{\hat{\beta}} = (\mathbb{Z}^{*T}\mathbb{Z}^*)^{-1}\mathbb{Z}^{*T}\underline{y}^* = (\mathbb{Z}^T W^{-1}\mathbb{Z})^{-1}\mathbb{Z}^T W^{-1}\underline{y}$$

And:

$$Cov(\underline{y}^*) = W^{-1/2} Cov(\underline{y}) W^{-1/2} = W^{-1/2} Cov(\underline{\epsilon}) W^{-1/2}$$

Example.

> Consider:
> $$\underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon} \qquad Cov(\underline{\epsilon}) = \sigma^2\Sigma$$
> With σ^2 unknown and Σ known.
> Now, take $W = \Sigma$, and:
> $$\underline{\hat{\beta}} = (\mathbb{Z}^T\Sigma^{-1}\mathbb{Z})^{-1}\mathbb{Z}^T\Sigma^{-1}\underline{y}$$

A special case is the **Weighted Least Squares (WLS)**. If $Sigma = diag(w_1, \dots, w_n)$, i.e., the covariance matrix is diagonal but the values are not weighted by the same value, we can use weighted least squares.
For instance, y_i average of n_i units, so:

$$Var(y_i) = \frac{1}{n_i}\sigma^2 \qquad \wedge \qquad w_i = \frac{1}{n_i}$$

Or, another example, y_i is the sum over n_i units. In this case:

$$Var(y_i) = n_i\sigma^2 \qquad \wedge \qquad w_i = n_i$$

14.21 Colinearity and Variable Selection

Note that the OLS fitted model goes through the training-set barycenter:

$$\underline{\hat{\beta}} = (\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}^T\underline{y} \qquad \text{OLS estimator of } \underline{\beta}$$

So, as $\underline{y} = \mathbb{Z}\underline{\beta} + \underline{\epsilon}$, the fitted model is:

$$y = \underline{z}^T\underline{\hat{\beta}}$$

with:
$$y \in \mathbb{R} \qquad \wedge \qquad \underline{z}^T = \begin{pmatrix} 1 & z_1 & \dots & z_r \end{pmatrix}^T \in \mathbb{R}^{r+1}$$

Now, take:

$$\underline{z}_0 = \begin{pmatrix} 1 & \bar{z}_1 & \dots & \bar{z}_r \end{pmatrix}^T = \frac{\mathbb{Z}\underline{1}}{\underline{1}^T\underline{1}}$$

So:

$$\begin{aligned} y_0 = \underline{z}_0^T\underline{\hat{\beta}} &= \frac{\underline{1}^T\mathbb{Z}}{\underline{1}^T\underline{1}}(\mathbb{Z}^T\mathbb{Z})^{-1}\mathbb{Z}\underline{y} = \frac{\underline{1}^T H\underline{y}}{n} \\ &= \frac{(H^T\underline{1})^T\underline{y}}{n} = \frac{\underline{1}^T\underline{y}}{n} = \bar{y} \end{aligned}$$

Now. let be:
$$\bar{y} = \underline{\hat{\beta}}_0 + \underline{\hat{\beta}}_1\bar{z}_1 + \dots + \underline{\hat{\beta}}_r\bar{z}_r$$

So isolating $\underline{\hat{\beta}}_0$ and replacing in:

$$y = \underline{\hat{\beta}}_0 + \underline{\hat{\beta}}_1 z_1 + \dots + \underline{\hat{\beta}}_r z_r$$

Then:

$$y - \bar{y} = \underline{\hat{\beta}}_1(z_1 - \bar{z}_1) + \cdots + \underline{\hat{\beta}}_r(z_r - \bar{z}_r)$$

So we can see that centering actually consists of, instead of working with $\underline{y}$ and $\mathbb{Z}$, doing it with:

$$\underline{y}^* = \underline{y} - \bar{y} \cdot \underline{1} = \begin{pmatrix} y_1 - \bar{y} \\ y_2 - \bar{y} \\ \dots \\ y_n - \bar{y} \end{pmatrix}$$

$$\mathbb{Z}^* = \begin{bmatrix} z_{11} - \bar{z}_1 & z_{12} - \bar{z}_2 & \dots & z_{1r} - \bar{z}_r & \\ z_{21} - \bar{z}_1 & z_{22} - \bar{z}_2 & \dots & z_{2r} - \bar{z}_r & \\ \dots & \dots & \dots & \dots & \dots \\ z_{n1} - \bar{z}_1 & z_{n2} - \bar{z}_2 & \dots & z_{nr} - \bar{z}_r & \end{bmatrix}$$

With the solution by OLS:

$$\begin{cases} \underline{\hat{\beta}}^* & = \arg\min_{\underline{\beta} \in \mathbb{R}^r} \|\underline{y}^* - \mathbb{Z}^* \underline{\beta}\|^2 \\ \hat{\beta}_0 & = \bar{y} - \hat{\beta}_1^* \bar{z}_1 - \cdots - \hat{\beta}_r^* \bar{z}_r \\ \hat{\beta}_i & = \hat{\beta}_i^* \\ i = 1, \dots, r & \end{cases}$$

From now on, assume $\underline{y}$ and $\mathbb{Z}$ to be centered and then recover $\underline{\hat{\beta}}$ by means of (*)

14.22 Principal Components Regression

Let be the design matrix:

$$\mathbb{Z} = \begin{bmatrix} z_{11} & z_{12} & \dots & z_{1r} \\ z_{21} & z_{22} & \dots & z_{2r} \\ \dots & \dots & \dots & \\ z_{n1} & z_{n2} & \dots & z_{nr} \end{bmatrix} \in \mathbb{R}^{n \times r}$$

Now, we can do PCA on $\mathbb{Z}$ with $k \leq r$:

$$PC_1, PC_2, \dots, PC_k$$

Now, we can use this as a design matrix:

$$\mathbb{Z}^* = \begin{bmatrix} z_{11}^* & z_{12}^* & \dots & z_{1k}^* \\ z_{21}^* & z_{22}^* & \dots & z_{2k}^* \\ \dots & \dots & \dots & \dots \\ z_{n1}^* & z_{n2}^* & \dots & z_{nk}^* \end{bmatrix} \in \mathbb{R}^{n \times k}$$

By construction, the columns are uncorrelated, and:

$$\underline{\hat{\beta}}^* = (\mathbb{Z}^{*T} \mathbb{Z}^*)^{-1} \mathbb{Z}^{*T} \underline{y}$$

And the fitted model:

$$y = \underline{z}^{*T}\underline{\hat{\beta}}^*, \qquad \underline{z}^* = (PC_1, \dots, PC_k)$$

And remember that:

$$\begin{aligned}
PC_1 &= e_{11}z_1 + e_{21}z_2 + \cdots + e_{r1}z_r \\
PC_2 &= e_{12}z_1 + e_{22}z_2 + \cdots + e_{r2}z_r \\
&\dots \\
PC_k &= e_{1k}z_1 + e_{2k}z_2 + \cdots + e_{rk}z_r
\end{aligned}$$

So the fitted model:

$$\begin{aligned}
y = z_1 &\underbrace{(e_{11}\hat{\beta}_1^* + e_{12}\hat{\beta}_2^* + \cdots + e_{1k}\hat{\beta}_k^*)}_{\hat{\gamma}_1} \\
&+z_2 \underbrace{(e_{21}\hat{\beta}_1^* + e_{22}\hat{\beta}_2^* + \cdots + e_{2k}\hat{\beta}_k^*)}_{\hat{\gamma}_2} \\
&\dots \\
&+z_r \underbrace{(e_{r1}\hat{\beta}_1^* + e_{r2}\hat{\beta}_2^* + \cdots + e_{rk}\hat{\beta}_k^*)}_{\hat{\gamma}_r} \\
= \hat{\gamma}_1 z_1 &+ \cdots + \hat{\gamma}_r z_r
\end{aligned}$$

Criticism:

- The model is not sparse, it is the result of the PCA, so we do no model selection.
- There's no guarantee that $PC_1, \dots PC_k$ is the linear combination most correlated with $\underline{y}$. Actually, we can choose a dimensional reduction that vanishes the variation that makes $\underline{y}$ vary across $\mathbb{Z}$.
  To solve this, we can calculate correlations that take into account y, two examples are canonical correlation, and partial least squares.

14.23 Ridge Regression

The ridge regression was created by Hoerl and Kennard in the 70' due to the collinearity problem, which gives a too high variability of $\hat{\beta}$.
The ridge optimization problem finds:

$$\begin{cases} \arg\min_{\underline{\beta}\in\mathbb{R}^r} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 \\ \|\underline{\hat{\beta}}\|^2 \leq s & s > 0 \end{cases}$$

Note that adding $\mathbb{Z}\underline{\hat{\beta}} - \underline{\hat{y}} = 0$ to the equation:

$$\begin{aligned}
\|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 &= \|\underline{y} - \underline{\hat{y}} - \mathbb{Z}(\underline{\beta} - \underline{\hat{\beta}})\|^2 \\
&= \|\underline{\hat{\epsilon}}\|^2 + \|\mathbb{Z}(\underline{\beta} - \underline{\hat{\beta}})\|^2
\end{aligned}$$

As the first term is not controllable, the problem becomes:

$$\begin{cases} \arg\min_{\underline{\beta}\in\mathbb{R}^r} \|\mathbb{Z}(\underline{\beta}-\underline{\hat{\beta}})\|^2 = \arg\min_{\underline{\beta}\in\mathbb{R}^r} (\underline{\beta}-\underline{\hat{\beta}})^T\mathbb{Z}^T\mathbb{Z}(\underline{\beta}-\underline{\hat{\beta}}) \\ \|\underline{\beta}\|^2 \leq s \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad s > 0 \end{cases}$$

$\underline{\hat{\beta}}_{ridge}$ is a shrinked estimator of $\underline{\beta}$.
To find $\underline{\hat{\beta}}_{ridge}$ consider the lagrangian and optimize:

$$\|\underline{y}-\mathbb{Z}\underline{\beta}\|^2 + \lambda\|\underline{\beta}\|^2$$

Where λ is a function of s, that penalizes the high $\underline{\beta}$'s.
To solve this, we use:

$$\underline{\hat{\beta}}_{ridge} = (\mathbb{Z}^T\mathbb{Z}+\lambda I)^{-1}\mathbb{Z}^T\underline{y}$$

Criticism:

- $\underline{\hat{\beta}}_{ridge}$ is a biased estimator, where $E[\underline{\hat{\beta}}_{ridge}] = (\mathbb{Z}^T\mathbb{Z}+\lambda I)^{-1}\mathbb{Z}^T\mathbb{Z}\underline{\beta}$
- H&K proved that:

$$\exists\lambda^* \quad st \quad E[\|\underline{y}-\mathbb{Z}\hat{\beta}_{ridge}(\lambda)\|^2] \leq E[\|\underline{y}-\mathbb{Z}\hat{\beta}_{OLS}\|^2]$$

- It is needed to find the right λ by cross-validation.
- Ridge Regression is not scale invariant, i.e., if we scale the $y's$, the result won't be just scaling the first fit.

  For this reason, it is recommended to standardize $\underline{y}$ and $\mathbb{Z}$ before fitting.
- If variables are standardized and regressors are orthogonal, then $\underline{\hat{\beta}}_{ridge}$ is just $\underline{\hat{\beta}}_{OLS}$ re-scaled. This is a very good property. This occurs due to $\mathbb{Z}^T\mathbb{Z}$ is a diagonal matrix.

We can make the constraint spiky to be more likely to have a solution that will be in one vertex that vanishes a coordinate i ($\hat{\beta}_i = 0$). This is called Lasso, created by Tibshirani in 1996. We must change the problem to:

$$\begin{cases} \arg\min_{\underline{\beta}\in\mathbb{R}^r} \|\underline{y}-\mathbb{Z}\underline{\beta}\|^2 \\ \|\underline{\beta}\|_1 \leq s \qquad\qquad\qquad s > 0 \end{cases}$$

Now, we can solve the Lagrangian:

$$\arg\min_{\underline{\beta}\in\mathbb{R}^d} \|\underline{y}-\mathbb{Z}\underline{\beta}\| + \lambda\|\underline{\beta}\|_1$$

But in this case there's no analytical solution. We can do the variable even more spiky, taking as norm norm $q < 1$:

$$\begin{cases} \arg\min_{\underline{\beta}\in\mathbb{R}^r} \|\underline{y}-\mathbb{Z}\underline{\beta}\|^2 \\ \|\underline{\beta}\|_q \leq s \qquad\qquad\qquad s > 0 \end{cases}$$

We can also add spikiness with a certain regularity. This is called elastic net:

$$\arg\min_{\underline{\beta}\in\mathbb{R}^d} \|\underline{y} - \mathbb{Z}\underline{\beta}\| + \lambda_1\|\underline{\beta}\|_1 + \lambda_2\|\underline{\beta}\|^2$$

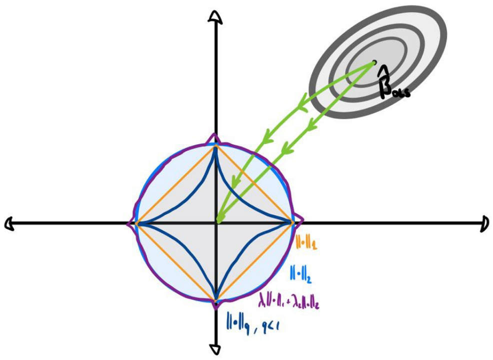

Figure 14: Example: Ridge Regression considering different types of norm. Green lines represent the orthogonal and non-orthogonal covariate cases.

Anyway, as a general model, we optimize:

$$\arg\min_{\underline{\beta}\in\mathbb{R}^d} \|\underline{y} - \mathbb{Z}\underline{\beta}\|^2 + \lambda \cdot \text{penalization}$$

Or more generally:

$$\arg\min_{\underline{\beta}\in\mathbb{R}^d} \|\underline{y} - f(\mathbb{Z})\|^2 + \lambda \cdot \text{penalization}$$

15 Linear Mixed Models (LMM)

This section will be based on Linear Mixed-Effects Model Using R: A Step-by-Step Approach, by Andrezej Galecki and Tomasz Burzykowski (Galecki & Burzykowski, 2013).

15.1 Introduction

A very important assumption of regression models is the independence and homoscedasticity of the observations. Anyway, this assumptions are not common in real world. For this reason, a more general model is needed to account for more complex structures.

Modeling general dependence and heteroscedasticity among observations concerns the design of the variance and covariance matrix. Linear Mixed Models (LMM), also called random effects models or multilevel/hierarchical models, are generalization of traditional regression models.

LMM adds a random component to the linear predictor, taking into account the contribution of different kind of dependencies among observations and denoise information contents, focusing on inference related to the presence of groups.

15.2 Classical Linear Model

Remember that, for independent observations with a homogeneous variance, we can use the classical LM. The model is usually notated as follows:

$$\underline{y} = \mathbb{X}\underline{\beta} + \underline{\epsilon}, \qquad \epsilon \sim N(0, R), \quad R = \sigma^2 I_n \in \mathbb{R}^{n \times n}$$

With the properties:

$$E[y_j] = \mu_j = \underline{x}_j^t \underline{\beta}$$
$$Var[y_j] = Var[\epsilon_j] = \sigma^2$$

The estimation for this problem can be done with OLS (even not needing normality assumption, only uncorrelated residual errors):

$$\underline{\hat{\beta}}_{OLS} = (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\underline{y}$$

And:

$$\hat{\sigma}^2_{OLS} = \frac{1}{n-(p+1)}(\underline{y} - \mathbb{X}\underline{\hat{\beta}}_{OLS})^T(\underline{y} - \mathbb{X}\underline{\hat{\beta}}_{OLS})$$

But in the normal case, we can do it with Maximum Likelihood as well, having the same result in the estimate of β:

$$\underline{\hat{\beta}}_{ML} = (\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\underline{y}$$

But a different and biased estimator for σ:

$$\hat{\sigma}^2_{ML} = \frac{1}{n}(\underline{y} - \mathbb{X}\underline{\hat{\beta}}_{ML})^T(\underline{y} - \mathbb{X}\underline{\hat{\beta}}_{ML}) = \frac{1}{n}\sum_{j=1}^{n}(\underline{y}_j - x\underline{\hat{\beta}}_{ML})$$

Anyway, if we use a Restricted Maximum Likelihood (REML), we can obtain an unbiased σ^2, having:

$$\hat{\sigma}^2_{REML} = \frac{1}{n-(p+1)}\sum_{j=1}^{n} r_j^2$$

15.3 Relaxing Homoscedasticity Assumption

We can now relax the homoscedasticity assumption, allowing for heteroscedastic observations. Still, we hold the independence and normal assumptions, but we now have LMs with heterogeneous variance.
Now, we relax the assumption so:

$$\epsilon_j \sim N(0, \sigma_j^2), \qquad \underline{\epsilon} \sim N(0, \underline{\sigma}^2 I), \quad \underline{\sigma}^2 = \begin{pmatrix}\sigma_1^2 & \dots\sigma_n^2\end{pmatrix}^T I$$

So now:

$$V[y_j] = \sigma_j^2$$

Plus, the model contains $n+p+1>n$ parameters, so the model is not identifiable. Now, we need to impose additional constraints on the residual variances to make it identifiable. The additional constraints are given by how we choose the variance, which is done with the Variance Function.

15.4 Variance Function

It is a way to represent variances more parsimoniously as a function of a small set of parameters:

$$\lambda^2(\underline{\delta}, \mu; \underline{v})$$

where the function is continuous and differentiable wrt $\underline{\delta}$ and:

- $\underline{v}_j$ is a vector of known covariates defining the variance function for observation j
- $\underline{\delta}$ contains a small set of variance parameters, common to all observations

In this case, the variance is:

$$V[\epsilon_j] = \sigma^2\lambda^2(\delta, \mu_j; v_j)$$

Definitely, the model errors would be:

$$\epsilon \sim N(0, R)$$

Where:

$$R = \sigma^2\Lambda$$

$$\Lambda = diag(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^{n\times n}$$

$$\lambda_j^2 = \lambda^2(\underline{\delta}, \mu_j; \underline{v}_j)$$

There are different types of variance functions:

1. Known weights:

   $$\lambda(\cdot) = \lambda(v)$$

   The estimation algorithm is called WLS.

2. Variance functions depending on δ but not on μ, noted as $\langle\delta\rangle-$group:

   $$\lambda(\cdot) = \lambda(\delta; v)$$

   The estimation algorithm is ML or REML.

3. Variance functions depending on δ and μ, noted as $\langle\delta, \mu\rangle-$group:

   $$\lambda(\cdot) = \lambda(\delta, \mu; v)$$

   The estimation algorithm is ML or REML-based GLS.

4. Variance functions depending on μ but not on δ, noted as $\langle\mu\rangle-$group:

$$\lambda(\cdot) = \lambda(\mu; v)$$

The estimation algorithm is IRLS.

The different algorithms can be found in (Galecki & Burzykowski, 2013).

15.5 Profiled Likelihood

Gne log-likelihood for the model using a variance function is:

$$\ell_{full} = (\underline{\beta}, \sigma^2, \underline{\delta}) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2}\sum_{j=1}^{n}\log(\lambda_j^2) - \frac{1}{2\sigma^2}\sum_{j=1}^{n}\lambda_j^{-2}(\underline{y}_j - \underline{x}_j^T\underline{\beta})^2$$

Note that $\lambda = \lambda(\underline{\delta})$ and that $\lambda = 1$ goes back to the canonical case.
To optimize this function, the usual procedure is to maximize wrt β for every value of $\underline{\delta}$, and get $\underline{\hat{\beta}}(\underline{\delta}) \quad wrt\ \underline{\beta}$, so we have: $\ell^*_{ML}(\sigma^2, \underline{\delta})$ and then, maximize wrt σ^2 to get $\hat{\sigma^2}_{ML}(\underline{\delta})$, and then, as $\underline{\delta}$ are supposedly known, obtain the maximum likelihood:

$$l^*_{ML}(\underline{\delta})$$

15.6 Relaxing Independence Assumption

Consider now more general Linear Models. Relax the assumptions of independence, and build LMs for correlated data.
Now, we can introduce correlated data for data clustered in N groups, each one with n_i elements. Then, for the group i:

$$y_i = \mathbb{X}_i\underline{\beta} + \epsilon_i \qquad \epsilon_i \sim N_{n_i}(0, \mathcal{R}_i), \quad \mathcal{R}_i = \sigma^2 R_i$$

In this case:

$$E[y_{ij}] = \mu_{ij} = z_{ij}^T\underline{\beta} \qquad \wedge \qquad V[y_i] = V[\epsilon_i] = \sigma^2 R_i$$

Example.

> n individuals receive MRI scan in the same hospital. 8 different scans are available, and people are randomly assigned to them.

15.7 Correlation Function

The correlation function models the correlation matrix, so it adds more restrictions and the system becomes unique.
The correlation function is as follows:

$$Corr(\epsilon_{ij}, \epsilon_{ij}^T) = h[d(t_{ij}, t_{ij}, \mathcal{Q})]$$

Where:

- $\mathcal{Q}$ is a vector of correlation parameters.
- d is a distance function.
- h is a continuous function with respect to $\mathcal{Q}$, ranging between -1 and 1, and subject to $h(0, \mathcal{Q}) = 1$.

The correlation structures can be classified in two main groups:

1. **Serial Structures:** Correlation structures which are defined in the context of time-series or longitudinal data. For example, autocorrelation.
2. **Spatial Structures:** Correlation structures which are defined in the context of spatial data. For example, semiovariogram.

The estimation in these cases is carried out by WLS or likelihood based methods mentioned before.

15.8 Disengage group effect

To disengage the effect of groups in the model, we can:

- Create categorical variables: The disadvantages are that it greatly increases the model complexity. Plus, it does not consider elements that are not in the dataset, we may want to analyze new groups.
- Add a random effect to the model

One approach that does not work the best is to separate each model for each group, since the elements may not be enough to perform the inference, and it becomes the models non-comparable, so we can't make inference on functions of two different groups. Averaging over groups does not work well either, since there's information loosing.
For hierarchical data with a single level of grouping ($i = 1, \ldots, N$), with n_i observations by group, we can formulate the classical LMM at a given level of a grouping factor as follows:

$$y_i = \underline{X}_i \underline{\beta} + Z_i \underline{b}_i + \epsilon_i, \qquad b_i \sim N_q(\underline{0}, D), \quad \epsilon_i \sim N_{n_i}(\underline{0}, R_i)$$

Where: $\underline{b}_i \perp \epsilon_i$ On the other hand, the model for the whole data is:

$$y = \mathbb{X}\underline{\beta} + \mathbb{Z}\underline{b} + \underline{\epsilon}, \qquad \underline{b} \sim N_{N(q+1)}(0, \sigma_b^2 D), \quad \underline{\epsilon} \sim N_n(\underline{0}, \sigma_\epsilon^2, R)$$

15.9 Unconditional Distribution of Random Effects

The unconditional distribution of the random effect is a multivariate normal distribution with zero mean and variance-covariance matrix D:

$$D(\sigma^2, \underline{\theta}_D) = \sigma^2 D(\underline{\theta}_D)$$

Where $\underline{\theta}_D$ is a vector of parameters, which represent the scaled σ^2 variances and covariances of elements of b.
There are two cases in θ_D

- **General Case:** Any two elements of $\underline{b}$ can be correlated, and there are no restrictions on the matrix D, besides it is positive-definite and symmetric.
  There are $q(q+1)/2$ elements in the matrix, q variances and $q(q-1)/2$ covariances. Although q is typically small, estimating all the parameters may be difficult if the sample size n is small.
- **Simplified Structure:** All elements on the vector $\underline{b}_i$ are independent, but there are q different variances (one for group).

15.10 Conditional Distribution of Random Effects

For classical LMMs, the conditional distribution $f_{y|b}(\underline{y}_i|\underline{b}_i)$ is a multivariate normal, with:

$$E[\underline{y}_i|\underline{b}_i] = \underline{\mu}_i = X_i\underline{\beta} + Z_i\underline{b}_i, \qquad \underline{\mu}_i = \begin{pmatrix}\mu_{i1} & \dots & \mu_{in_i}\end{pmatrix}$$
$$V[\underline{y}_i|\underline{b}_i] = V[\underline{\epsilon}_i|\underline{b}_i] = \sigma^2 R_i$$

Therefore, mean value of the dependent variable is defined by a linear combination of the fixed effects and random effects, respectively.

In the most general form, LMMs are not identifiable, so constraints like the following are needed:

$$V[\epsilon_{ij}|\underline{b}_i] = \sigma^2 R_{ij} = \sigma^2\lambda^2(\mu_{ij}, \underline{\delta}; \underline{v}_{ij})$$

The marginal distribution of $f_y(\underline{y}_i)$ of the observations is obtained by "integrating out" the random effects:

$$f_y(\underline{y}_i) = \int f_{y,b}(\underline{y}_i, \underline{b}_i) d\underline{b}_i = \int f_{y,b}(\underline{y}_i|\underline{b}_i) f_b(\underline{b}_i) d\underline{b}_i$$

Given that $f_{y,b}(\underline{y}_i, \underline{b}_i)$ and $f_b(\underline{b}_i)$ are densities of multivariate normal distributions, the marginal distribution of $\underline{y}$ is also a multivariate normal, and it can be derived analytically:

$$E[\underline{y}_i] = X_i\underline{\beta} \qquad \wedge \qquad V[\underline{y}_i] = \sigma^2[Z_i D Z_i^T + R_i] = \sigma^2[Z_i D(\underline{\theta}_D) Z_i^T + R_i(\underline{\theta}_R, v_i)]$$

The marginal mean value is defined by a linear combination of the vectors of covariates, as for the LMs.

On the other hand, the marginal covariance matrix, the global variance, consists of two components:

- The random effects
- The residuals

15.11 Estimation

Definitely, the model fitting has 3 different components:

- Estimates of fixed effects: How the entire population behaves.
- Estimates of random effects: How each group deviates from the fixed effects.
- Estimates of variance parameters: Variability that is not specific among groups, including residual variance.

Remark.

> Variance components are non-negative by definition. Nevertheless, methods for estimating them may underestimate variance values close to 0, leading to negative estimates. This could happen when the number of random effects (q) is small, or when the number of observations per random effect (n_i) is small.

Let be the simplest model (same variance and uncorrelated):

$$y_{ij} = \beta_0 + \beta_1 X_{j1} + \cdots + \beta_p X_{jp} + b_i + \epsilon_{ij}, \qquad i = 1, \ldots, N, \quad j = 1, \ldots, n_i$$

Where $b_i \sim N(0, \sigma_b^2) \quad \forall i$.
Profiled confidence intervals (a confidence interval where we focus on only one parameter) for the fixed effects, random effects, and variance parameters can be calculated. Specifically for random effects, we can know when they are different from 0. To compute the amount of variability the presence of the groups accounts for, we can use the Proportion of Variance due to Random Effect (PVRE):

$$\text{PVRE} = \frac{\hat{\sigma}_b^2}{\hat{\sigma}_b^2 + \hat{\sigma}_\epsilon^2}$$

We can use the estimate of σ_b^2 to perform scenario analysis, for example, what if belonging to a $-2\hat{\sigma}_b^2$ group (or $+2\hat{\sigma}_b^2$ group), given the same unit conditions.

We may cluster random effect estimates of similar groups and use the group medoids as representative elements for quantifying the effect of the communities of groups.

16 Ensemble Methods

16.1 The problem

Let be a supervised problem and its model:

$$y = f(\underline{x}) + \epsilon, \qquad E[\epsilon] = 0, \epsilon \perp\!\!\!\perp \underline{x}$$

Let be a fitted model $\hat{f}_{\mathbb{X}}$.
Recall the bias-variance tradeoff: Then, given $\underline{x}_0 \in \mathbb{R}^p, \underline{x}_0 \notin \mathbb{X} \quad st \quad :$

$$y_0 = f(\underline{x}_0) + \epsilon_0$$

Then, the expected prediction error is:

$$E[(y_0 - \hat{f}_{\mathbb{X}}(\underline{x}_0))^2] = \underbrace{[f(\underline{x}_0) - E[f_{\mathbb{X}}(\underline{x}_0)]]^2}_{bias^2} + \underbrace{Var(\hat{f}_{\mathbb{X}}(x_0))}_{Var} + \underbrace{Var(\epsilon_0)}_{\text{irreductible}}$$

And remember the bias-variance trade-off.

Example.

> CART's tends to overfit the data. Then, it has low bias and high variance.

Our objective is to keep the bias low and reduce the variance of the predictor.

16.2 Ensemble Technique

Illustrate the ensemble technique with the following example:

Example.

> $$\mathbb{X} = \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix}, y_1, \dots, y_n \ iid \sim \mu, \sigma^2$$
>
> Then, a model $y = \mu + \epsilon$, makes for each $i = 1, \dots, n$:
>
> $$E[y_i] = \mu$$
>
> Then, define the predictor $f_{\mathbb{X}}(x_0) = y_1$, it is unbiased, and:
>
> $$(\mu - E[f_{\mathbb{X}}])^2 + Var(f_{\mathbb{X}}) + Var(\epsilon_0) = 0 + \sigma + \sigma = 2\sigma$$
>
> Now take a different predictor $f_{\mathbb{X}} = \frac{1}{n}\sum_{i=1}^n y_i = \bar{y}_n$. It is still unbiased, and:
>
> $$E[(y_0 - f_{\mathbb{X}})^2] = \frac{1}{n}\sigma^2 + \sigma^2 \leq 2\sigma$$
>
> This implies that if $n \to \infty$, $E[(y_0 - f_{\mathbb{X}})^2] \to \sigma$

Then the idea is that if we have independent copies of the same predictor:

$$\hat{f}_{\mathbb{X}_1}, \hat{f}_{\mathbb{X}_2}, \dots, \hat{f}_{\mathbb{X}_B} \ iid \sim \hat{f}_{\mathbb{X}}$$

Then $f_B = \frac{1}{B}\sum_{b=1}^{B}\hat{f}_{\mathbb{X}_b}$:

$$E[\hat{f}_B] = E[\hat{f}_{\mathbb{X}}] \qquad \wedge \qquad Var[\hat{f}_B] = \frac{1}{B}Var[\hat{f}_{\mathbb{X}}]$$

Now, note that:

$$E[(y_0 - \hat{f}_B(\underline{x}_0))^2] = [f(\underline{x}_0) - E[\hat{f}_{\mathbb{X}}(x_0)]]^2 + \frac{1}{B}Var(\hat{f}_{\mathbb{X}}(\underline{x}_0)) + Var(\epsilon_0) \leq E[(y_0 - \hat{f}_{\mathbb{X}}(x_0))^2]$$

This is done by having training sets:

$$\mathbb{X}_1, \dots, \mathbb{X}_B \in \mathbb{R}^{n\times(p+1)}$$

The simplest option to obtain these datasets is to repeat n experiments, by generating $\mathbb{X}$ B times, but this is very expensive.
Anyway, in this case, we have to evaluate what's better, doing a single experiment with all this data, or effectively splitting it:

$$\hat{f}_{\mathbb{X}_1\cup\dots\cup\mathbb{X}_B} \qquad vs \qquad \frac{1}{B}\sum_{b=1}^{n}\hat{f}_{\mathbb{X}_b}$$

What is really usual, is using bootstrap.

16.3 Bagging (Bootstrap and Aggregating)

Bootstrap is an algorithm introduced by Efron in 1979. It consists of sampling units from $\mathbb{X}$ with replacement: $(\underline{x}_i^*, y_i^*)$, n times, so we generate:

$$\mathbb{X} = \begin{bmatrix}\underline{x}_1^* \\ \dots \\ \underline{x}_n^*\end{bmatrix} = \begin{bmatrix}y_1^* \\ \dots \\ y_n^*\end{bmatrix}$$

Instead, Bagging is an algorithm introduced by Breiman in 1994. It solves the problem when we have, from the beginning, only one training set $\mathbb{X}$. It consists of generating by bootstrap B training sets:

$$\mathbb{X}_1^*, \dots, \mathbb{X}_B^*$$

Then, the predictor proposed by this method is:

$$\hat{f}_B = \frac{1}{B}\sum_{i=1}^{B}\hat{f}_{\mathbb{X}_i^*}$$

The problem with this method is that our variance reduction lies in the independence of the samples. In this case, the samples are not independent.
Let be $\mathbb{X}_1^*, \dots, \mathbb{X}_B^*$, indeed, fix a unit i in $\mathbb{X}$: $(\underline{x}_i, y_i)$, then:

$$P[(\underline{x}_i, y_i) \in \mathbb{X}^*] = 1 - P[(\underline{x}_i, y_i) \notin \mathbb{X}^*] = 1 - (1 - \frac{1}{n})^n \to 1 - e^{-1}$$

So the expected proportion of data shared by $\mathbb{X}_i$ and $\mathbb{X}_j$ is $1 - \frac{1}{e} \approx 0.63$ for n large.

16.4 Random Forest (RF)

It is an algorithm introduced by Breiman in 2001. It reduces the variance of the predictor of Bagging by doing a feature selection (Plus the unit selection of Bagging).

When introducing a CART, at each split, we choose a feature X^* among $(X_1, \dots, X_p)$ and a threshold such that it maximizes the decrease in the Average Evaluation Function/Gini (see section 14.2).

When building a tree of an RF, instead, at each split we choose X^* with the same criteria. but among $(X_1^*, \dots, X_m^*)$, randomly selected from $(X_1, \dots, X_p)$. In other words we don't look for the best split in every column, but in a selection of columns, that on top of that changes in every split.

We can choose m in different ways:

- m=p: It is bagging.
- In regression problems, it is usual to choose
$$m = \frac{1}{3}p$$
- In classification problems, it is usual to choose:
$$m = \sqrt{p}$$

The random forest predictor is:

- For Regression:
$$\hat{f}_B(\underline{x}_0) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_{\mathbb{X}_b^*}(\underline{x}_0)$$
- For classification:
$$\hat{f}_B(\underline{x}_0) = \text{mode}\left\{\hat{f}_{\mathbb{X}_1^*}(\underline{x}_0), \dots, \hat{f}_{\mathbb{X}_B^*}(\underline{x}_0)\right\}$$

16.5 Random Forest Explainability

The problem with RF is that it is not that easy to explain why the model predicted what it predicted.
An option to explain it is use **Variable Importance:** Every time X_i is used for generating a split. Record the reduction in RSS/Gini, and eventually, the sum of these reductions will create an index for X_i.

Another option is **Shapley Value:** It was introduced by Shapley in 1951 in the field of game theory. The idea consists of having $\{1, \dots, p\}$ players, and defining the set of coalitions:

$$\mathcal{P} = \{S \quad st \quad S \subseteq \{1, \dots, p\}\}$$

Let $V : \mathcal{P} \to \mathbb{R}$ be a worth function. It is said $V(S)$ is the worth of coalition S and $V(i)$ the worth of a player, $i \in \{1, \dots, p\}$.
If S is a coalition and $i \notin S$ a player, we want to know how the worth of the coalition increases due to the addition of i to S:

$$V(S \cup \{i\}) - V(S)$$

And how many coalitions with $|S|$ without i can we make:

$$p\binom{p-1}{|S|} = \frac{p!}{|S|!(p-|S|-1)!}$$

Then, for each player $i \in \{1, \dots, p\}$, we can calculate the shapley value:

$$\phi_i = \sum_{S \subseteq \{1,\dots,p\} \setminus \{i\}} \frac{p!}{|S|!(p-|S|-1)!} \left(V(S \cup \{i\}) - V(S)\right)$$

It has some properties:

1. **Efficiency:**
$$\sum_{i=1}^{p} \phi_i = V(\{1, \dots, p\})$$

2. **Null Player:**
$$\text{If } V(S \cup \{i\}) - = V(S) \quad \forall S \in \mathcal{P} \qquad \Longrightarrow \qquad \phi_i = 0$$

3. **Symmetry:** Let be $i, j \in \{1, \dots, p\}$ and
$$V(S \cup \{i\}) = V(S \cup \{j\}) \quad \forall S \in \mathcal{P} \qquad \Longrightarrow \qquad \phi_i = \phi_j$$

4. **Linearity:** Let be $v; w : \mathcal{P} \to \mathbb{R}$ worth features, and $a, b \in \mathbb{R}$:
$$\phi_i^{av+bw} = a\phi_i^v + b\phi_i^w \qquad \forall i = 1, \dots, p$$

In the context of statistics, the players are features. Then, the prediction through $\hat{f}$ is $\hat{f}_{\mathbb{X}|S}$, a predictor using only features belong to $S \in \{1, \dots, p\}$.

So, given $\underline{x}_0 \in \mathbb{R}^p$ and $y_0 = f(\underline{x}_0) + \epsilon$, we need to specify:

$$V(S) = \hat{f}_{\mathbb{X}|S}(\underline{x}_0|S) - \bar{y}$$

Note that $\bar{y}$ is the prediction part that can be done without the features.

Finally, the contribution of X_i to predict $\hat{f}_{\mathbb{X}}(\underline{x}_0)$ becomes:

$$\phi_i = \sum_{S\subseteq\{1,\ldots,p\}\setminus\{i\}} \frac{p!}{|S|!(p-|S|-1)!}\left(\hat{f}_{\mathbb{X}|S\cup\{i\}}(\underline{x}_0) - \hat{f}_{\mathbb{X}|S}(\underline{x}_0)\right)$$

Anyway, estimating these values is extremely expensive. There are some algorithms used:

- SHAP (S. Lundberg & Lee, 2017)
- TreeSHAP: Specific for random forest (S. M. Lundberg, Erion, & Lee, 2019)

Remark.

> For a fixed feature i, we can calculate $\phi_i(\underline{x}_k)$ for every $k = 1, \ldots, n$ unit in the dataset. This can give an idea of how the Shapley value changes over different units. Different Shapley values could indicate that it is a very important feature for the prediction.
> We can normalize these numbers with $\sum_{k=1}^{n} |\phi_i(\underline{x}_k)|$ to compare them.

Remark.

> More details can be found in Interpretable Machine Learning: A Guide for Making Black Box Models Explainable, C. Molnar (2025). It can be found on GitHub for free.

17 Spatial Statistics

17.1 The Problems

- The **Geostatistical Data** consists of $D \subseteq R^d, d = 1, 2, 3, 4$ a space. Inside this space, there are fixed sampling locations $s_1, \ldots, s_n$. There are $Z_{s_1}, \ldots, Z_{s_n}$ observations of each location. This observations have a distribution:

  $$\{z_s, \quad s \in D\} \qquad (*)$$

  We will focus on the Euclidean domain, since it is the simplest case.
  The typical goal is to characterize the distribution of (*), which involves estimating the mean and the spatial dependence of the data.

- The **Lattice Data** or **Areal Data** consists of $D \subseteq R^d, d = 1, 2, 3, 4$ a space. D is partitioned into $s_1, \ldots, s_n \subseteq D$. Z_{s_i} are the values observed on the area s_i

- The **Spatial Point Processes** consists of $D \subseteq R^d, d = 1, 2, 3, 4$ a space. Inside this space, there are random locations $s_1, \ldots, s_n$. In some cases, there are observations Z_{s_i} at each s_i. In particular:
  - **Standard Spatial Point Process:** If only $s_1, \ldots, s_n$ are available
  - **Marked Spatial Point Process:** If $Z_{s_1}, \ldots, Z_{s_n}$ are available with $s_1, \ldots, s_n$

17.2 Geostatistics

Geostatistics most important goal is to estimate $\{Z_s, \quad s \in D\}$.
We say that a process Z_s is stationary if there is regularity and homogeneity of distributions across the domain, formally:

Definition 17.1: Strong Stationarity

Let be $s_1, \ldots, s_n \in D$. Then a process $\{Z_s\}$ is strong stationary if $\forall h \in \mathbb{R}^d$:

$$(Z_{s_1}, Z_{s_2}, \ldots, Z_{s_n})^T \sim (Z_{s_1+h}, Z_{s_2+h}, \ldots, Z_{s_n+h})^T$$

Assume that:

$$E[Z_s] < \infty \qquad \wedge \qquad Var[Z_s] < \infty \qquad \forall s \in D$$

Define:

Definition 17.2: Second Order Stationarity

Let D be a domain and C a function. Then a process $\{Z_s\}$ is second order stationary if:

- $E[Z_s] = m \qquad \forall s \in D$
- $Cov(Z_{s_i}, Z_{s_j}) = C(s_1 - s_2) \qquad \forall s_i, s_j \in D$

Remark.

> As a consequence:$Var(Z_s) = C(\underline{0})$ is constant in space.

17.3 Covariogram

The function $C : \mathbb{R}^d \to \mathbb{R}$ that models the variance is called the covariogram.

It has some properties:

- C is a positive semi-definite function: $\forall \lambda_i, \lambda_j \in \mathbb{R}, \forall s_i, s_j \in D$:

$$\sum_i \sum_j \lambda_i \lambda_j C(s_i - s_j)$$

- C is a symmetric function: $\forall h \in \mathbb{R}^d$:

$$C(-h) = C(h)$$

- C is bounded: $\forall h \in \mathbb{R}^d$:

$$|C(h)| \leq C(\underline{0})$$

Structural Properties of the covariogram (see also figure15):

- **Sill:** $= \lim_{\|h\| \to \infty} \gamma(\|h\|)$
- **Range:** $R \quad st \quad \gamma(\|h\|) = C(\underline{0}) \quad \forall \|h\| \geq R.$

  In practice, it is used 90 or 95% of $C(\underline{0})$
- **Nugget:** $\tau^2 = \lim_{\|h\| \to 0^+} \gamma(\|h\|)$

  Note that there is a discontinuity at 0 if $\tau^2 > 0$

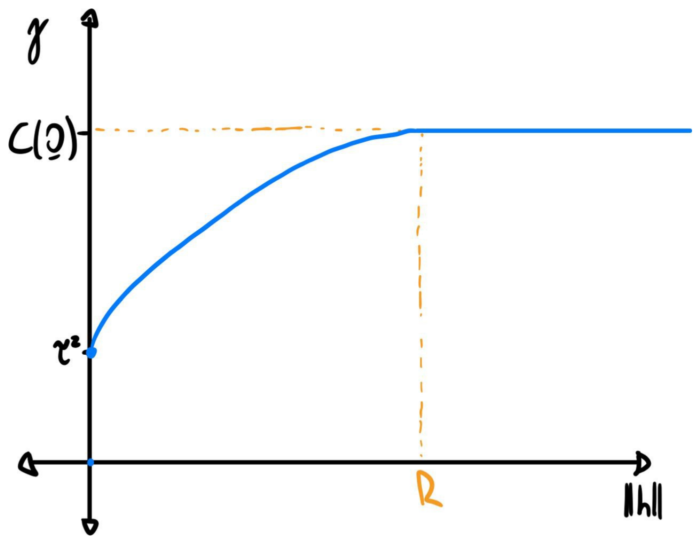

Figure 15: Example: Semivariogram and its elements

17.4 Variogram

Definition 17.3: Isotropy

Let C be the covariogram function. A process $\{Z_s \in D\}$ second order stationary is isotropic if:

$$Cov(Z_{s_1}, Z_{s_2}) = C(\|s_1 - s_1\|) \qquad \forall s_1, s_2 \in D$$

Definition 17.4: Semivariogram

Assume 2nd order stationarity and Isotropy The semiovariogram is defined as:

$$\gamma(h) := Var(Z_{s_1} - Z_{s2}), \qquad \forall s_1, s_2 \in D \quad st \quad h = \|s_1 - s_2\|$$

Definition 17.5: Variogram

The variogram is defined as $2\gamma(s_1 - s_2)$

Remark.

$$2\gamma(s_1 - s_2) := Var(Z_{s_1}) + Var(Z_{s_2}) - 2Cov(Z_{s_1}, Z_{s_2}) = 2C(\underline{0}) - 2C(s_1 - s_2)$$

Then, the semivariogram can be defined as:

$$\gamma(h) = C(\underline{0}) - C(h)$$

Proposition 17.6

The semivariogram satisfies the following properties:

- Conditional negative definite: $\forall s_i, s_j \in D, \quad \forall \lambda_i, \lambda_j \in \mathbb{R} \quad st \quad \sum_i \lambda_i = 0$

$$\sum_i \sum_j \lambda_i \lambda_j \gamma(s_i - s_j) \leq 0$$

- Symmetry: $\forall h \in \mathbb{R}^d \qquad \gamma(h) = \gamma(-h)$
- Non-negative: $\forall h \in \mathbb{R}^d \qquad \gamma(h) \geq 0$
- Null at 0: $\gamma(\underline{0}) = 0$
- Subquadratic growth:

$$\lim_{\|h\| \to \infty} \frac{\gamma(h)}{\|h\|^2} = 0$$

Proposition 17.7

Under 2nd order stationarity and isotropy:

$$2\gamma(h) = E[(Z_{s_1} - Z_{s_2})^2] - \underbrace{(E[Z_{s_1}] - E[Z_{s_2}])^2}_{E[Z_{s_1}] = E[Z_{s_2}]} = E[(Z_{s_1} - Z_{s_2})^2]$$

17.5 Variogram Empirical Estimator under conditions

As we have seen, spatial dependence can be modeled with the variogram. Then, to estimate spatial dependence, our goal is to estimate $2\gamma(h)$. We will do it **under second-order stationarity and isotropy conditions.**

Remark.

$$\gamma(h) = C(0) - C(h) \implies C(h) = C(0) - \gamma(h) = \lim_{\tilde{h} \to \infty} \gamma(\tilde{h}) - \gamma(h)$$

We will use an empirical estimator:

Fix $h = h_0$. Define:

$$N(h_0) = \{(s_i, s_j) \quad st \quad h_0 = \|s_i - s_j\|\}$$

Now estimate:

$$2\hat{\gamma}(h_0) = \frac{1}{|N(h_0)|} \sum_{(s_i,s_j)\in N(h_0)} (Z_{s_i} - Z_{s_j})^2$$

In practical applications, it is unlikely to have a representative number of elements for each h, so it is calculated by balls or radius ϵ. This is called binned variogram.

17.6 Valid Model for Empirical Estimate

Suppose the semiovariogram is modeled by a parametric valid model $\gamma(h;\theta)$.
The goal is to find $\theta \in \Theta$ that best fits

$$\underline{\hat{\gamma}} = (\hat{\gamma}_1, \ldots, \hat{\gamma}_L)^T \qquad \hat{\gamma}_i = \hat{\gamma}(h_i), i = 1, \ldots, L$$

For instance:

$$\hat{\theta} = \arg\min_{\theta\in\Theta} \sum_{\ell=1}^{L} w_\ell(\hat{\gamma}_\ell - \gamma(h_\ell;\theta))$$

Now, some typical parametric families:

- **Pure Nugget:** Let $\tau^2 > 0$

$$\gamma(h;\tau) = \begin{cases} 0 & h = 0 \\ \tau^2 & h > 0 \end{cases}$$

- **Exponential model:** Let $\sigma^2 > 0, a > 0$:

$$\gamma(h;\sigma^2 > 0, a > 0) = \begin{cases} 0 & h = 0 \\ \sigma^2(1 - e^{-\frac{h}{a}}) & h > 0 \end{cases}$$

- **Spherical model:** Let $\sigma^2 > 0, a > 0$:

$$\gamma(h;\sigma^2 > 0, a > 0) = \begin{cases} 0 & h = 0 \\ \sigma^2 & 0 < h \leq a \\ \sigma^2\left(\frac{3}{2}\frac{h}{a} - \frac{1}{2}\left(\frac{h}{a}\right)^3\right) & a > 0 \end{cases}$$

- Mastérn Model: It uses a ν smoothness parameter.

Proposition 17.8

The valid models have the following properties:

- If γ_1, γ_2 are both valid $\implies \gamma_1 + \gamma_2$ is valid.
- If $c > 0$ andγ_1 is valid $\implies c\gamma_1$ is valid.

Remark.

Recall that $\lim_{\|h\|\to\infty} \frac{\gamma(\|h\|)}{\|h\|^2} = 0$, so the variogram cannot grow faster than the parabola. Anyway, the empirical variogram can grow faster.
In this case, under the previous assumptions, we estimate the variogram by estimating:

$$\gamma(h) = E[(Z_{s_1} - Z_{s_2})^2]$$

Instead, if stationarity does not hold, we estimate:

$$\gamma(h) = Var(Z_{s_1} - Z_{s_2}) = E[(Z_{s_1} - Z_{s_2})^2] - (E[Z_{s_1}] - E[Z_{s_2}])^2$$

Note that if we do not consider the second part, there will be a bias equal to the second part of the equation. If this grows fast, it can be the reason why the empirical variogram is growing faster than the parabola.
So when we estimate $2\hat{\gamma}$:

$$E[(Z_{s_1} - Z_{s_2})^2] = 2\gamma(h) + (E[Z_{s_1}] - E[Z_{s_2}])^2$$

17.7 Krigging

Let $\{Z_s, s \in D\}, D \subseteq \mathbb{R}^d, d = 2, 3$ a process. Let $Z_{s_1}, \ldots, Z_{s_n}$ observations. Let $s_0 \in D$ be a target location, the objective now is to predict the value of this location.
Then, we need to find a measurable function $f(\underline{Z})$ with $\underline{Z} = \begin{pmatrix} Z_{s_1} & \ldots & Z_{s_n} \end{pmatrix}$ that best approximate Z_{s_0} in the sense:

$$E[(Z_{s_0} - f(\underline{Z}))^2]$$

So we take:

$$f^*(\underline{Z}) = E[Z_{s_0} | \underline{Z}]$$

This predictor has the following properties:

- **Unbiasedness:**
$$E[E[Z_{s_0} | \underline{Z}]] = E[Z_{s_0}]$$
- **Orthogonality:**
$$Cov\,(E[Z_{s_0} | \underline{Z}] \quad , \quad Z_{s_0} - E[Z_{s_0} | \underline{Z}]) = 0$$
- **Interpolation:**
$$s_0 \in \{s_1, \ldots, s_n\} \implies E[Z_{s_0} | \underline{Z}] = Z_{s_0}$$

Plus, adding the Gaussianity assumption:

- **Linearity on data:**

$$\exists \lambda_0, \dots, \lambda_n \in \mathbb{R} \qquad E[Z_{s_0} | \underline{Z}] = \lambda_0 + \sum_{i=1}^{n} \lambda_i Z_{s_i}$$

  In particular, the chosen parameters must be linear.

Krigging looks with the **best linear unbiased predictor (BLUP)**:

$$Z_{s_0}^* = \lambda_0^* + \sum_{i=1}^{n} \lambda_i^* Z_{s_i}$$

By the following optimization problem:

$$\lambda^* = \arg\min_{\lambda} E\left[\left(Z_{s_0} - \left(\lambda_0 + \sum_{i=1}^{n} \lambda_i Z_{s_i}\right)\right)^2\right] \quad st \quad E\left[\lambda_0 + \sum_{i=1}^{n} \lambda_i Z_{s_i}\right] = E[Z_{s_0}]$$

Under the assumption of the covariance (or variogram) known ($Cov(Z_{s_1}, Z_{s_2}) \quad \forall s_1, s_2 \in D$)

Remark.

> Note that this is a different optimization process than OLS.

Anyway, we simplify the problem by adding extra assumptions. There are different ones we can add:

- Simple Kriging: Known mean:

$$E[Z_s] = m_s \quad \forall s \in D$$

- Ordinary Kriging: Second order assumptions and isotropy:

$$E[Z_s] = m \quad \forall s \in D \qquad \wedge \qquad Cov(Z_{s_1}, Z_{s_2}) = C(\|s_1 - s_2\|)$$

- Universal Kriging: Non-stationarity:

$$E[Z_s] = m_s \quad \forall s \in D \qquad \wedge \qquad Cov(Z_{s_1}, Z_{s_2}) = C(\|s_1 - s_2\|)$$

17.8 Ordinary Kriging

By linearity:

$$E\left[\lambda_0 + \sum_{i} \lambda_i Z_{s_i}\right] = E[Z_{s_0}]$$

So with ordinary kriging, to have uniform unbiasedness, we need:

$$\lambda_0 + \sum_i \lambda_i m = m$$

That is satisfied if:

$$\begin{cases} \lambda_0 = 0 \\ \sum_{i=1}^n \lambda_i = 1 \end{cases}$$

Now, to solve the optimization problem under these assumptions, we use Lagrange:

$$\phi(\underline{\lambda}, \xi) := E\left[\left(Z_{s_0} - \sum_{i=1}^n \lambda_i Z_{s_i}\right)^2\right] + 2\xi\left(\sum_{i=1}^n \lambda_i - 1\right)$$

We can rewrite this problem:

$$\begin{aligned}
\phi(\underline{\lambda}, \xi) &= Var\left[Z_{s_0} - \sum_{i=1}^n \lambda_i Z_{s_i}\right] + 2\xi\left(\sum_{i=1}^n \lambda_i - 1\right) \\
&= Var[Z_{s_0}] + Var\left[\sum_{i=1}^n \lambda_i Z_{s_i}\right] - 2Cov\left(Z_{s_0} \quad , \quad \sum_i \lambda_i Z_{s_i}\right) + 2\xi\left(\sum_{i=1}^n \lambda_i - 1\right) \\
&= C(0) + \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j C(\|s_i - s_j\|) - 2\sum_i \lambda_i C(\|s_0 - s_i\|) + 2\xi\left(\sum_i \lambda_i - 1\right)
\end{aligned}$$

Then we take:

$$\begin{aligned}
\frac{\partial \phi}{\partial \lambda_i} &= 2\sum_{j=1}^n \lambda_j C(\|s_i - s_j\|) - 2C(\|s_0 - s_i\|) + 2\xi \qquad\qquad i = 1, \dots, n \\
\frac{\partial \phi}{\partial \xi} &= 2(\sum_i \lambda_i - 1)
\end{aligned}$$

And with the first-order conditions are:

$$\begin{cases} \frac{\partial \phi}{\partial \lambda_i} = 0 \\ \frac{\partial \phi}{\partial \xi} = 0 \end{cases} \implies \begin{cases} \sum_{j=1}^n \lambda_j C(\|s_i - s_j\|) + \xi = C(\|s_0 - s_i\|) \\ \sum_i \lambda_i = 1 \end{cases}$$

And as a system:

$$\begin{bmatrix} \Sigma & 1 \\ \underline{1} & 0 \end{bmatrix} \begin{pmatrix} \underline{\lambda} \\ \xi \end{pmatrix} = \begin{pmatrix} \underline{\sigma}_0 \\ 1 \end{pmatrix}$$

Where:

- $\Sigma_{ij} = C(\|s_i - s_j\|) \quad \forall i, j = 1, \dots, n$
- $\sigma_{0,i} = C(\|s_0 - s_j\|) \quad \forall i = 1, \dots, n$

17.9 Universal Kriging (UK)

The universal Kriging does not take stationary assumptions. Then, we need to model:

$$Z_s = m_s + \delta_s, \qquad m_s = E[Z_s] \quad \wedge \quad \delta_s = Z_s - m_s$$

m_s is called mean or drift, and δ_s is called residual (Note $E[\delta_s] = 0$).

Plus, we assume

1. We assume a linear model for m_s:

$$m_s = \sum_{\ell=0}^{L} a_\ell f_\ell(s), \qquad s \in D, \qquad f_0(s) = 1$$

   Where $\{f_\ell\}$ is a regressor depending on $s \in D$, and they are known everywhere (at least in $s_1, \dots, s_n$ and s_0).

   Remark.

   If $L = 0 \implies m_s = m \quad \forall s$

2. δ_s is a second-order statistic with known covariogram C (under isotropy)

To have this, universal kriging finds:

$$Z^*_{s_0} = \lambda^*_0 + \sum_{i=1}^{n} \lambda^*_i Z_{s_i}$$

With the following optimization problem, kriging, but now with the previous assumptions:

$$\lambda^* = \arg\min_\lambda E\left[\left(Z_{s_0} - \left(\lambda_0 + \sum_{i=1}^{n} \lambda_i Z_{s_i}\right)\right)^2\right] \quad st \quad E\left[\lambda_0 + \sum_{i=1}^{n} \lambda_i Z_{s_i}\right] = E[Z_{s_0}]$$

We can reframe unbiasedness ($E[\lambda_0 + \sum_{i=1}^{n} \lambda_i Z_{s_i}] = E[Z_{s_0}]$) for every possible value of m_S in the model:

$$\lambda_0 + \sum_{i=1}^{n} \lambda_i \left(\sum_{\ell=0}^{L} a_\ell f_\ell(s_i)\right) = \sum_{\ell=0}^{L} a_\ell f_\ell(s_0)$$
$$\lambda_0 + \sum_{\ell=0}^{L} a_\ell \left(\sum_{i=1}^{n} \lambda_i f_\ell(s_i)\right) = \sum_{\ell=0}^{L} a_\ell f_\ell(s_0)$$

So the solution must satisfy: $\begin{cases} \lambda_0 = 0 \\ \sum_i \lambda_i f_\ell(s_i) = f_\ell(s_0), \quad \ell = 0, \dots, L \end{cases}$

Now, using Lagrange with these conditions:

$$\phi(\underline{\lambda}, \underline{\xi}) = E\left[\left(Z_{s_0} - \sum_{i=1}^{n} \lambda_i Z_{s_i}\right)^2\right] + 2\sum_{\ell=0}^{L} \xi_\ell \left(\sum_{i=1}^{n} \lambda_i f_\ell(s_i) - f_\ell(s_0)\right)$$

$$= C(0) + \sum_i \sum_j \lambda_i \lambda_j C(\|s_i - s_j\|) - 2\sum_i \lambda_i C(\|s_0 - s_i\|) + 2\sum_{\ell=0}^{L} \xi_\ell \left(\sum_{i=1}^{n} \lambda_i f_\ell(s_i) - f_\ell(s_0)\right)$$

Therefore, solving the first-order equations, the Universal Kriging system is:

$$\begin{cases} 2\sum_j \lambda_j C(\|s_i - s_j\|) + 2\sum_\ell \xi_\ell f_\ell(s_0) = 2C(\|s_0 - s_i\|), & i = 1, \dots, n \\ \sum_i \lambda_i f_\ell(s_i) = f_\ell(s_0), \quad \ell = 0, \dots, L \end{cases}$$

In a matrix form:

$$\begin{bmatrix} \Sigma & \mathbb{F} \\ \mathbb{F} & \underline{0} \end{bmatrix} \begin{pmatrix} \underline{\lambda} \\ \underline{\xi} \end{pmatrix} = \begin{pmatrix} \underline{\sigma}_0 \\ \underline{f}_0 \end{pmatrix}$$

where:

- Σ is the covariance of the observations:

$$(\Sigma)_{ij} = C(\|s_i - s_j\|)$$

- $\mathbb{F}$ is the design matrix of the linear model:

$$\mathbb{F}_{i\ell} = f_\ell(s_i), \quad i = 1, \dots, n, \quad \ell = 0, \dots, L$$

- $\underline{f}_0$ is the design vector at s_0:

$$f_{0,\ell} = f_\ell(s_0)$$

17.9.1 Estimating variogram without stationarity

At the beginning, we assumed having a known variogram (see item 2). Nonetheless, our method for estimating the variogram (see section 17.5) was based on stationarity assumptions.

To estimate our covariogram, we need the residuals of the fixed model. Anyway, we can't fit the model without the known covariogram. We can see that there is a circular dependence on these problems.

Note that we know how to get $\hat{\gamma}, \hat{C}, \hat{\Sigma}$ under stationarity. In this case, we can't use the same estimators on $Z_{s_1}, \dots, Z_{s_n}$ since we don't have stationarity anymore.

Now, we could estimate γ, C, Σ from $\hat{\delta}_{s_1}, \dots, \hat{\delta}_{s_n}$.

Model:

$$\begin{cases} Z_{s_i} = \sum_{\ell=0}^{L} a_\ell f_\ell(s_i) + \delta_{s_i}, & i = 1, \dots, n \\ \underline{Z} = F\underline{a} + \underline{\delta}, \quad Cov(\underline{\delta}) = \Sigma \end{cases}$$

And it can be noted that we can solve it with GLS:

$$\begin{aligned}
\underline{\hat{a}} &= \arg \min_{\underline{a} \in \mathbb{R}^{L\times 1}} (\underline{Z} - \mathbb{F}a)^T \Sigma^{-1} (\underline{Z} - \mathbb{F}a) \\
&= \arg \min_{\underline{a} \in \mathbb{R}^{L\times 1}} \underline{\hat{\delta}}^{a^T} \Sigma^{-1} \underline{\hat{\delta}}^{a} \\
&= (\mathbb{F}^T \Sigma^{-1} \mathbb{F})^{-1} \mathbb{F}^T \Sigma^{-1} \underline{Z}
\end{aligned}$$

But, we need a heuristic since there are no $\hat{\delta}_{s_1}, \ldots, \hat{\delta}_{s_n}$:

Algorithm 3 Iterative Procedure for Estimating Variogram

Require: $Z_{s_1}, \ldots, Z_{s_n}$
Ensure: $\hat{\gamma}$
  Initialize: $\hat{a}^{OLS} \to \hat{\delta}^{OLS} \to \hat{\Sigma}$
  for i=1,..,n **do**
    Estimate $\underline{a}$ as $\underline{\hat{a}} = \underline{\hat{a}}^{GLS}$ knowing $\hat{\Sigma}$
    Get $\hat{\Sigma}$ from $\underline{\hat{\delta}} = \underline{Z} - \mathbb{F} - \underline{\hat{a}}$
  end for
  return $\hat{\gamma}$

18 Functional Data Analysis

18.1 Introduction

Functional Data is data that can model a function. It consists of a sample of functional observations. Even though observations are discrete and affected by noise, the data reflect a smooth variation of the phenomenon.

18.2 Notions of Hilbert Spaces

Hilbert Spaces generalize the concept of Euclidean Space to any, even infinite, dimension, having a vectorial structure, angles, projections, and distance, among other structures. An example is L^2 space.

Remark.

> Hilbert Spaces allow Pythagoras' theorem.

18.3 Functional Data

Let H be a hilbert space with functions on a fixed closed interval $T = [t_{\min}, t_{\max}]$

Definition 18.1: Functional Random Variable

A functional random variable is a random element defined on a probability space with values in H: $f : \Omega \to H$

Definition 18.2: Functional Datum

A functional datum is a realization of a functional random variable. i.e., given $\omega \in \Omega$:

$$x = X(\omega) \quad st \quad T = [t_{\min}, t_{\max}] \to \mathbb{R}$$

Definition 18.3: Functional Dataset

A functional dataset is a collection of functional data

From now, let $X : \Omega \to H$ be a functional random variable.

Definition 18.4: Frechet Mean

The Frechet mean of X is the unique element $\mu \in H$ that solves:

$$\arg \inf_{x \in H} E[\|X - x\|_H^2]$$

Remark.

> If $H = L^2$ the frechet mean coincides with the point-wise mean: $E[X(t)] = \mu(t) \quad \forall t \in T$

Remark.

> In any H, we can estimate the frechet mean via the sample estimator:
>
> $$\bar{X} = \frac{1}{n} \sum_i^n X_i$$

From now, assume $E[\|X\|_H^4] < \infty$

Definition 18.5: Covariance Operator

The Covariance operator of X, $C : H \to H$ is defined as:

$$Cx = E[\langle X, x \rangle X], \qquad x \in H$$

Remark.

If $H = L^2$, the covariance operator can be defined equivalently through a kernel operator:

$$[Cx](t) = \int_T c(s,t)x(s)\ ds, \qquad x \in L^2$$

Where the covariance kernel is precisely the point-wise covariance:

$$c(s,t) = E[X(s)X(t)]$$

18.4 Smoothing

Let $Y \in \mathcal{F}$ be a functional variable. The smoothing model consists of a non parametric model:

$$Y_i = f(t_i) + \epsilon_i \qquad \epsilon_i \sim indep \qquad (0, \sigma^2)$$

Let $\{(Y_1, t_1), \dots, (Y_n, t_n)\}$ be a sample.
Then, the estimation problem is:

$$\hat{f} = \arg\min_{f \in \mathcal{F}} RSS(f) = \arg\min_{f \in \mathcal{F}} \left\{ \sum_{i=1}^{n} (Y_i - f(t_i))^2 \right\}$$

But this problem is not well-posed. To solve this, we can:

- Restrict search to F_K , with $dim(F_K) = K \ll n$:

$$\hat{f} = \arg\min_{f \in \mathcal{F}_K} RSS(f) = \arg\min_{f \in \mathcal{F}_K} \left\{ \sum_{i=1}^{n} (Y_i - f(t_i))^2 \right\}$$

- Do not restrict, but add a roughness penalty:

$$\hat{f} = \arg\min_{f \in \mathcal{F}} RSS(f) = \arg\min_{f \in \mathcal{F}} \left\{ \sum_{i=1}^{n} (Y_i - f(t_i))^2 + \mathcal{P}_f \right\}$$

18.5 Approximating with finite dimension

$$\begin{aligned}\hat{f} &= \arg\min_{f \in \mathcal{F}_K} RSS(f) = \arg\min_{f \in \mathcal{F}_K} \left\{ \sum_{i=1}^{n} (Y_i - f(t_i))^2 \right\} \\ &= \arg\min_{c \in \mathbb{R}^k} RSS(c) = \arg\min_{c \in \mathbb{R}^k} \left\{ \sum_{i=1}^{n} (Y_i - \sum_{j=1}^{K} c_j \psi_j(t_i))^2 \right\}\end{aligned}$$

Where $\psi_i, \dots, \psi_K$ are K basis functions such that $F_K = span(\psi_1, \dots, \psi_K)$.
Finally, the smoothing model is:

$$Y_i = f(t_i) + \epsilon_i \qquad \Longrightarrow \qquad Y_i = \sum_{j=1}^{K} c_j \psi_j(t_i) + \epsilon_i$$

There are different bases we can use. We use B-spline basis: It is computationally efficient. It also has local support and the band-limited structure of key matrices.
The points we define these functions (called knots) can be placed with the percentiles (or equally spaced).
The matrix formulation is:

$$\underline{Y} = \underline{f} + \underline{\epsilon} = \Psi c + \underline{\epsilon}, \qquad \underline{f} = \underline{\psi}^T \underline{c}, \quad E[\underline{Y}] = 0, \quad Var[\underline{Y}] = \sigma^2 I_n$$

where:

- $\Psi = \begin{bmatrix} \psi_1(t_1) & \psi_2(t_1) & \dots & \psi_K(t_1) \\ \psi_1(t_2) & \psi_2(t_2) & \dots & \psi_K(t_2) \\ \dots & \dots & \dots & \dots \\ \psi_1(t_n) & \psi_2(t_n) & \dots & \psi_K(t_n) \end{bmatrix} \qquad \underline{\Psi} = (\psi_1, \dots, \psi_K)^T$
- $\underline{Y} = (Y_1, \dots, Y_n)^T$
- $\underline{f} = (f(t_1), \dots, f(t_n))^T$
- $\underline{c} = (c_1, \dots, c_K)^T$
- $\underline{\epsilon} = (\epsilon_1, \dots, \epsilon_n)$

And the estimators are:

$$\begin{aligned}
&\hat{\underline{c}} = \arg\min_{\underline{c}\in\mathbb{R}^K} RSS(\underline{c}) = \arg\min_{\underline{c}\in\mathbb{R}^K} \left\{(\underline{Y} - \Psi\underline{c})^T(\underline{Y} - \Psi\underline{c})\right\} = (\Psi^T\Psi)^{-1}\Psi^T\underline{Y} \\
&\hat{\underline{f}} = \Psi^T\hat{\underline{c}} \\
&\hat{\underline{Y}} = \hat{\underline{f}} = \Psi\hat{\underline{c}} = \Psi(\Psi^T\Psi)^{-1}\Psi^T\underline{Y} = S\underline{Y} \\
&S = \Psi(\Psi^T\Psi)^{-1}\Psi^T \qquad \text{(Projection Matrix, with property } S^TS = S) \\
&df = K = \text{trace}(S) = \text{trace}(S^TS) = \text{trace}(2S - S^TS) \\
&\hat{\sigma}^2 = \frac{1}{n-K}(\underline{Y} - \hat{\underline{Y}})^T(\underline{Y} - \hat{\underline{Y}}) = \frac{1}{n - \text{trace}(S)}(\underline{Y} - \hat{\underline{Y}})^T(\underline{Y} - \hat{\underline{Y}})
\end{aligned}$$

Some properties of the estimators are:

$$\begin{aligned}
\hat{f}(t) &= \underline{\psi}(t)^T\hat{\underline{c}} = \psi(t)^T(\Psi^T\Psi)^{-1}\Psi^T\underline{Y} \\
E[\hat{\underline{f}}(t)] &= E\left[\psi(t)^T(\Psi^T\Psi)^{-1}\Psi^T\underline{Y}\right] = \psi(t)^T(\Psi^T\Psi)^{-1}\Psi^T\Psi\underline{c} = \underline{\psi}(t)^T\underline{c} \\
Bias[\hat{\underline{f}}(t)] &= f(t) - E[\hat{\underline{f}}(t)] = f(t) - \underline{\psi}(t)^T\underline{c} \\
Var[\hat{\underline{f}}(t)] &= E[(\hat{f}(t) - E[\hat{f}(t)])^2] = \sigma^2\underline{\psi}(t)^T(\Psi^T\Psi)^{-1}\underline{\psi}(t) \\
\hat{Var}[\hat{\underline{f}}(t)] &= \hat{\sigma}^2\underline{\psi}(t)^T(\Psi^T\Psi)^{-1}\underline{\psi}(t)
\end{aligned}$$

The number of bases K controls the model bias-variance trade-off. K can be selected by AIC, C_p, mallows, or cross-validation.
It can also be selected by generalized cross-validation:

Definition 18.6: GCV

The Generalized Cross Validation (GCV(K)) is:

$$\text{GCV}(K) = \frac{n}{n-K}\frac{1}{n-K}(\underline{Y} - \underline{\hat{Y}})^T(\underline{Y} - \underline{\hat{Y}}) = \frac{n}{(n - \text{trace}(s))^2}(\underline{Y} - \underline{\hat{Y}})^T(\underline{Y} - \underline{\hat{Y}})$$

The idea is to choose the value of K close to that minimizing:

$$MSE[\hat{f}(t)] = E[(\hat{f}(t) - f(t))^2] = Bias^2[\hat{f}(t)] + Var[\hat{f}(t)]$$

Under regularity conditions, as $n \to \infty$ and $K(n) \to \infty$:

$$Bias^2[\hat{f}(t)] \to 0 \quad \wedge \quad Var[\hat{f}(t)] \to 0 \quad \wedge \quad MSE[\hat{f}(t)] \to 0 \quad \wedge \quad \hat{f}(t) \approx \text{Gaussian}$$

If we limit to the gaussian distribution, we can do CI and tests on f(t).

Remark.

> At the most basic version CIs are computes one-at-a-time, and even more, for a specific t.

18.6 Approximating with Penalization

The second approach consists of do not restrict $\mathcal{F}$, so $\dim(\mathcal{F}) = K \approx n$, but add a roughness penalty:

$$\begin{aligned}\hat{f} &= \arg\min_{f\in\mathcal{F}} RSS_\lambda(f) = \arg\min_{f\in\mathcal{F}_K}\left\{\sum_{i=1}^{n}(Y_i - f(t_i))^2 + \lambda\mathcal{P}(f)\right\}\\ &= \arg\min_{\underline{c}\in\mathbb{R}^K}\left\{(\underline{Y} - \Psi\underline{c})^T(\underline{Y} - \Psi\underline{c}) + \lambda\underline{c}^T P\underline{c}\right\}, \qquad P \in \mathbb{R}^k \times \mathbb{R}^k\end{aligned}$$

Theorem 18.7

If (a,b) s.t. $a < t_{[1]} < t_{[2]} < \cdots < t_{[n]} < b$, $\mathcal{F} = H^2(a,b)$, and $\mathcal{P}(f) = \int_a^b (f''(t))^2\, dx$, then $\hat{f}$ is a natural cubic splines over (a,b) with knots at $t_{[1]}, \ldots, t_{[n]}$

Remark.

> For large datasets, it may be convenient to use a mixed approach, reducing the dimensionality to a still large K, with a roughness penalty.

Expressions of the estimators are:

$$\begin{aligned}
&\underline{\hat{c}} = (\underline{\Psi}^T\underline{\Psi} + \lambda P)^{-1}\underline{\Psi}^T\underline{Y} \\
&\underline{\hat{Y}} = \underline{\hat{f}} = \Psi\underline{\hat{c}} = \Psi(\underline{\Psi}^T\underline{\Psi} + \lambda P)^{-1}\underline{\Psi}^T\underline{Y} = S\underline{Y} \\
&S = \Psi(\underline{\Psi}^T\underline{\Psi} + \lambda P)^{-1}\underline{\Psi}^T \qquad \text{Sub-projection operator } (S^TS \neq S) \\
&df = \text{trace}(S) < K \qquad (\text{or } df = \text{trace}(S^TS) \text{ or } df = \text{trace}(2S - S^TS)) \\
&\underline{\hat{\sigma}}^2 = \frac{1}{n - \text{trace}(S)}(\underline{Y} - \underline{\hat{Y}})^T(\underline{Y} - \underline{\hat{Y}})
\end{aligned}$$

Some properties of the estimators are:

$$\begin{aligned}
\hat{f}(t) &= \underline{\psi}(t)^T\underline{\hat{c}} = \underline{\psi}(t)^T(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\Psi^T\underline{Y} \\
E[\underline{f}\hat{(t)}] &= E\left[\underline{\psi}(t)^T(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\Psi^T\underline{Y}\right] = \underline{\psi}(t)^T(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\Psi^T\Psi\underline{c} \\
Bias[\underline{\hat{f}}(t)] &= f(t) - E[\underline{\hat{f}}(t)] = f(t) - \underline{\psi}(t)^T(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\Psi^T\Psi\underline{c} \\
Var[\underline{f}\hat{(t)}] &= \sigma^2\underline{\psi}(t)^T(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\Psi(t)^T\Psi(t)(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\underline{\psi}(t) \\
V\hat{a}r[\underline{f}\hat{(t)}] &= \hat{\sigma}^2\underline{\psi}(t)^T(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\Psi(t)^T\Psi(t)(\Psi^T\Psi + \lambda\mathcal{P})^{-1}\underline{\psi}(t)
\end{aligned}$$

Under regularity conditions, as $n \to \infty$, $K(n) \to \infty$, and $\lambda(n) \to 0$:

$$Bias^2[\hat{f}(t)] \to 0 \qquad \wedge \qquad Var[\hat{f}(t)] \to 0 \qquad \wedge \qquad \hat{f}(t) \approx \text{Gaussian}$$

If we limit to the gaussian distribution, we can do CI and tests on f(t).

18.7 Derivatives Analysis

Smoothing requires special care when the curve estimate is asked, not only to provide a good smoothing of the data, but also to reflect the features of the curve.

We can improve our results by studying the derivatives of curves (or their functions).
We can modify SSE to take into account the derivatives:

$$\text{SSE}_\lambda = SSE + \lambda \int (f^{[d]}(s))^2 \, ds$$

Or even more general:

$$\text{SSE}_\lambda = SSE + \lambda \int (Lf(s))^2 \, ds$$

18.8 Window Smoothings

The first way to use windows to fit functions locally is to fit polynomials in fixed windows separately, while ensuring continuity at the knots.

A more complex way is to fit the polynomial in a moving window, using a kernel to aggregate the overlapping results smoothly:

$$\arg\min_{\underline{c}\in\mathbb{R}^L} \sum_{i=1}^{n} \ker_h(s_0, s_i)[z_i - \sum_{\ell=0}^{L} c_\ell (s_0 - s_i)^\ell]^2, \qquad \text{where } \ker_h(s_0, s_i) = D\left(\frac{|s_0 - s_i|}{h}\right)$$

18.9 Adding constraints

In addition, we may want to model specific type of functions, so in this case we can simply change the formulations:

- **Positive functions:**

$$f(s) = e^{W(s)}, \qquad \text{Where } W(s) = \sum_k c_k \psi_k(s)$$

  Estimating f by minimizing:

$$\sum_{i=1}^{n} (z_i - e^{W(s_i)})^2 + \lambda \int W''(s)\ ds$$

- **Increasing functions:**

$$f(s) = C + \int_{s_0}^{s} e^{W(s)}\ dt, \qquad \text{Where } W(s) = \sum_k c_k \psi_k(s)$$

In these specific cases, we can use a logarithm too.

18.10 Basis adaptive to data

If we want to model the basis adaptive to the data. If the expression of the basis function does not exist, this is a good alternative. We can use one of these options:

- Free-knot regression splines
  - Unidimensional curves (Zhou & Shen, 2001).
  - Multidimensional curves (Sangalli, Secchi, Vantini, & Veneziani, 2009).
- Wavelets
  - Unidimensional curves (Hastie, Tibshirani, & Friedman, 2009)
  - Multidimensional curves (Pigoli & Sangalli, 2012)
- Functional Principal Components Analysis

This is good for modeling sharp local features, or functions in a specific space and frequency location

18.11 Phase Variability

Phase Variability: Different curves exhibit similar features, but at different times or space locations for different statistical units. This regularity has to be taken into account.

We can align similar functions by **Registration/Alignment/Warping:** Finding suitable functions $h_1(t), \ldots, h_N(t)$ such that $c_1(h_1(t)), \ldots, c_2(h_N(t))$ are the most similar.

For example, in measurements that start at random points, it is natural to use $h_i(t) = t + \delta_i$. In other situations, it is natural to use $h_i(t) = \alpha_i t$.

- **Landmark approach** Known landmarks along the curves that are aligned so that landmarks occur at the same abscissa points. e.g., crossing for zero, peaks, valleys, points of inflection.

  Let $c_1, \ldots, c_N, \qquad c_i : [0, T] \to \mathbb{R}^d$. Suppose having L landmarks for the i-th curve, at $t_{i1}, \ldots, t_{iL}$. Let $t_{01}, \ldots, t_{0L}$ landmarks in a template curve c_0 (we can take one or define it as the average of t_{ij}'s). Then, the landmark warping functions for the i-th curve is any increasing function h_i such to:

  - $h_i(0) = 0$
  - $h_i(t_{0j}) = t_{ij} \qquad \forall j = 1, \ldots, L$
  - $h_i(T) = T$

  Then $(0, 0), (t_{0,1}, t_{i1}), \ldots, (t_{0L}, t_{iL}), (T, T)$ are interpolated by a piece-wise line, a polygon or higher order monotone splines.

  A possible model for warping functions is:

  $$h_i(t) = C_{0i} + C_{1i} \int_0^t e^{W_i(u)} \, du$$

  The landmark-based registration may require significant user input and can be sensitive to the accuracy of the landmark identification. Moreover, in some applications it is not possible to identify well-defined features that can be used as landmarks.

- **Continuous Approach** It consists of defining a measure of similarity/dissimilarity between curves that are aligned to maximize/minimize their similarity/dissimilarity. Then, the curves are aligned by warping their space abscissa parameters, choosing the optimal warping function from a function class, such that it minimizes the final distance among the curves.

Remark.

> Instead, in amplitude variability, what changes is the intensity of the signal

18.12 Dissimilarity choose

The problem of decoupling amplitude and phase variability is not univocally defined.

Different measures of similarity and function classes can be used, leading to different results. The choice of these two defines the variability taken into account in the specific problem.

In any case, the dissimilarity space (ρ, W) chosen must satisfy properties that ensure the alignment problem is well-posed and the procedure is coherent:

- ρ must be bounded, reflexive, symmetric, and transitive.
- W must be a convex vector space, with a group structure wrt function composition.
- The pair (ρ, W) must satisfy:

$$\rho(\underline{c}_1, \underline{c}_2) = \rho(\underline{c}_1 \circ h, \underline{c}_2 \circ h) \qquad \forall h \in W$$

  And:

$$\rho(\underline{c}_1 \circ h_1, \underline{c}_2 \circ h_2) = \rho(\underline{c}_1 \circ h_1 \circ h_2^{-1}, \underline{c}_2) = \rho(\underline{c}_1, \underline{c}_2 \circ h_2 \circ h_1^{-1})$$

(ρ, W) defines on the considered set of functions $\mathcal{C}$ a partition in equivalence classes. Some classical (W, ρ)'s are:

- $W = W_{\text{shift}}$

$$\rho = \|c_1 - c_2\|$$

- $W = W_{\text{shift}}$

$$\rho = \|c_1' - c_2'\|$$

- $W = W_{\text{shift}}$

$$\rho = \|(c_1 - \bar{c}_1) - (c_2 - \bar{c}_2)\|$$

- $W = W_{\text{shift}}$

$$\rho = \|(c_1' - \bar{c_1'}) - (c_2 - \bar{c_2'})\|$$

- $W = W_{\text{affinity}}$

$$\rho = \left\| \frac{c_1}{\|c_1\|} - \frac{c_2}{\|c_2\|} \right\|$$

- $W = W_{\text{affinity}}$

$$\rho = \left\| \frac{c_1'}{\|c_1'\|} - \frac{c_2'}{\|c_2'\|} \right\|$$

- $W = W_{\text{diffeomorphism}}$

$$\rho = \|sign(c_1')\sqrt{|c_1'|} - sign(c_2')\sqrt{|c_2'|}\|$$

18.13 Curve Alignment

If a template curve φ is known, it is enough to align each curve to the template.

If the template is unknown, it must be estimated from the data, leading to a complex optimization problem:

$$\text{Find } \underline{\varphi} \in \mathcal{C} \wedge \underline{h} \subseteq W \quad st \quad \frac{1}{N}\sum_{i=1}^{N} \rho(\underline{\varphi}, \underline{c}_i \circ h_i) \geq \frac{1}{N}\sum_{i=1}^{N} \rho(\underline{\psi}, \underline{c}_i \circ \underline{g}_i), \quad \forall \psi \in \mathcal{C} \quad \forall \underline{g} \subseteq W$$

$\underline{h} = \{h_1, \ldots, h_N\}$ *and* $\underline{g} = \{g_1, \ldots, g_N\}$

There are some procedures that iterates the template estimation step and the alignment step.

18.14 Functional Principal Component Analysis (fPCA)

18.14.1 PCA

Given a dataset of N zero-mean multivariate observations in $\mathbb{R}^p$ $(X_1, \ldots, X_N)$, PCA finds the orthonormal directions $\underline{a}_1, \ldots, \underline{a}_p$ of maximum variability for the dataset. Equivalently:

$$\forall k = 1, \ldots, p \quad \text{find } \underline{a}_k = \arg\max_{\underline{a} \in \mathbb{R}^p} Var(a^T X) \quad st \quad \underline{a}^T \underline{a} = 1, \quad \underline{a}_j^T \underline{a} = 0 \quad \forall j < k$$

We can rewrite the problem:

$$\forall k = 1, \ldots, p \quad \text{find } a_k = \arg\max_{\underline{a} \in \mathbb{R}^p} \frac{1}{N}\sum_{i=1}^{N} (\underline{a}^T \underline{X}_i)^2 \quad st \quad \underline{a}^T \underline{a} = 1, \quad \underline{a}_j^T \underline{a} = 0 \quad \forall j < k$$

or:

$$\forall k = 1, \ldots, p \quad \text{find } a_k = \arg\max_{\underline{a} \in \mathbb{R}^p} \frac{1}{N}\sum_{i=1}^{N} \langle a, X_i \rangle^2 \quad st \quad \|\underline{a}\| = 1, \quad \langle \underline{a}_j, \underline{a} \rangle = 0 \quad \forall j < k$$

Now, we assume $N > p$ and the absence of collinearity, so the matrix is full-rank.

The solution is, given S the sample covariance matrix of $X_1, \ldots, X_N$, the principal components are found as the eigenvectors of the matrix S. So, it is equivalent to solve the equation:

$$Se_k = \lambda_k e_k$$

Where u_{ik} is called score, the projection of the observation X_i along e_k direction:

$$u_{ik} = \langle X_i, e_k \rangle = X_i' e_k$$

18.14.2 fPCA adaptation

$$\forall k = 1, \ldots, p \quad \text{find } \xi_k = \arg\max_{\underline{\xi}_k \in H} \frac{1}{N}\sum_{i=1}^{N} \langle \xi, X_i \rangle_H^2 \quad st \quad \|\xi\| = 1, \quad \langle \underline{\xi}_j, \underline{a} \rangle_H = 0 \quad \forall j < k$$

As in multivariate PCA, fPC are related with the eigen decomposition of the functional counterpart of the sample covariance matrix:

$$S\xi_k = \lambda_k \xi_k$$

We only have discrete and noisy realizations of $X_1, \dots, X_N$. For the i-th statistical unit $\{x_{i1}, \dots, x_{in}\} : x_{ij}$ is the value observed at feature s_j.

To choose the p, we look for an elbow in the cumulative percentage of total variance explained by the first functional components:

$$\text{CPV}(p) = \frac{\sum_{k=1}^{p} \hat{\lambda}_k}{\sum_{k=1}^{N} \hat{\lambda}_k}$$

18.14.3 Regularized fPCA

An alternative way to compute fPC is find the K best approximation property (for zero-centered data):

$$(\xi_k)_{j=1}^{K} = \arg \min_{(\{f_k\} st \langle f_k, f_l \rangle_H = \delta_{kl})} E\left[\left\| X - \sum_{k=1}^{K} \langle X, f_k \rangle_H f_k \right\|\right]$$

If $H \equiv L^2$:

$$(\xi_k)_{j=1}^{K} = \arg \min_{(\{f_k\} st \langle f_k, f_l \rangle_H = \delta_{kl})} E\left[\int \left(X - \sum_{k=1}^{K} \langle X, f_k \rangle_H f_k \right)^2\right]$$

An empirical version for the first component would be:

$$\frac{1}{n}\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{n}(x_{ij} - u_i f(s_j))^2$$

This formulation allows us to add a regularized term

$$(\underline{\hat{u}}, \hat{\xi}_1) = \arg\min_{\underline{u}, f} \left\{ \frac{1}{n}\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{n}(x_{ij} - u_i f(s_j))^2 + \lambda \int_{\mathcal{D}} (L_{\mathcal{D}} f - \underline{u})^2 \right\}$$

We can use a generic L_D, for example having a penalization:

$$+\lambda \underline{u}^T \underline{u} \int (f'')^2$$

But we can also use specific physical information about the data (e.g. PDE equation) that we can use to use an specific L_D operator.

References

(2024). (Accessed: 2025-02-15)

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, *57*(1), 289–300. Retrieved 2025-03-16, from http://www.jstor.org/stable/2346101

Benjamini, Y., & Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, *29*(4), 1165–1188. Retrieved 2025-03-17, from http://www.jstor.org/stable/2674075

Cabassi, A., Pigoli, D., Secchi, P., & Carter, P. A. (2017). Permutation tests for the equality of covariance operators of functional data with applications to evolutionary biology. *Electronic Journal of Statistics*, *11*(2), 3815 – 3840. Retrieved from https://doi.org/10.1214/17-EJS1347 doi: 10.1214/17-EJS1347

Galecki, A., & Burzykowski, T. (2013). *Linear mixed effects models using r. : A step-by-step approach.* doi: 10.1007/978-1-4614-3900-4

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning* (2nd ed., Vol. 1). Springer New York, NY. Retrieved from https://link.springer.com/book/10.1007/978-0-387-84858-7 (54 b/w illustrations, 604 illustrations in colour) doi: 10.1007/978-0-387-84858-7

Johnson, R. A., & Wichern, D. W. (2007). *Applied multivariate statistical analysis* (6th ed.). Pearson Prentice Hall.

Lundberg, S., & Lee, S.-I. (2017). *A unified approach to interpreting model predictions.* Retrieved from https://arxiv.org/abs/1705.07874

Lundberg, S. M., Erion, G. G., & Lee, S.-I. (2019). *Consistent individualized feature attribution for tree ensembles.* Retrieved from https://arxiv.org/abs/1802.03888

Pigoli, D., & Sangalli, L. M. (2012). Wavelets in functional data analysis: Estimation of multidimensional curves and their derivatives. *Computational Statistics & Data Analysis*, *56*(6), 1482-1498. Retrieved from https://ideas.repec.org/a/eee/csdana/v56y2012i6p1482-1498.html doi: 10.1016/j.csda.2011.12.016

Sangalli, L. M., Secchi, P., Vantini, S., & Veneziani, A. (2009). Efficient estimation of three-dimensional curves and their derivatives by free-knot regression splines, applied to the analysis of inner carotid artery centrelines. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, *58*(3), 285-306. Retrieved from https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9876.2008.00653.x doi: https://doi.org/10.1111/j.1467-9876.2008.00653.x

Sohil, F., Sohail, M., & Shabbir, J. (2021, 09). An introduction to statistical learning with applications in r: by gareth james, daniela witten, trevor hastie, and robert tibshirani, new york, springer science and business media, 2013, 41.98, $eisbn : 978-1-4614-7137-7. Statistical\ Theory\ and\ Related\ Fields, 6, 1-1. doi :$

Wasserstein, R. L., & Lazar, N. A. (2016). The asa statement on p-values: Context, process, and purpose. *The American Statistician*, *70*(2), 129–133. Retrieved from https://doi.org/10.1080/00031305.2016.1154108 doi: 10.1080/00031305.2016.1154108

Zhou, S., & Shen, X. (2001). Spatially adaptive regression splines and accurate knot selection schemes. *Journal of the American Statistical Association*, *96*(453), 247–259. Retrieved 2025-06-08, from http://www.jstor.org/stable/2670363